Supplementary Material

What does a Bayes-filtered transformer believe? A predictive Monte Carlo approach

This document collects additional PMC reliability diagnostics and the complete set of PMC-recovered marginal densities and CDFs for all three task families across all task diversities. It accompanies the main paper; the main paper reproduces a representative subset. All figures are generated by the code at https://github.com/afiq-aswadi/bft-pmc.

1 Additional PMC reliability diagnostics

These figures supplement the main paper's reliability appendix. They report prior and posterior recovery against both references as rollout length or count varies, including task diversities near the main experiments' transitions. Green denotes the memorizing reference and blue the generalizing reference. Each posterior run uses one in-distribution prompt of length 32; prior runs use empty prompts. The fixed reference sample contains 1024 draws for each reference.

Length curves use $R = 1024$ and recompute latent estimates from prefixes of the same saved rollouts. The dashed line marks $N = 223$. Count curves use the indicated terminal horizon and 50 subsets without replacement for $R < 1024$; bars show conditional subset standard deviations, and the $R = 1024$ endpoint is a single full-set estimate. Dotted curves show fresh reference-sample benchmarks. Negative unbiased energy-distance estimates are retained. A logarithmic vertical axis is used for length curves when all distances are positive and span a large range.

LR lengths count covariate–response pairs; BAU and Markov lengths count observations excluding BOS. Posterior runs beyond $N = 223$ exceed the training context. The empty-prompt prior boundary is 256 pairs for LR and 255 observations for BAU/Markov. Separate horizon runs may use different realized prompts and sample sets and are not paired comparisons. NoPE has no positional signal; learned embeddings repeat their last trained slots beyond the context (separate covariate/response slots in LR), and RoPE uses previously unseen rotary positions. These are sensitivity analyses of the recorded runs, not controlled causal comparisons of positional encoding.

Table 1: Index of supplementary reliability curves. Each group includes prior/posterior results and both distance metrics.

Family / encoding	Task diversities	Horizon groups
LR / NoPE	$32, 128, 256$	$223, 500$
BAU / NoPE	$32, 256, 512$	$223, 500$
Markov	$16, 32, 64, 128$	223
LR and BAU / learned, RoPE	32	Length to 500; counts at $223, 500$

1.1 Linear regression: horizon 223

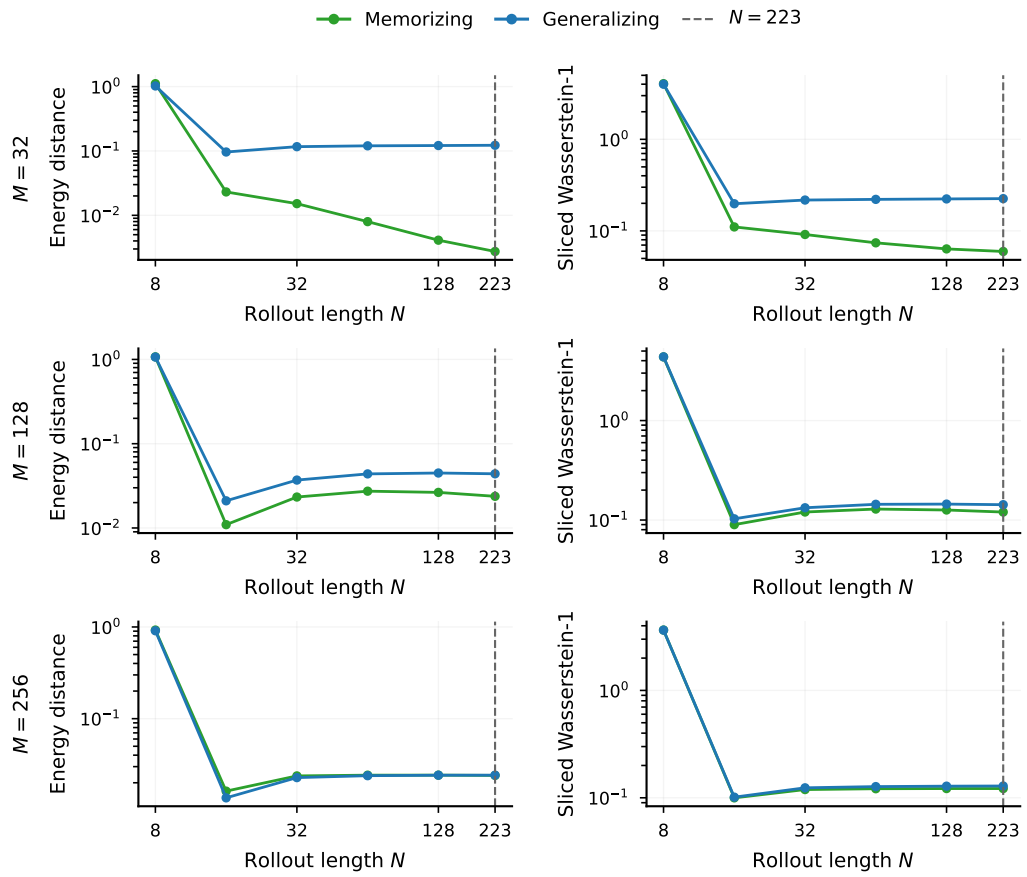

Figure 1: **Prior recovery versus rollout length.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 223$ at $R = 1024$; the dashed line marks $N = 223$. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

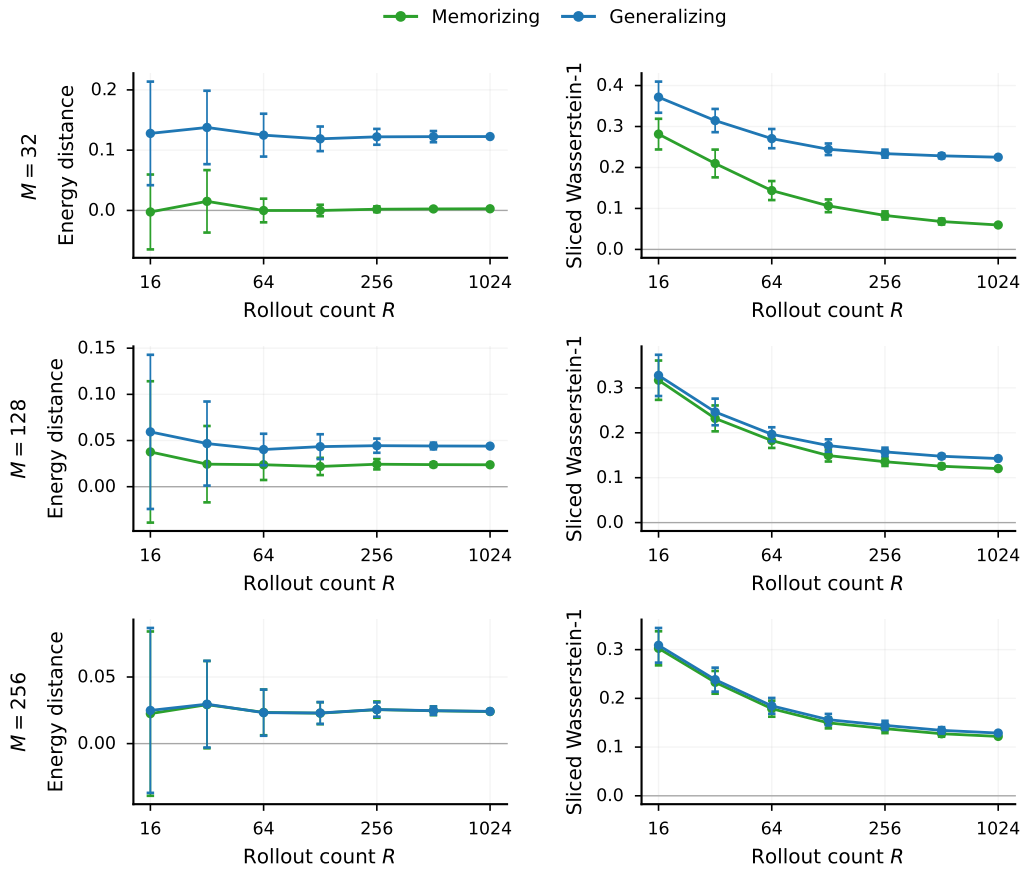

Figure 2: **Prior recovery versus rollout count.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

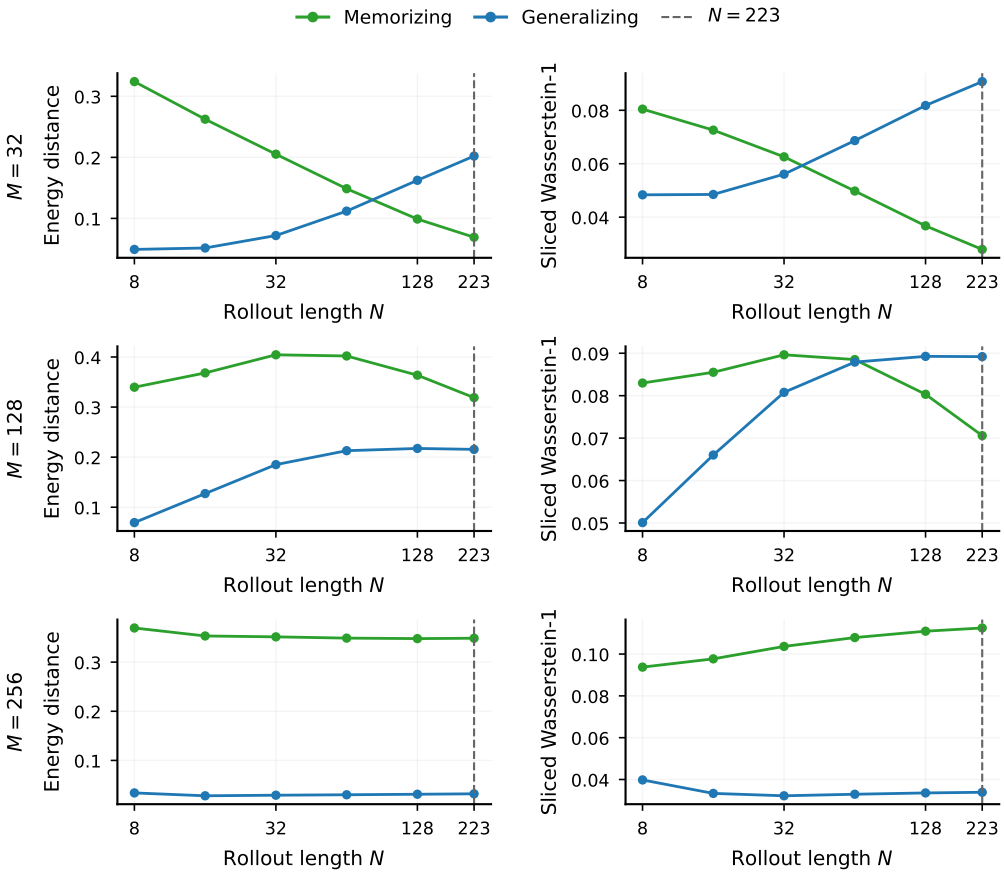

Figure 3: **Posterior recovery versus rollout length.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 223$ at $R = 1024$; the dashed line marks $N = 223$. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

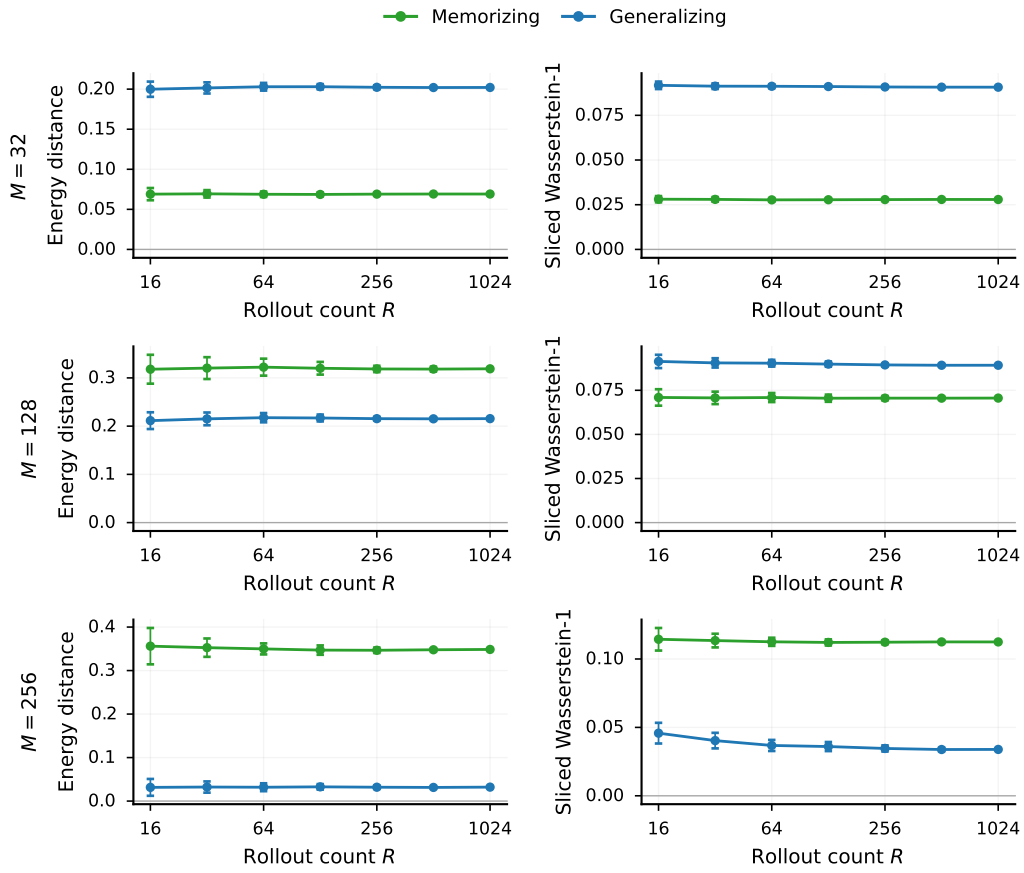

Figure 4: **Posterior recovery versus rollout count.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

1.2 Linear regression: horizon 500

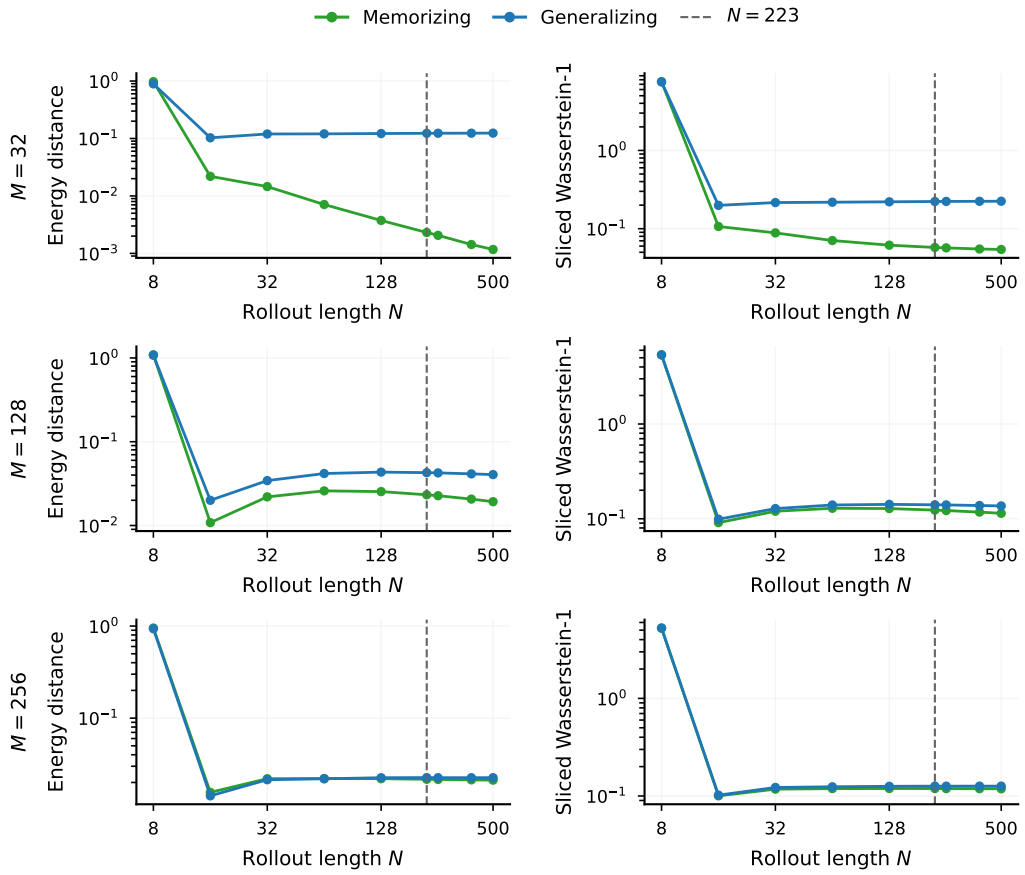

Figure 5: **Prior recovery versus rollout length.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 500$ at $R = 1024$; the dashed line marks $N = 223$. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

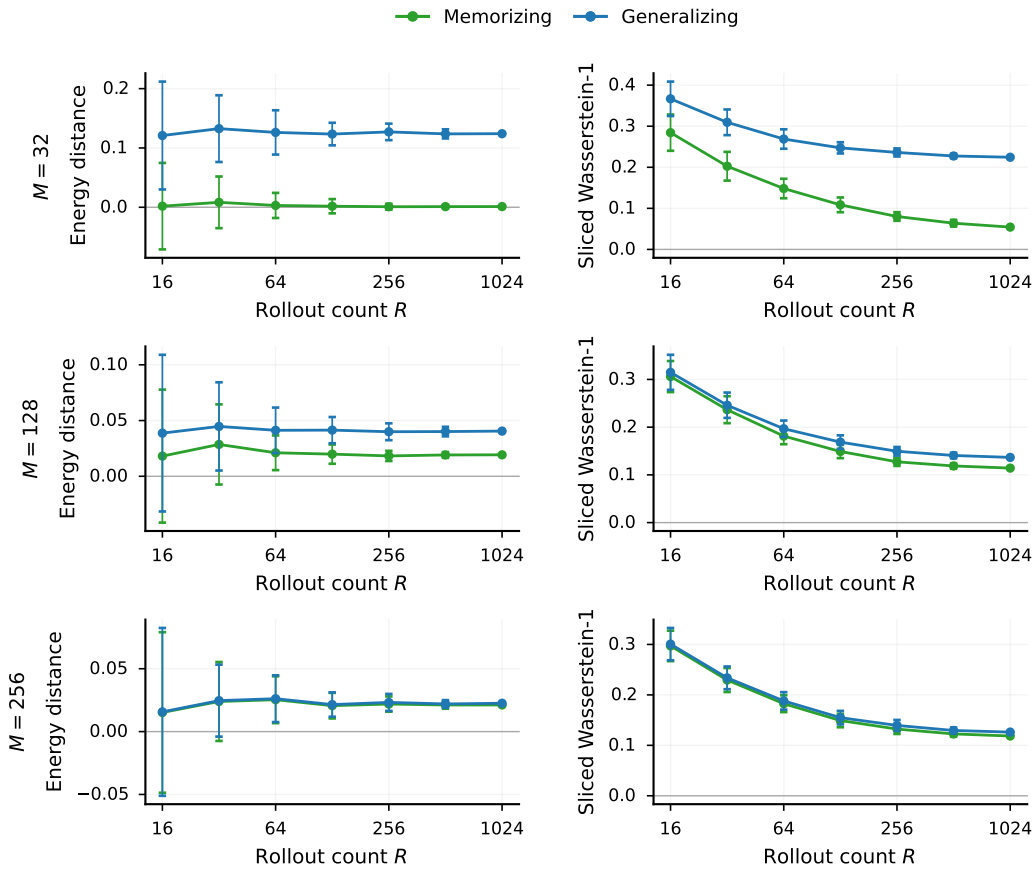

Figure 6: **Prior recovery versus rollout count.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 500$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

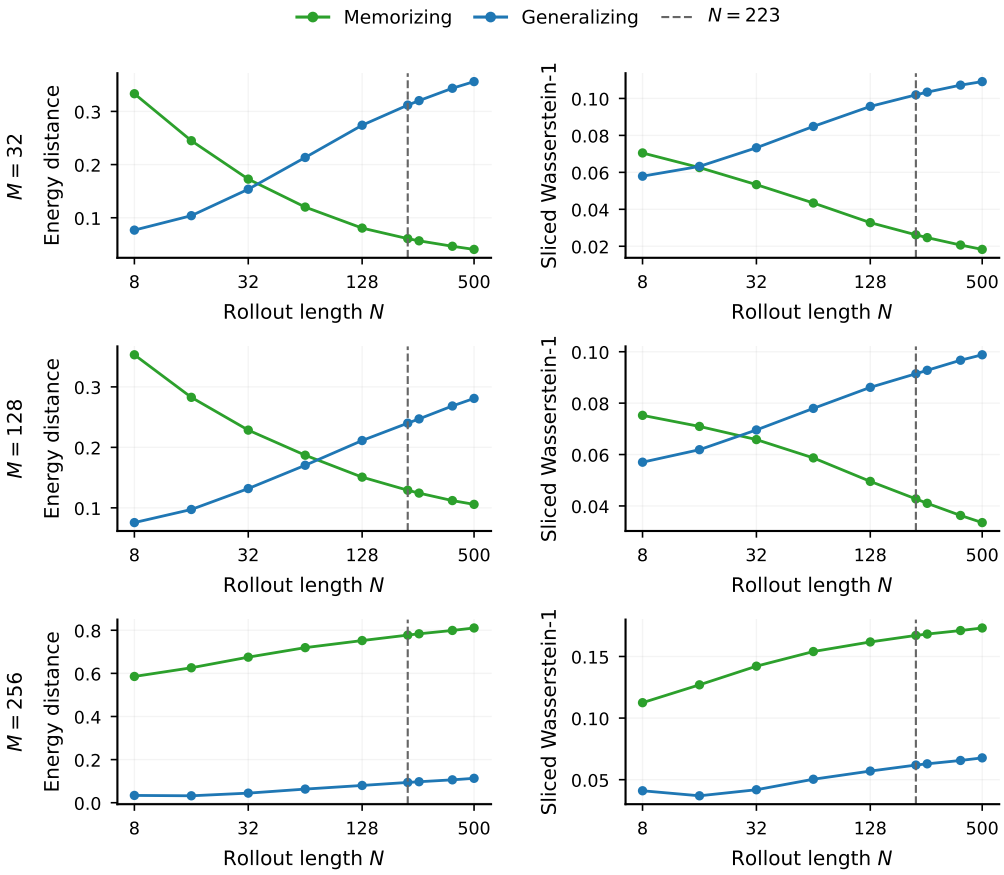

Figure 7: **Posterior recovery versus rollout length.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 500$ at $R = 1024$; the dashed line marks $N = 223$. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

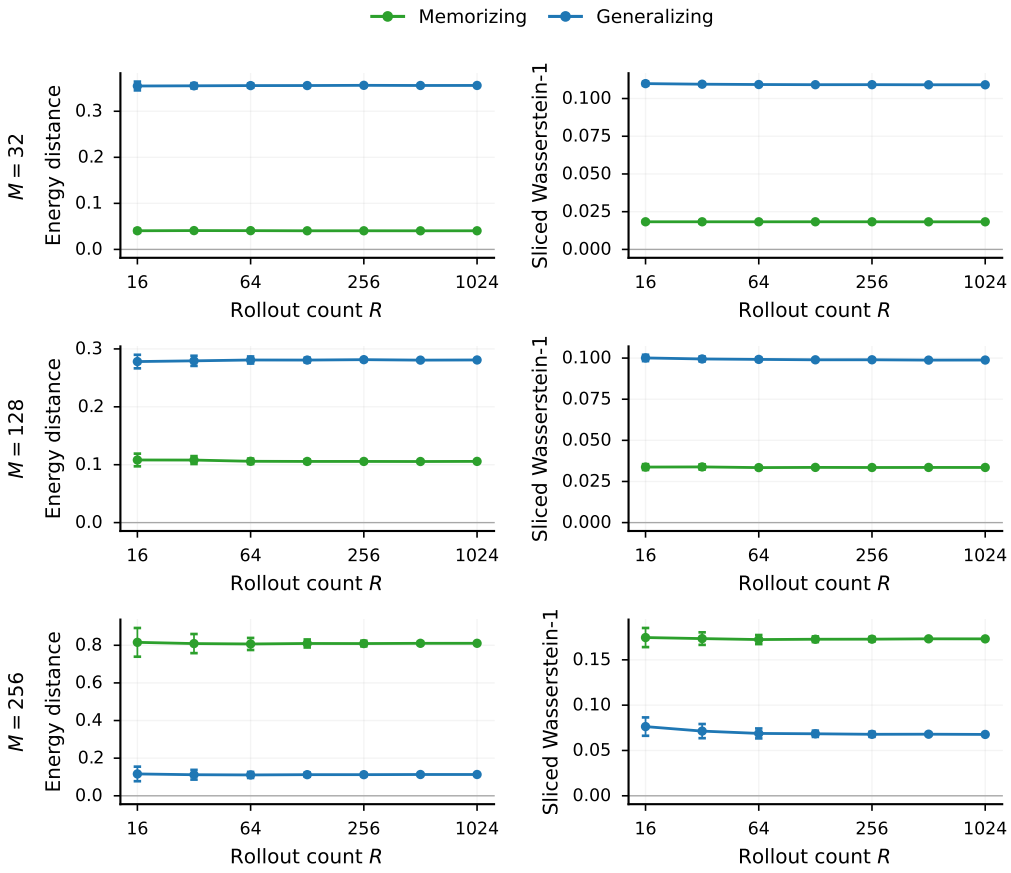

Figure 8: **Posterior recovery versus rollout count.** Rows: $M = 32$, $M = 128$, $M = 256$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 500$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

1.3 Balls-and-urns: horizon 223

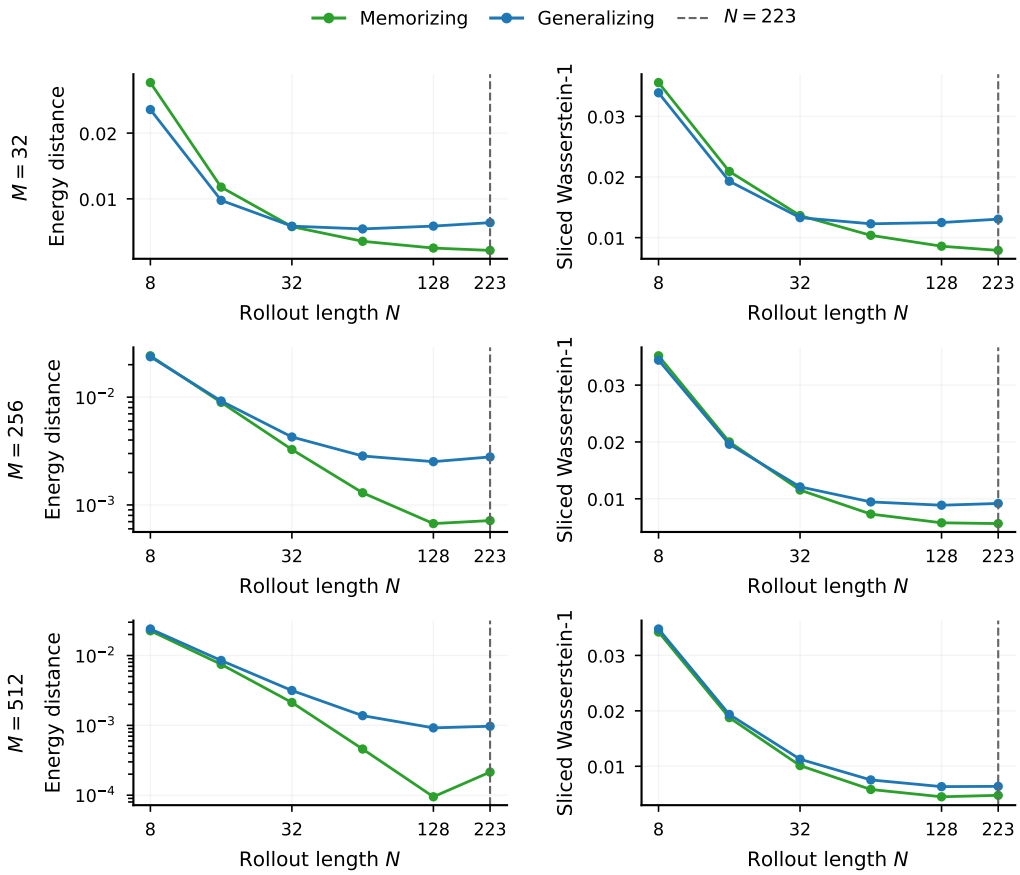

Figure 9: **Prior recovery versus rollout length.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 223$ at $R = 1024$; the dashed line marks $N = 223$. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

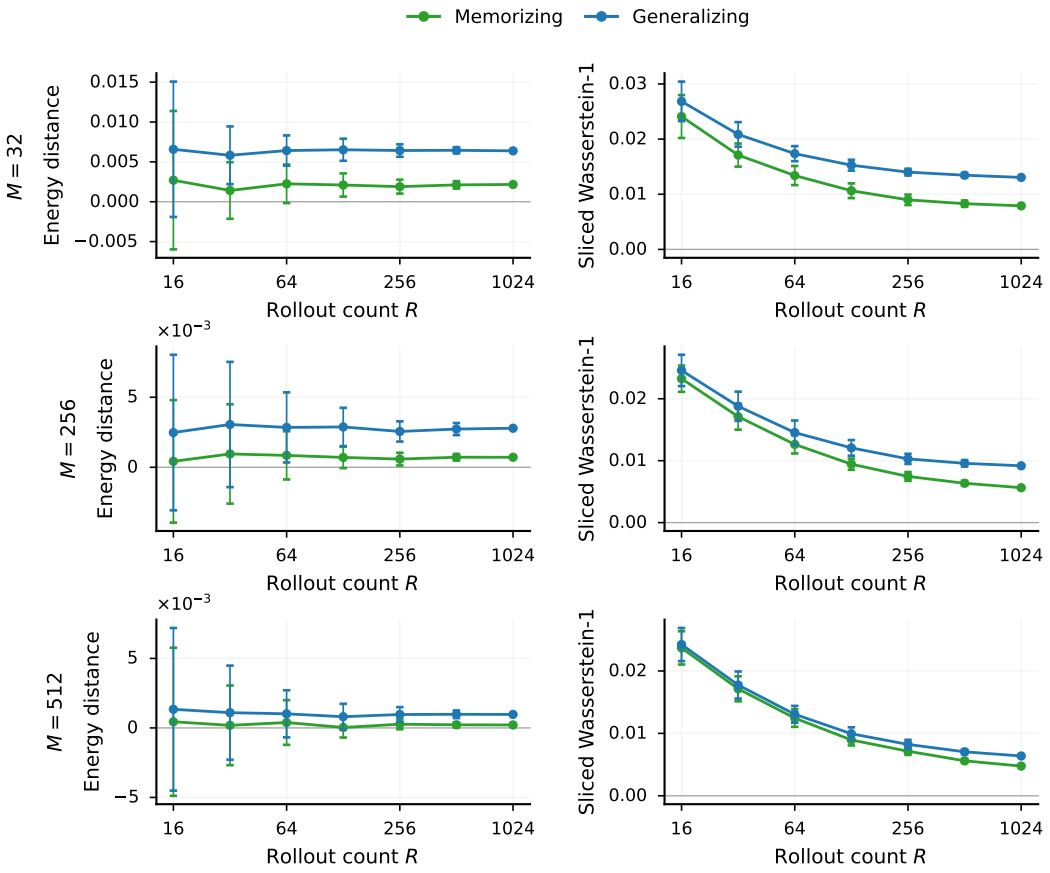

Figure 10: **Prior recovery versus rollout count.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

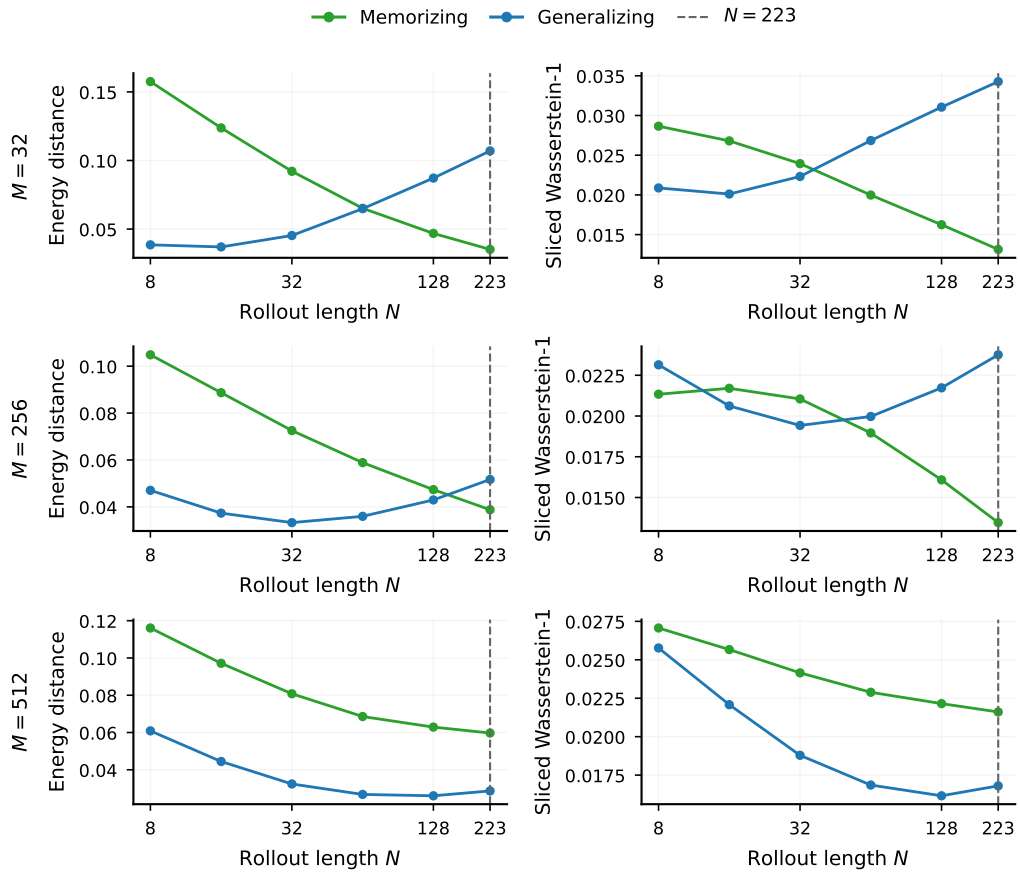

Figure 11: **Posterior recovery versus rollout length.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 223$ at $R = 1024$; the dashed line marks $N = 223$. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

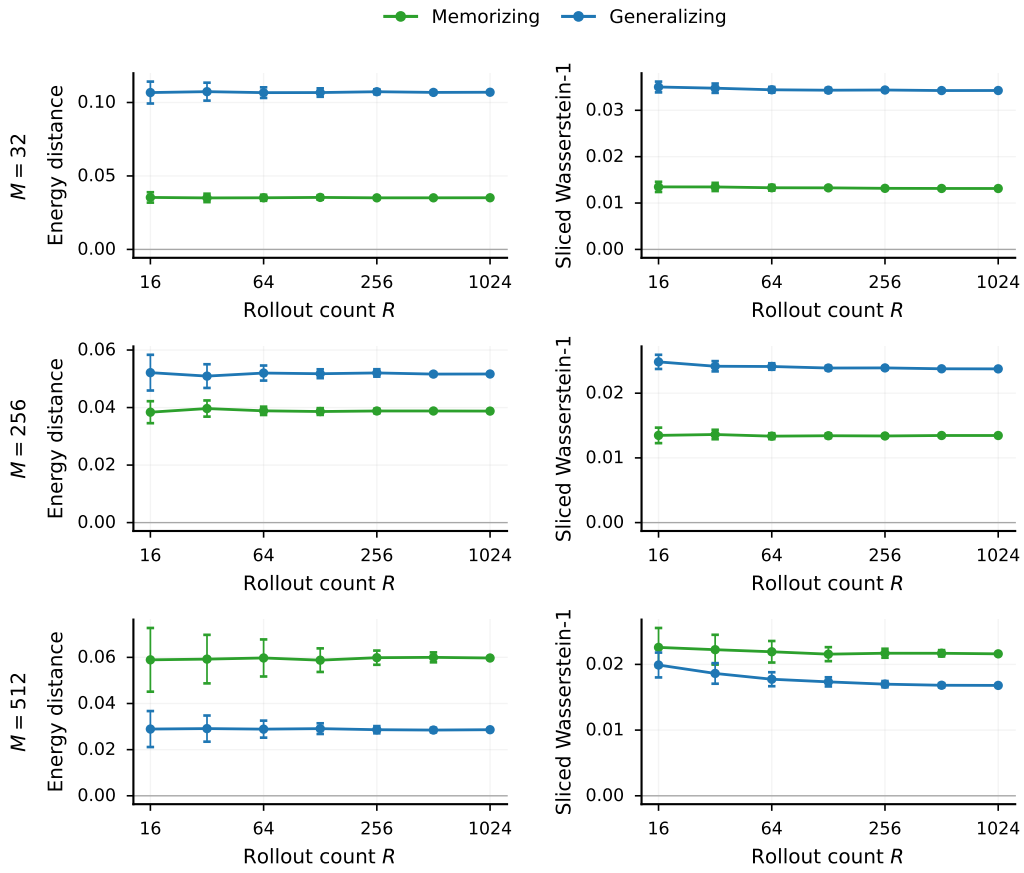

Figure 12: **Posterior recovery versus rollout count.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

1.4 Balls-and-urns: horizon 500

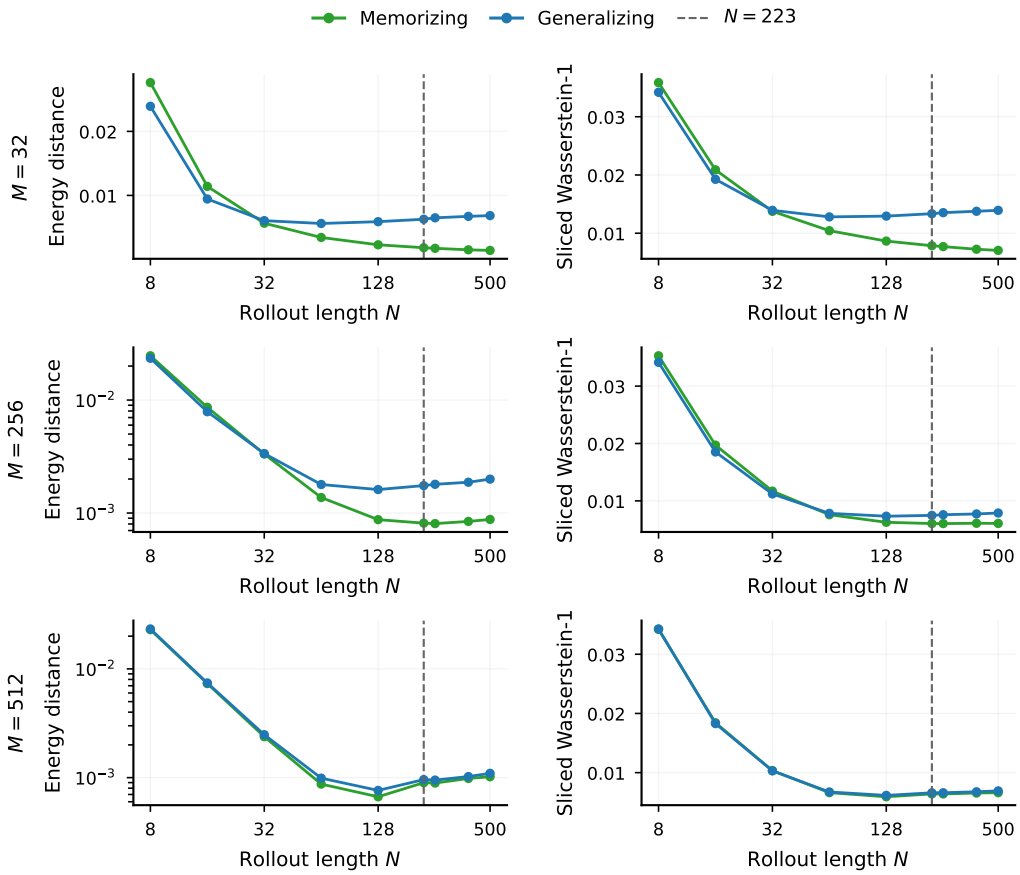

Figure 13: **Prior recovery versus rollout length.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 500$ at $R = 1024$; the dashed line marks $N = 223$. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

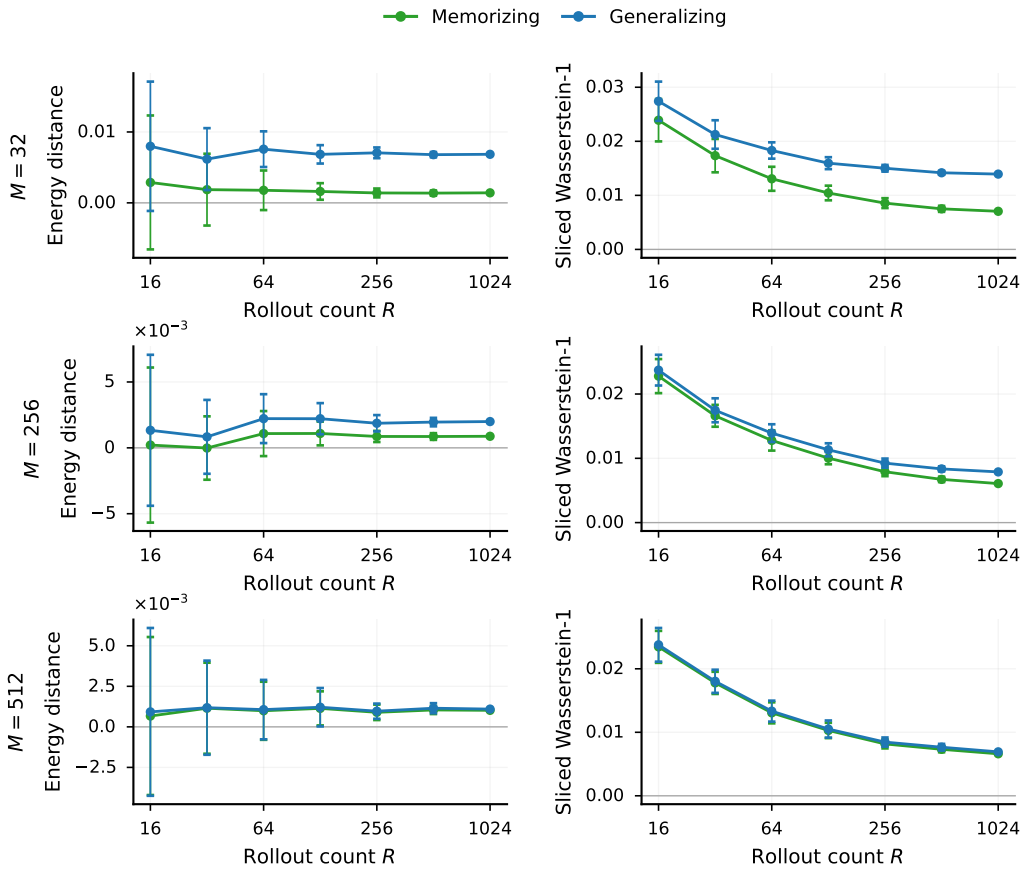

Figure 14: **Prior recovery versus rollout count.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 500$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

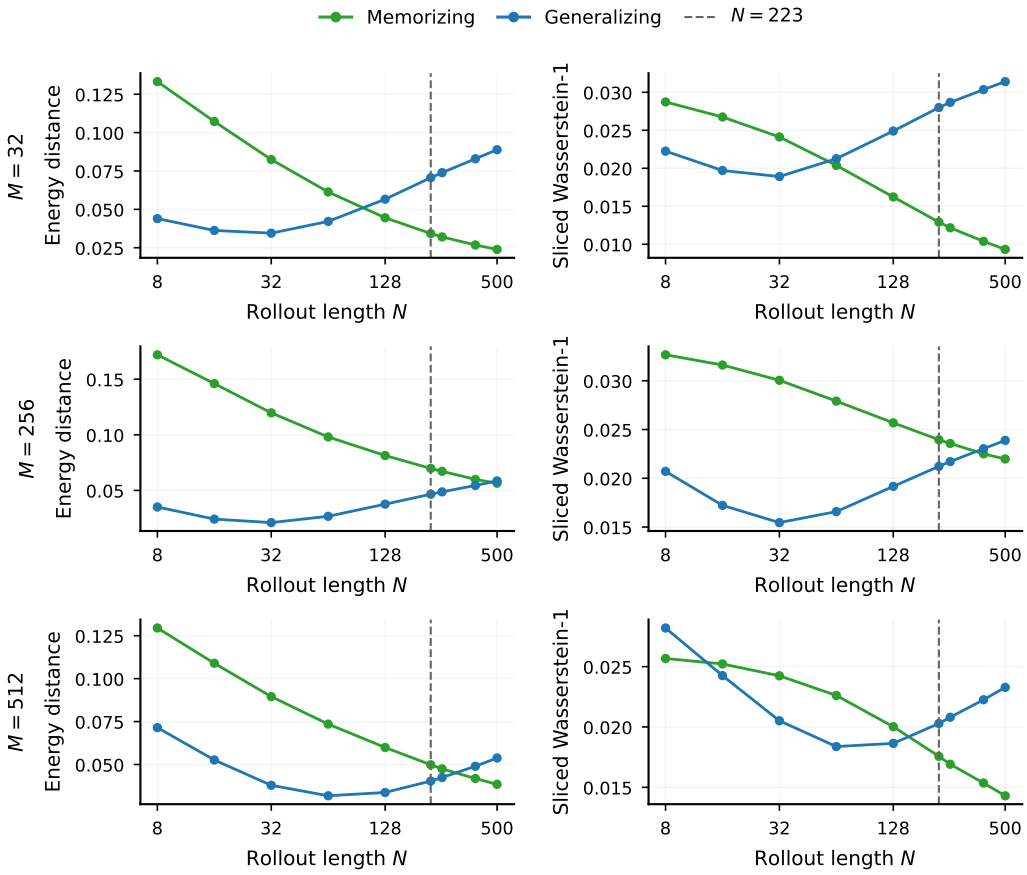

Figure 15: **Posterior recovery versus rollout length.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 500$ at $R = 1024$; the dashed line marks $N = 223$. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

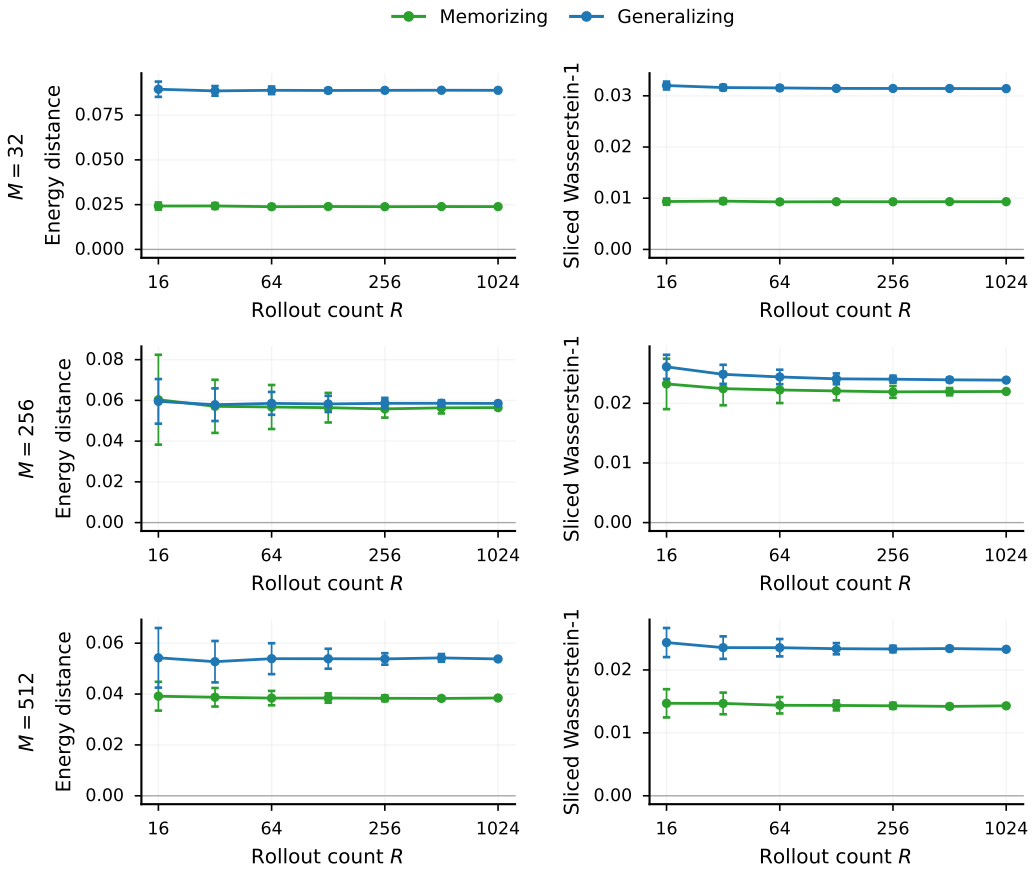

Figure 16: **Posterior recovery versus rollout count.** Rows: $M = 32$, $M = 256$, $M = 512$. No positional encoding. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 500$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

1.5 Markov chains: horizon 223

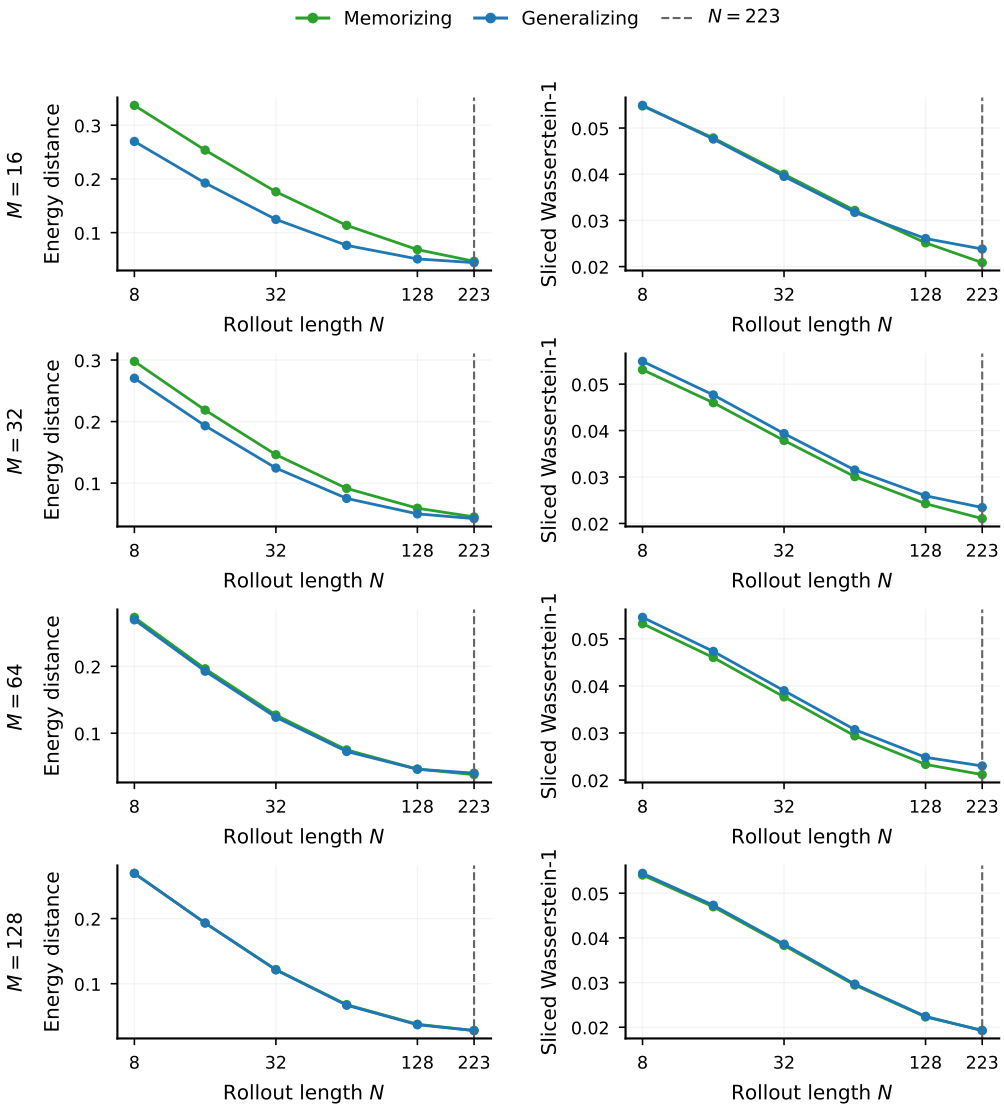

Figure 17: **Prior recovery versus rollout length.** Rows: $M = 16$, $M = 32$, $M = 64$, $M = 128$. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 223$ at $R = 1024$; the dashed line marks $N = 223$. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

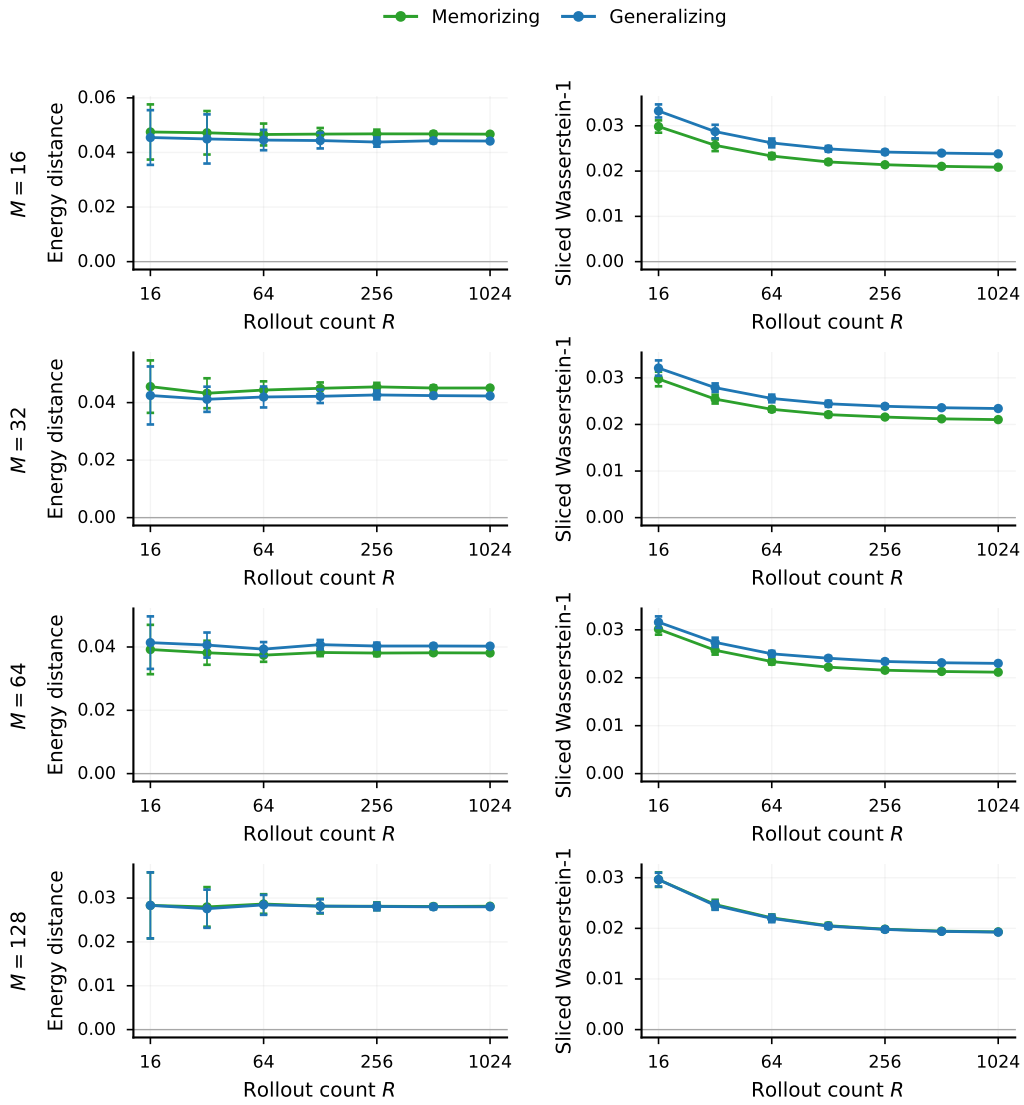

Figure 18: **Prior recovery versus rollout count.** Rows: $M = 16$, $M = 32$, $M = 64$, $M = 128$. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

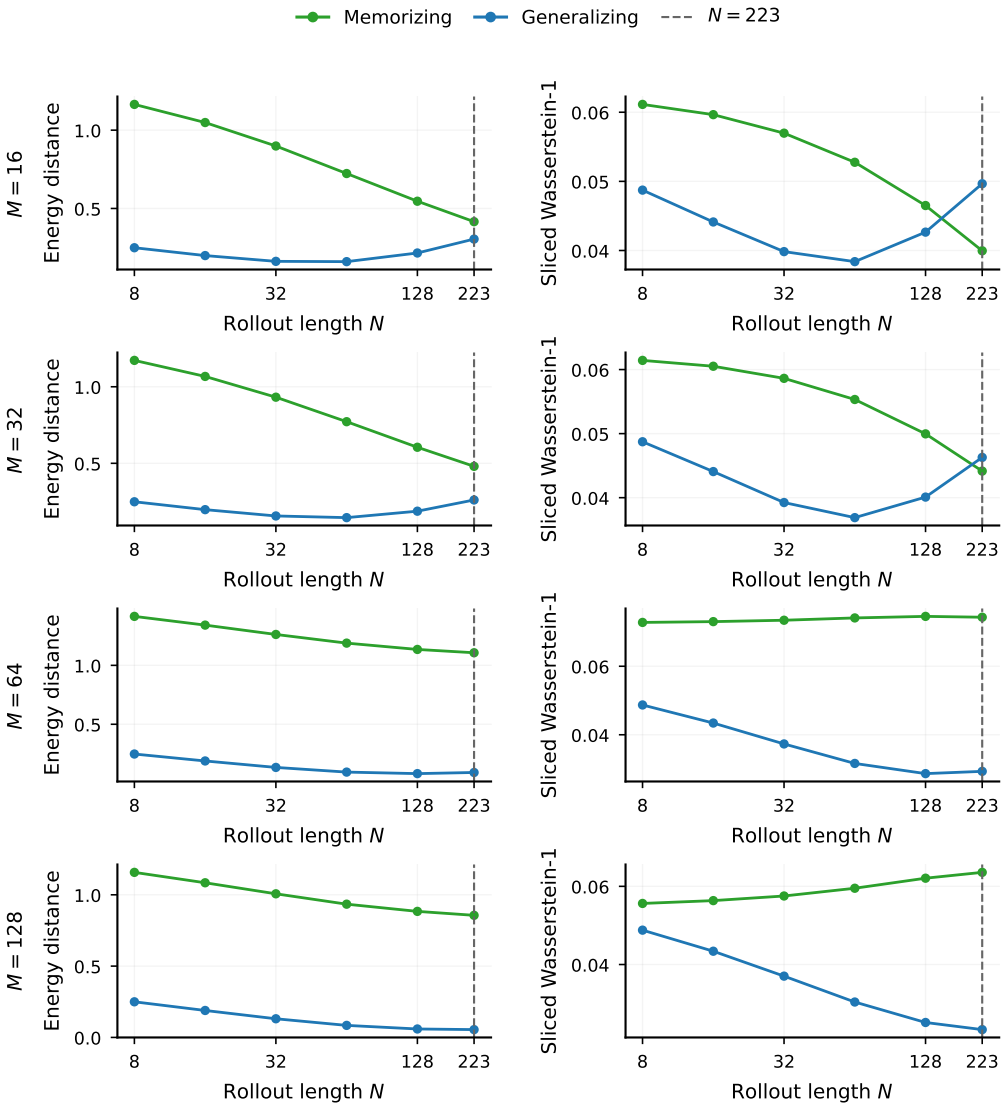

Figure 19: **Posterior recovery versus rollout length.** Rows: $M = 16$, $M = 32$, $M = 64$, $M = 128$. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 223$ at $R = 1024$; the dashed line marks $N = 223$. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

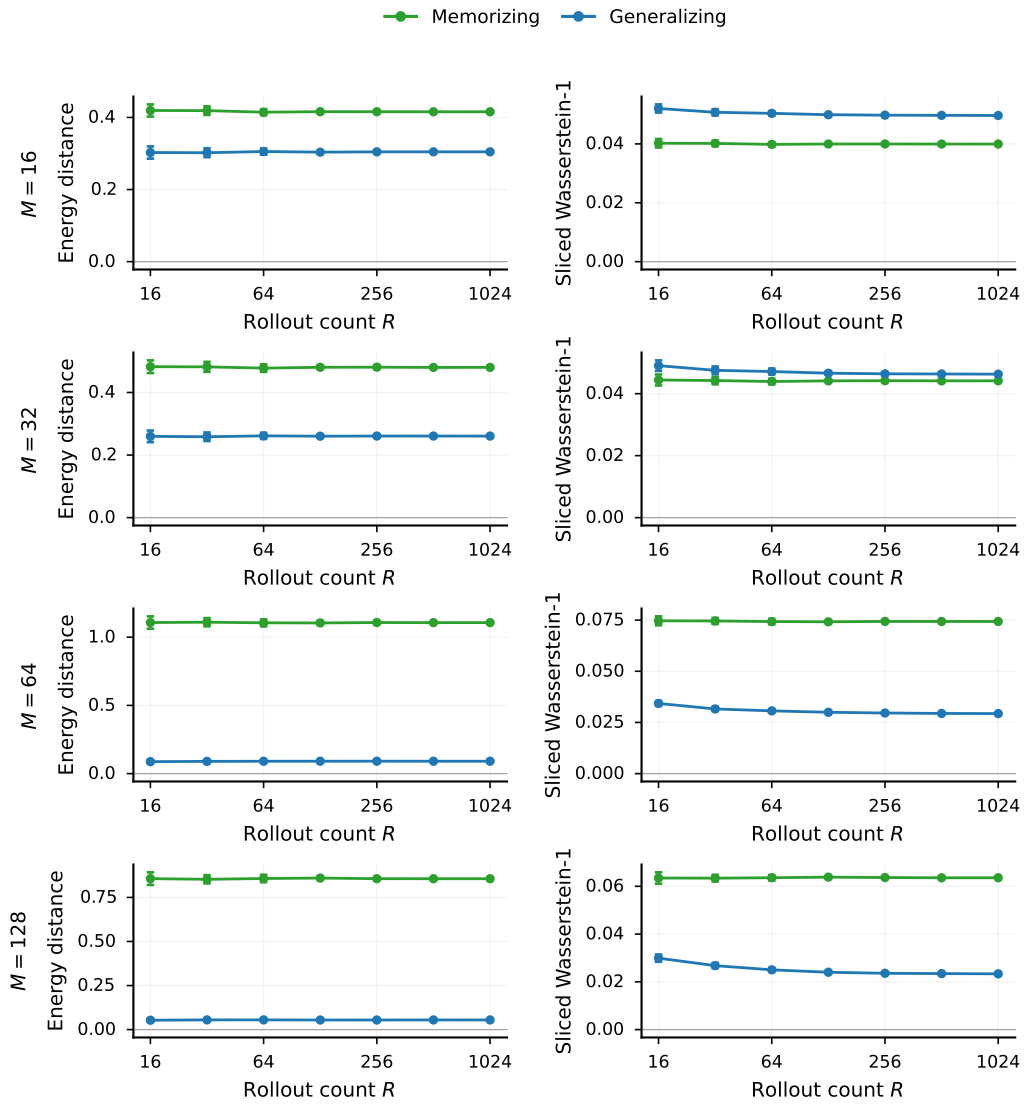

Figure 20: **Posterior recovery versus rollout count.** Rows: $M = 16$, $M = 32$, $M = 64$, $M = 128$. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

1.6 Learned and rotary positional encodings

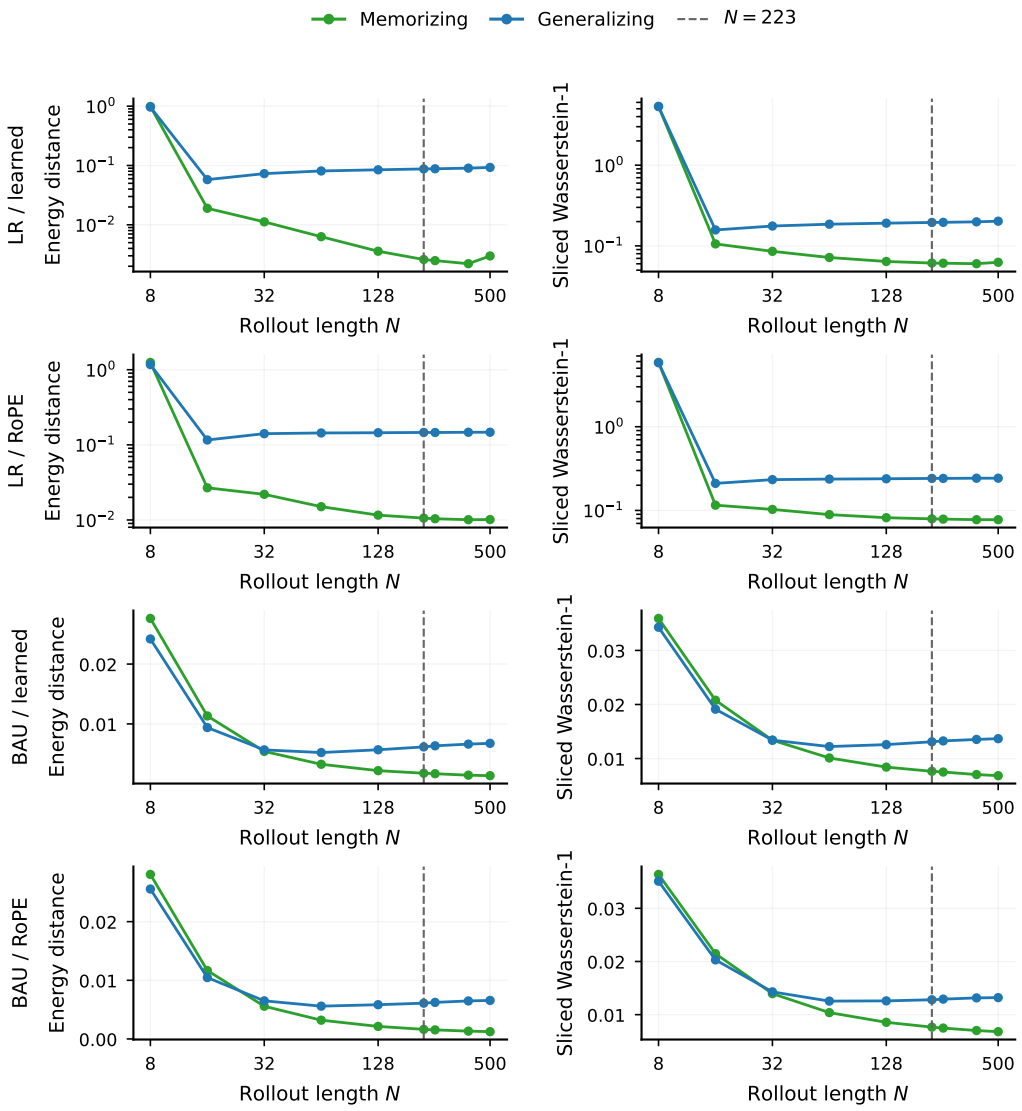

Figure 21: **Prior recovery versus rollout length.** Rows: LR learned, LR RoPE, BAU learned, BAU RoPE, all at $M = 32$. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 500$ at $R = 1024$; the dashed line marks $N = 223$. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

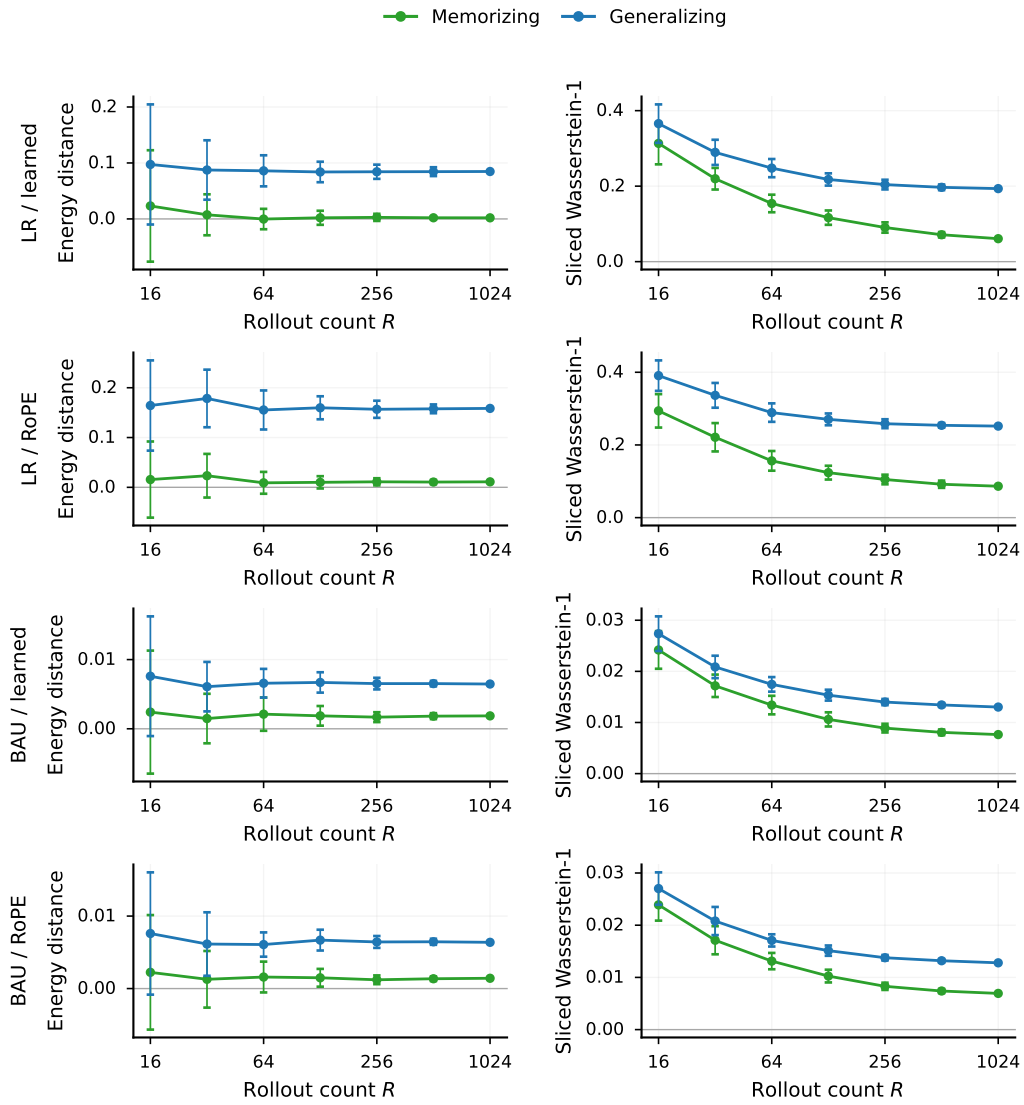

Figure 22: **Prior recovery versus rollout count.** Rows: LR learned, LR RoPE, BAU learned, BAU RoPE, all at $M = 32$. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

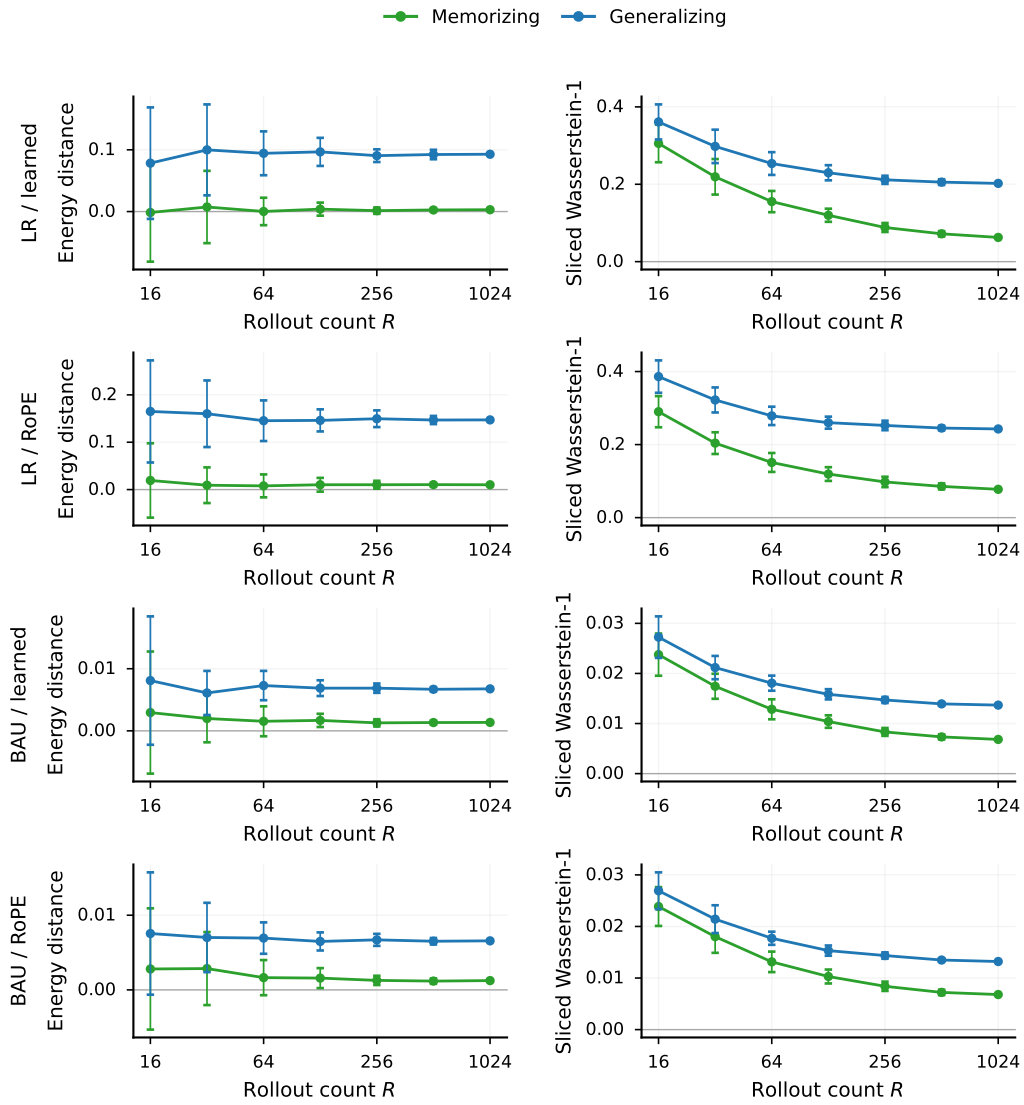

Figure 23: **Prior recovery versus rollout count.** Rows: LR learned, LR RoPE, BAU learned, BAU RoPE, all at $M = 32$. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 500$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

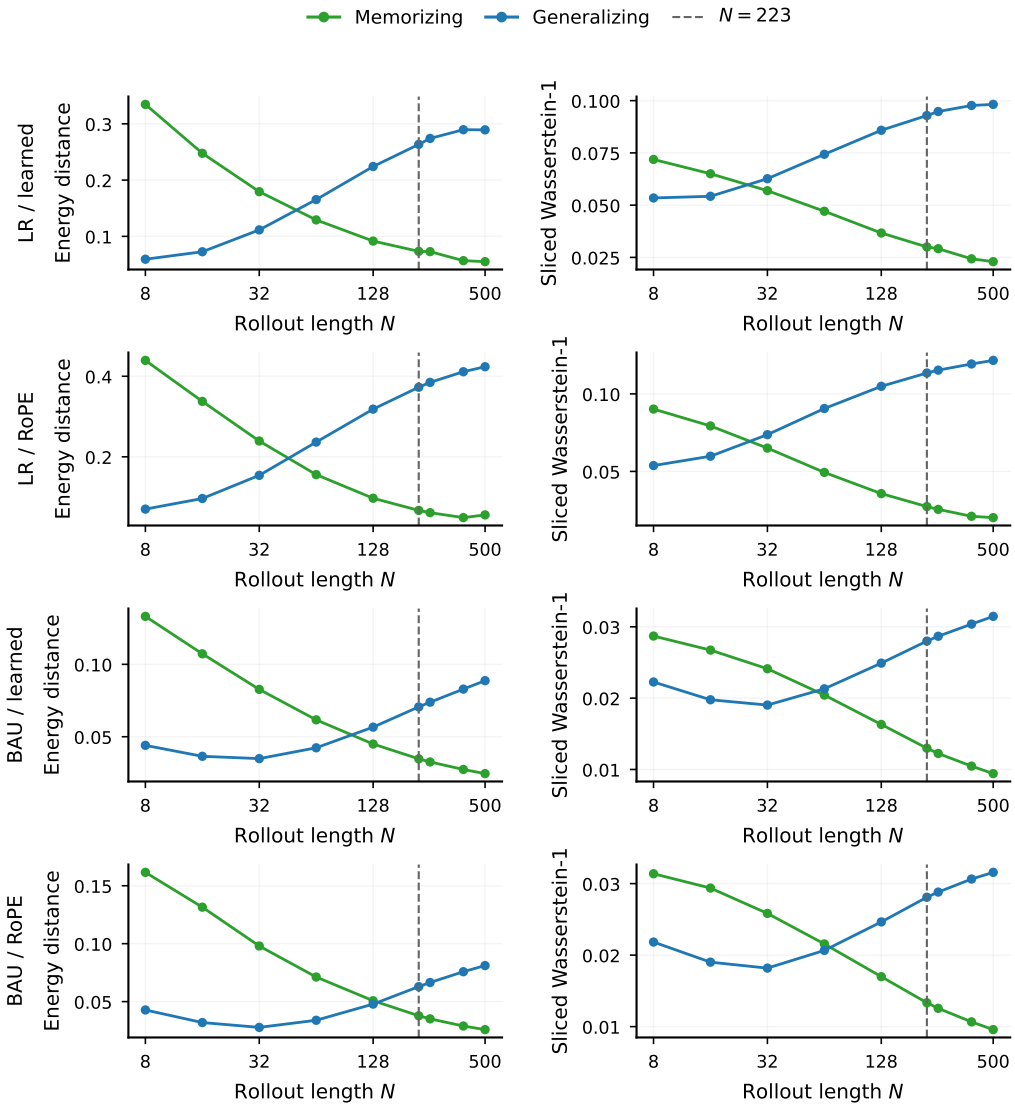

Figure 24: **Posterior recovery versus rollout length.** Rows: LR learned, LR RoPE, BAU learned, BAU RoPE, all at $M = 32$. Columns: energy distance and sliced Wasserstein-1. Rollouts extend to $N = 500$ at $R = 1024$; the dashed line marks $N = 223$. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

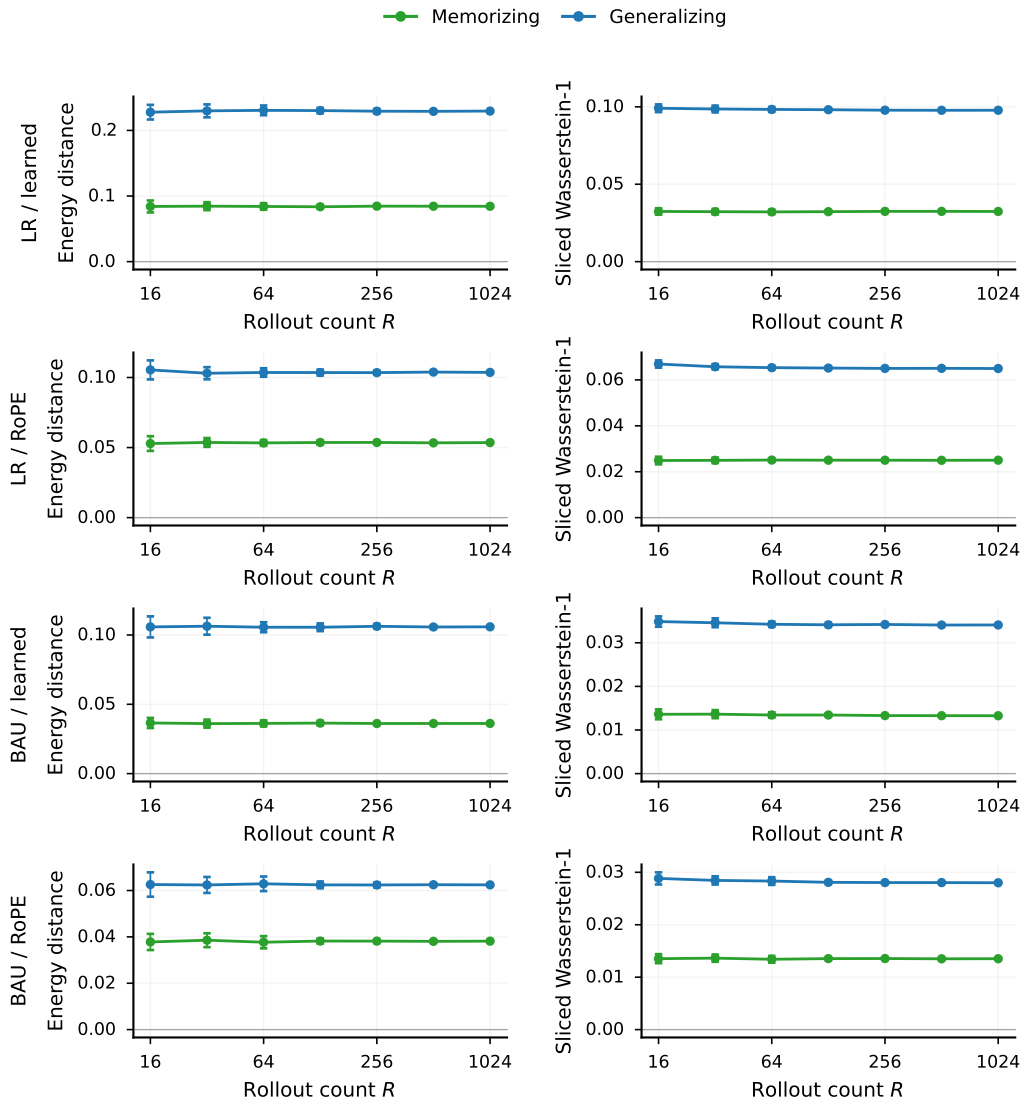

Figure 25: **Posterior recovery versus rollout count.** Rows: LR learned, LR RoPE, BAU learned, BAU RoPE, all at $M = 32$. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 223$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

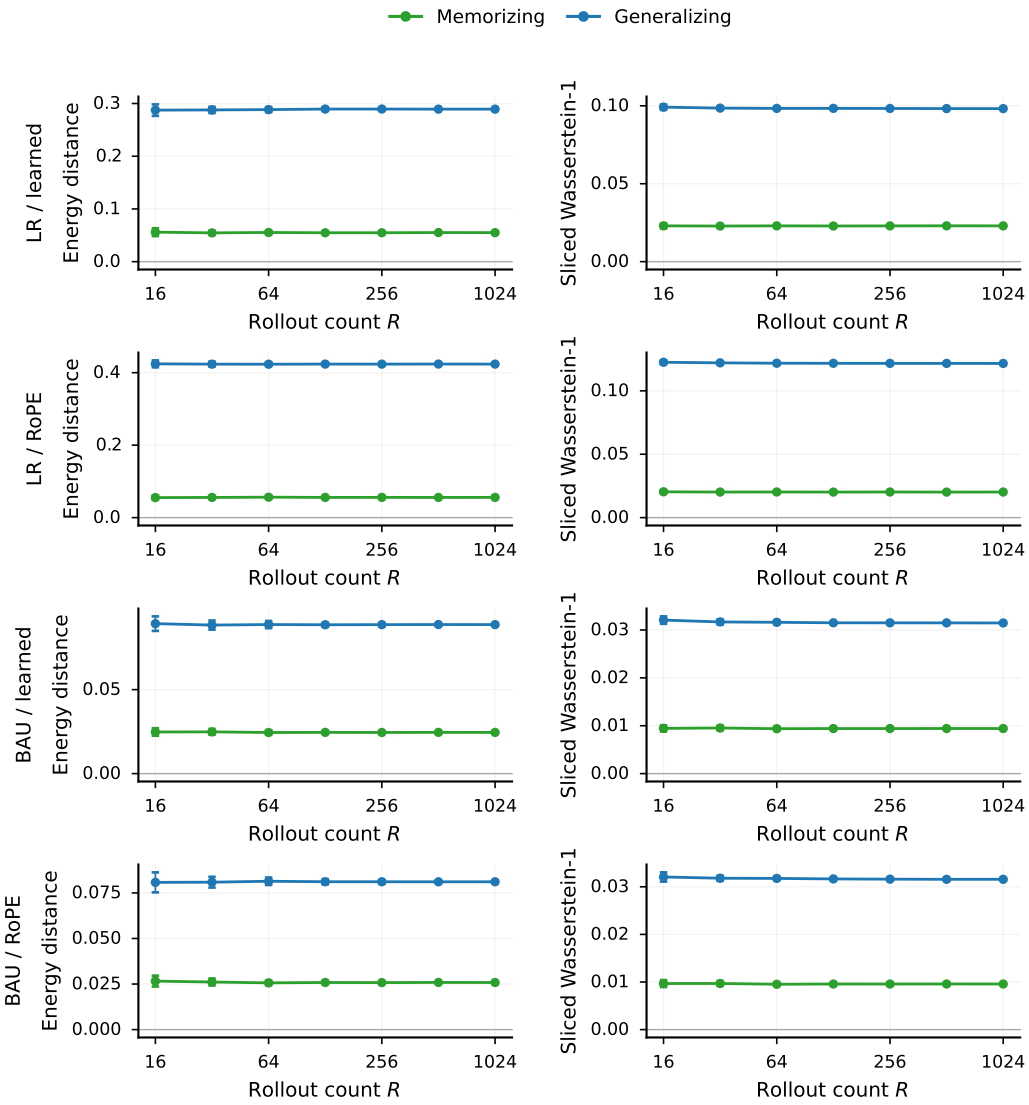

Figure 26: **Posterior recovery versus rollout count.** Rows: LR learned, LR RoPE, BAU learned, BAU RoPE, all at $M = 32$. Columns: energy distance and sliced Wasserstein-1. Fixed horizon $N = 500$. Bars show subset standard deviations; dotted curves show reference benchmarks. The full-set endpoint has no estimated spread. The extended run exceeds the training context at its longest horizons. Each posterior run uses one prompt; these curves do not independently locate a task-diversity threshold.

2 Marginal densities and CDFs from PMC

How to read the panels. This appendix collects the full marginal distributions recovered by PMC for each task family. Every panel overlays three objects, colored consistently across all figures in this paper: the PMC samples (yellow), the memorizing reference (green), and the generalizing reference (blue). In prior panels the green and blue objects are the priors themselves, and the legend labels them Π_M and Π_∞. In posterior panels they are the posteriors induced by those priors, and the legend labels them $\Pi^{\mathrm{mem}}(\cdot \mid c)$ and $\Pi^{\mathrm{gen}}(\cdot \mid c)$. PMC densities are histograms with kernel density overlays, PMC CDFs are empirical CDFs, and the generalizing reference is always an analytic curve.

The memorizing reference is a discrete distribution on the M training tasks, and it is rendered in two ways. In *prior* density panels at low task diversity ($M \leq 16$ for linear regression, $M \leq 8$ for balls-and-urns and Markov chains), each training task is drawn as a green vertical line spanning the full panel height. The prior weights are uniform, so these lines are identical. In every other density panel (priors at larger M, and all posteriors), the memorizing reference is a normalized histogram over the training tasks, weighted by their prior or posterior weights. In a posterior panel at low M the histogram bars are narrow and can resemble vertical lines of varying heights: the height of each bar is proportional to the posterior weight of that training task given the prompt, so the tallest bar marks the task the posterior favors. In CDF panels the memorizing reference is always the closed-form weighted step function.

Reading the figures for Markov chains. For Markov chains the latent task is a 10×10 transition matrix, so each figure is a matrix of panels mirroring the transition matrix: the panel in row $\mathrm{src} = a$ and column $\mathrm{dst} = b$ shows the marginal distribution of the transition probability $Q_{a,b}$ from state a to state b. All panels of one matrix share a single x-range so that identical underlying distributions render identically across panels.

Table **??** indexes which figure corresponds to which combination of family, quantity, prompt source, and plot type. Each figure shows marginal densities (or CDFs) of the PMC-recovered distribution against the memorizing and generalizing reference laws. The linear regression and balls-and-urns figures stitch together marginal plots across task diversity. For Markov chains, the index table points to the $M = 4$ figures, and the equivalent figures for the remaining task diversities ($M = 8$ through $M = 2048$) follow in this appendix.

Table 2: **Index of marginal figures.** Each entry points to a figure in this appendix. Posterior figures condition on an out-of-distribution (ood) or in-distribution (id) prompt. Entries for Markov chains show results for a fixed task diversity $M = 4$.

	Marginal density			**Marginal CDF**		
		Posterior			Posterior	
Family	Prior	ood	id	Prior	ood	id
Linear regression	**??**	**??**	**??**	**??**	**??**	**??**
Balls-and-urns	**??**	**??**	**??**	**??**	**??**	**??**
Markov chains	**??**	**??**	**??**	**??**	**??**	**??**

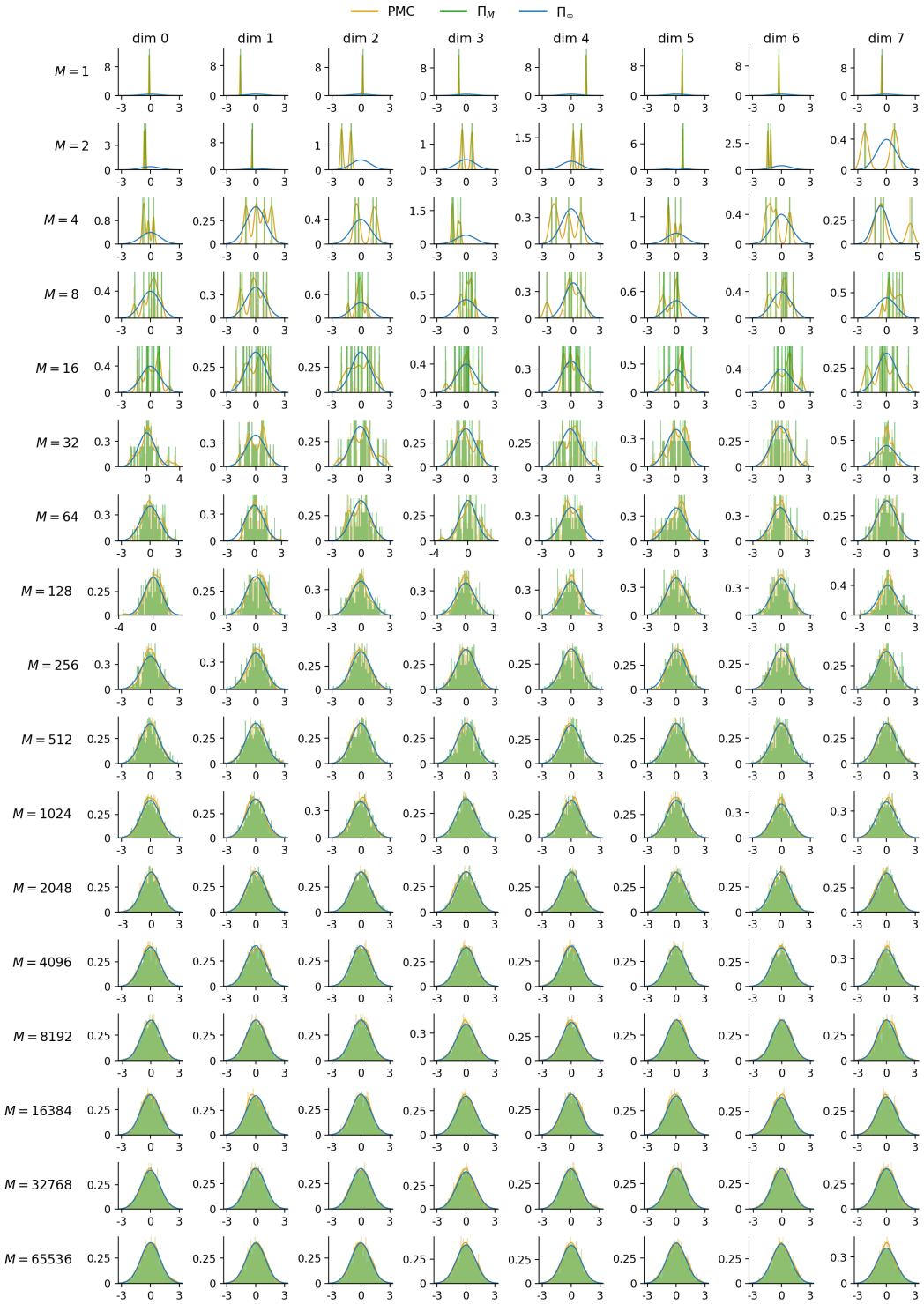

Figure 27: **Linear regression: PMC prior, marginal density.** Rows sweep task diversity M.

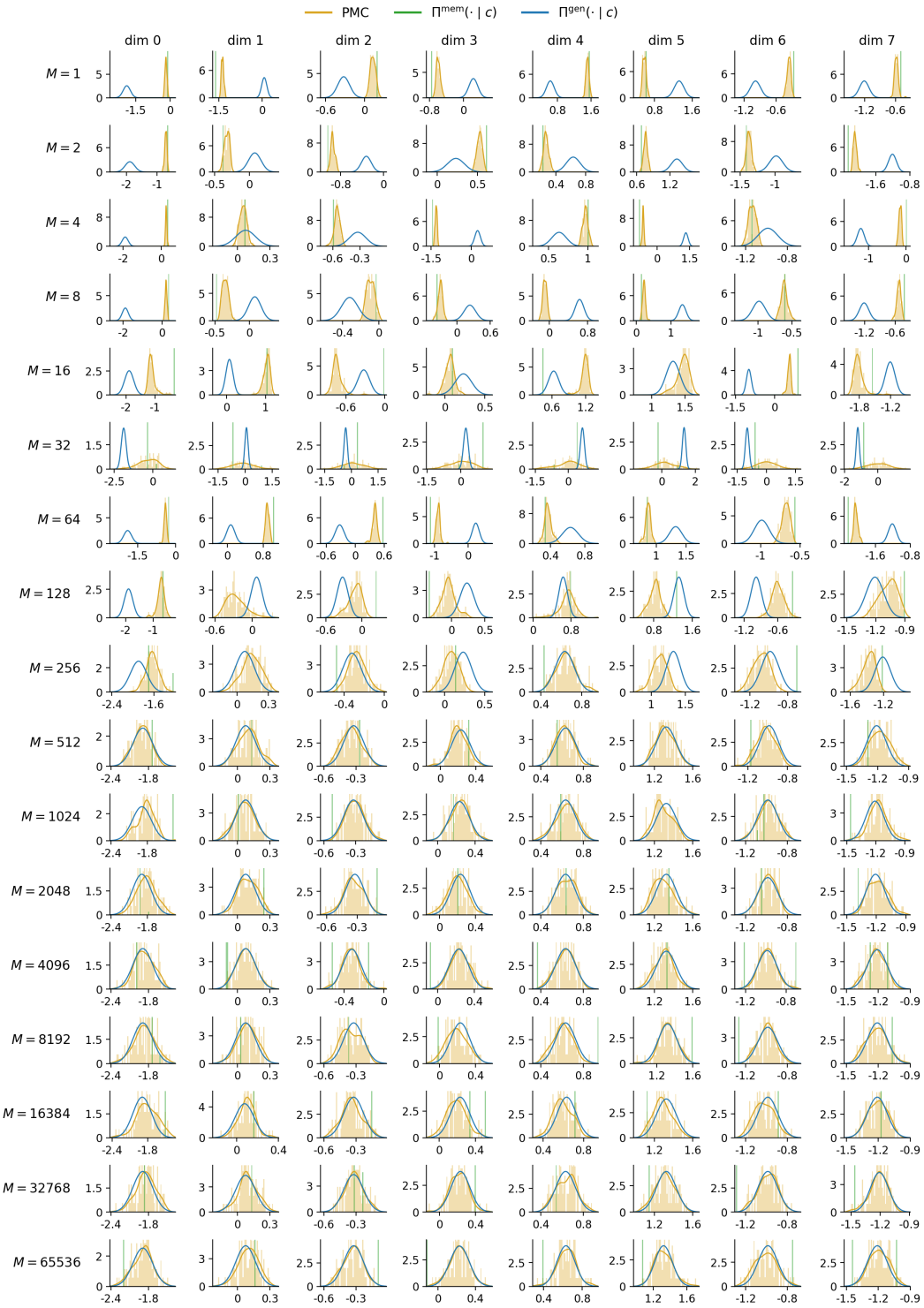

Figure 28: **Linear regression: PMC posterior, marginal density.** Out-of-distribution prompt of length 8. Rows sweep task diversity M.

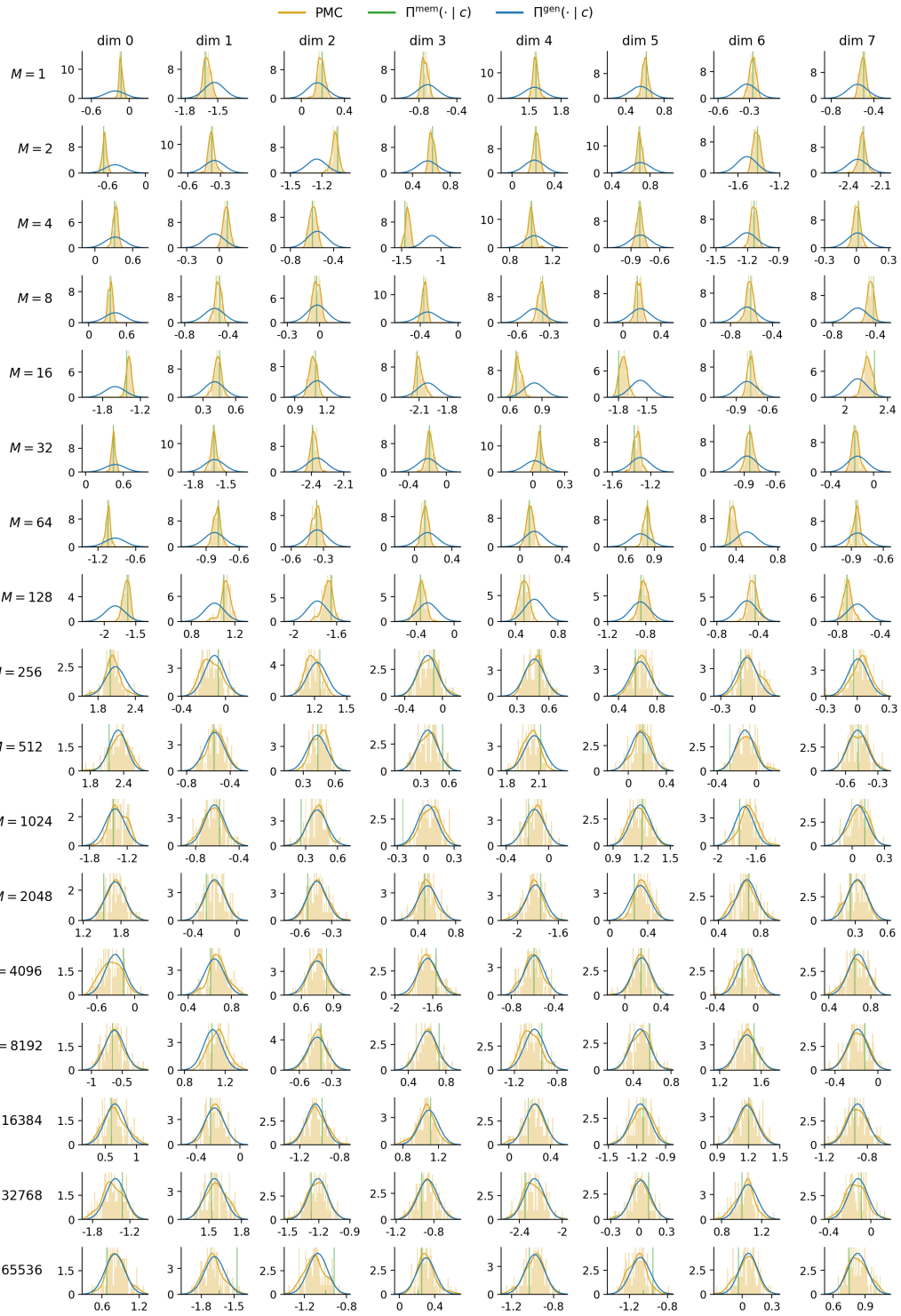

Figure 29: **Linear regression: PMC posterior, marginal density.** In-distribution prompt of length 8. Rows sweep task diversity M.

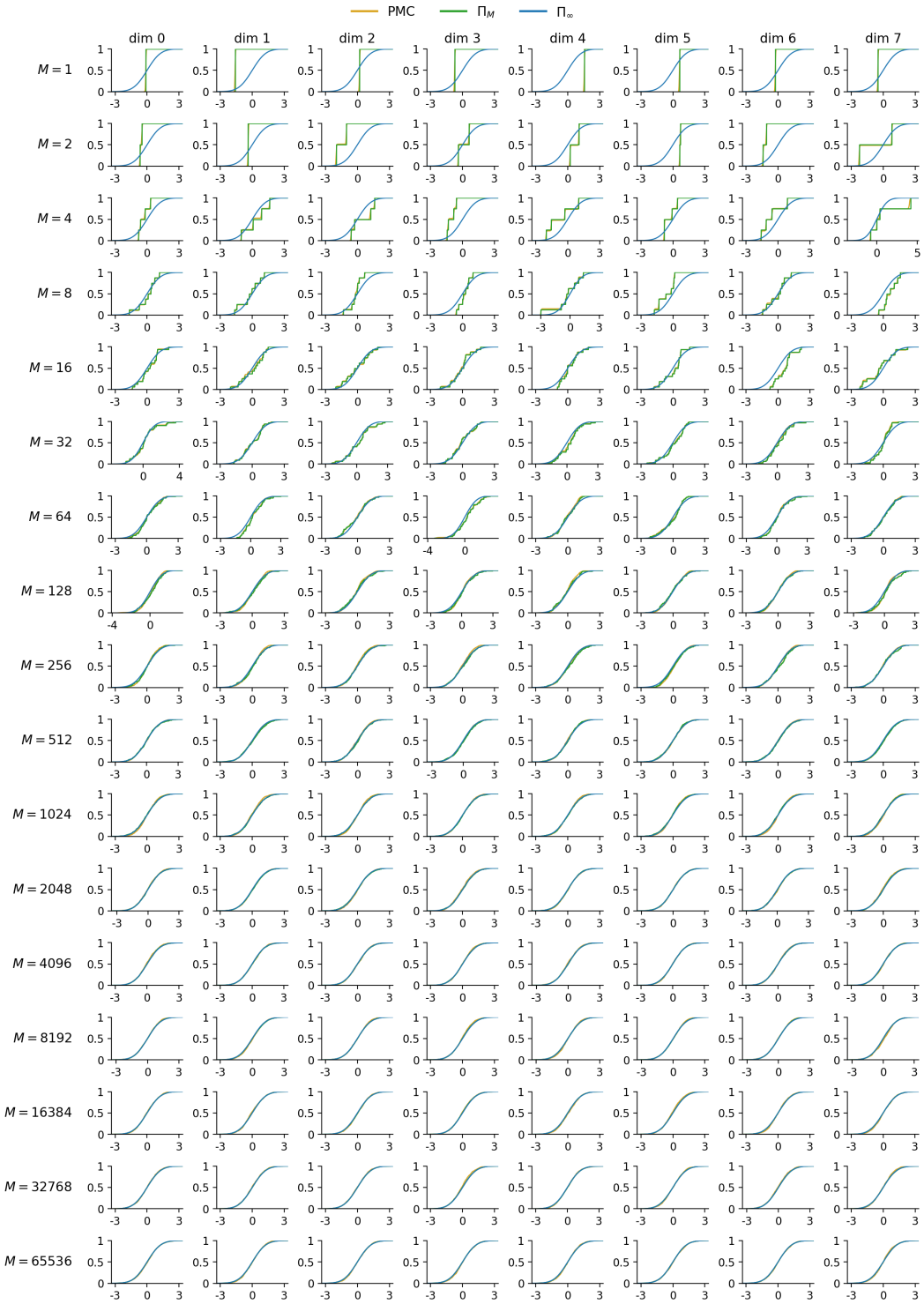

Figure 30: **Linear regression: PMC prior, marginal CDF.** Rows sweep task diversity M.

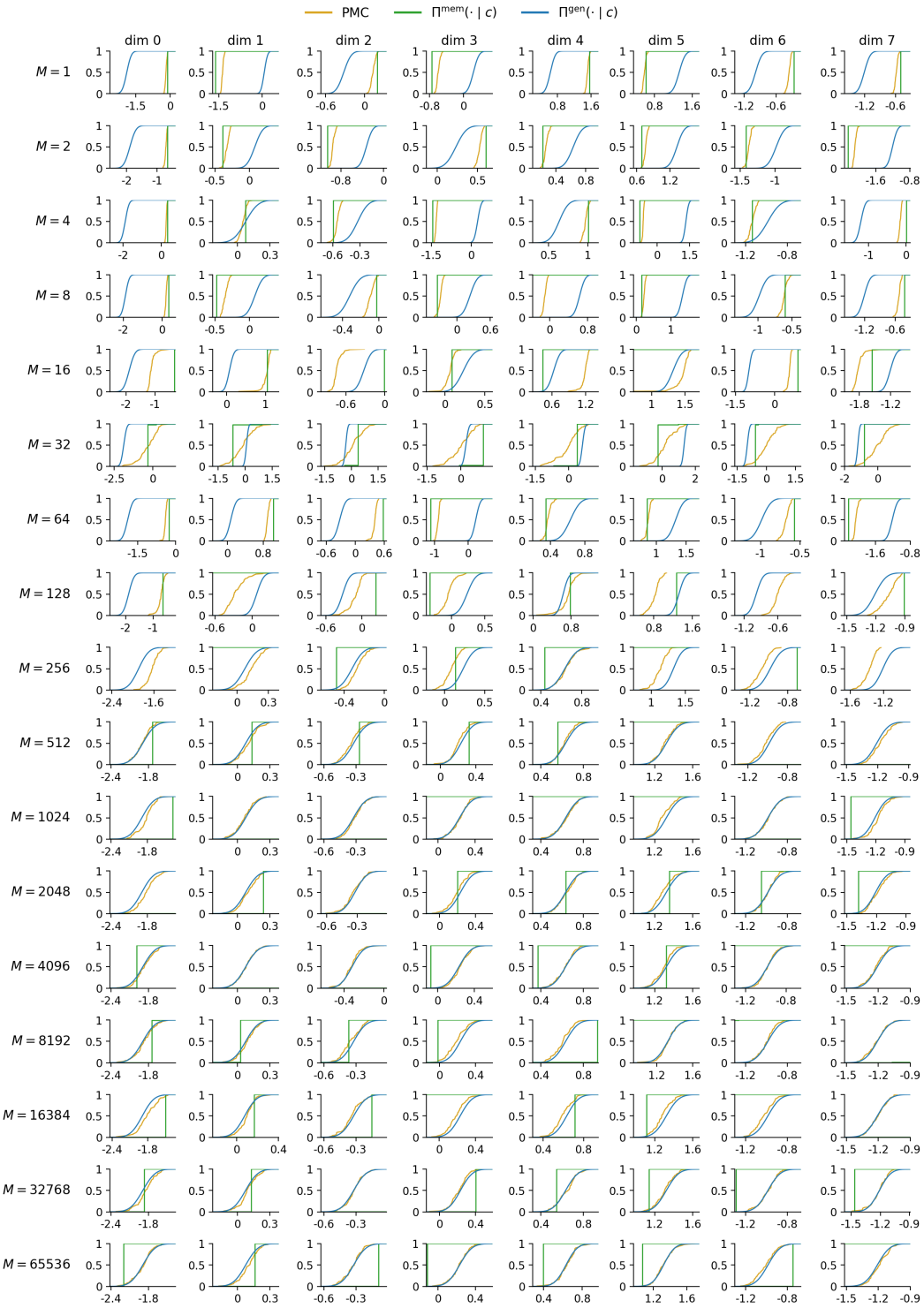

Figure 31: **Linear regression: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8. Rows sweep task diversity M.

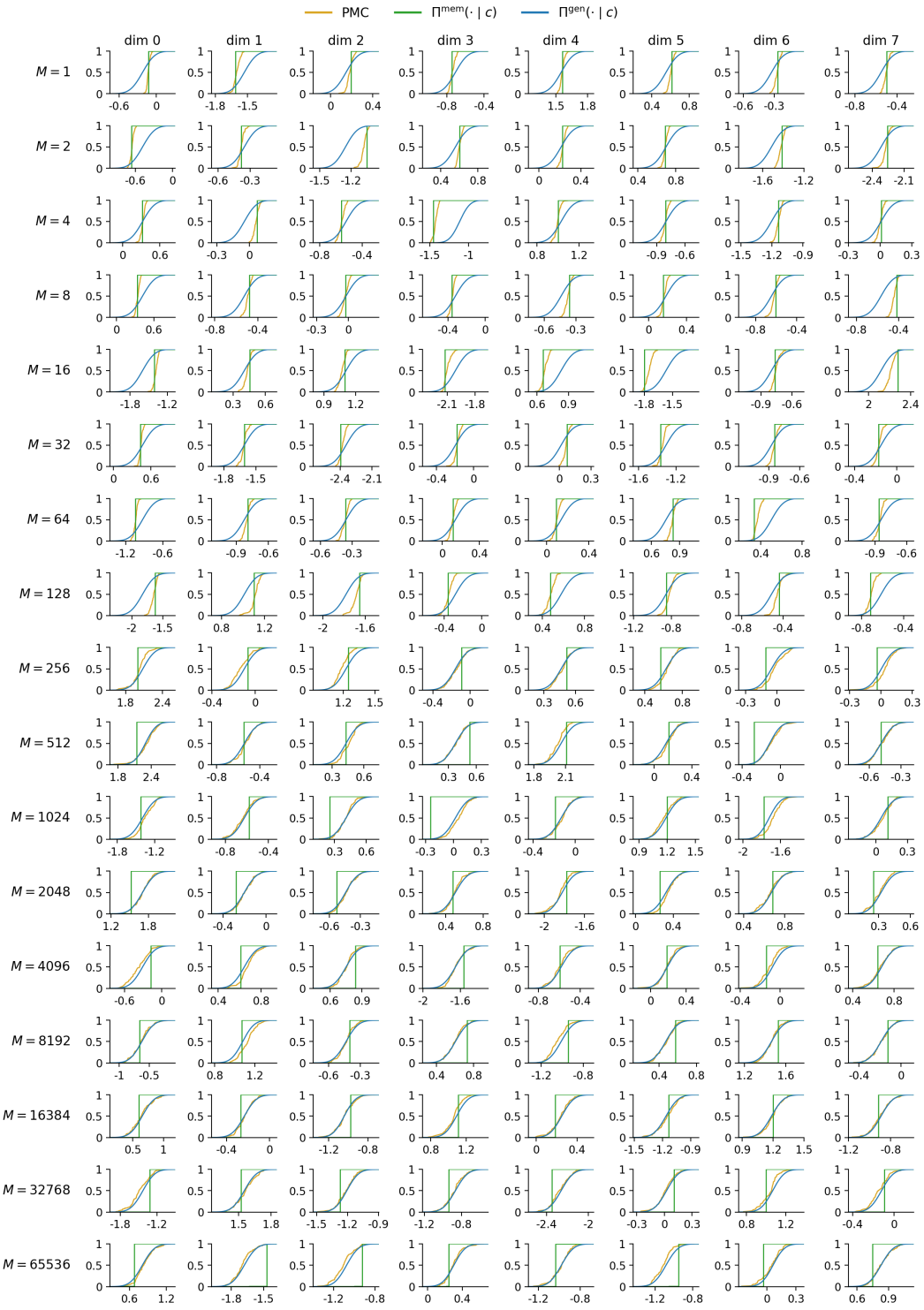

Figure 32: **Linear regression: PMC posterior, marginal CDF.** In-distribution prompt of length 8. Rows sweep task diversity M.

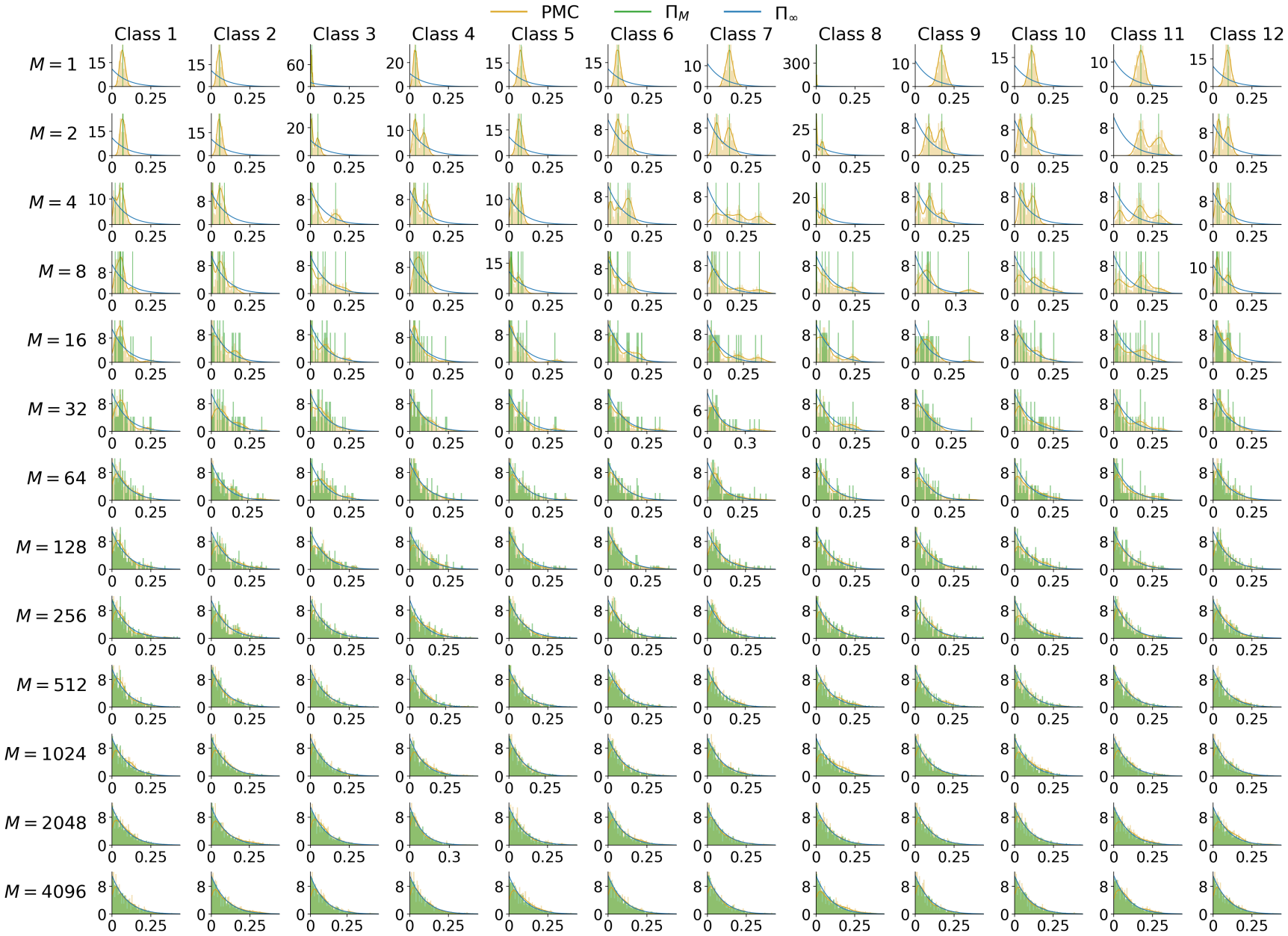

Figure 33: **Balls-and-urns: PMC prior, marginal density.** Rows sweep task diversity M.

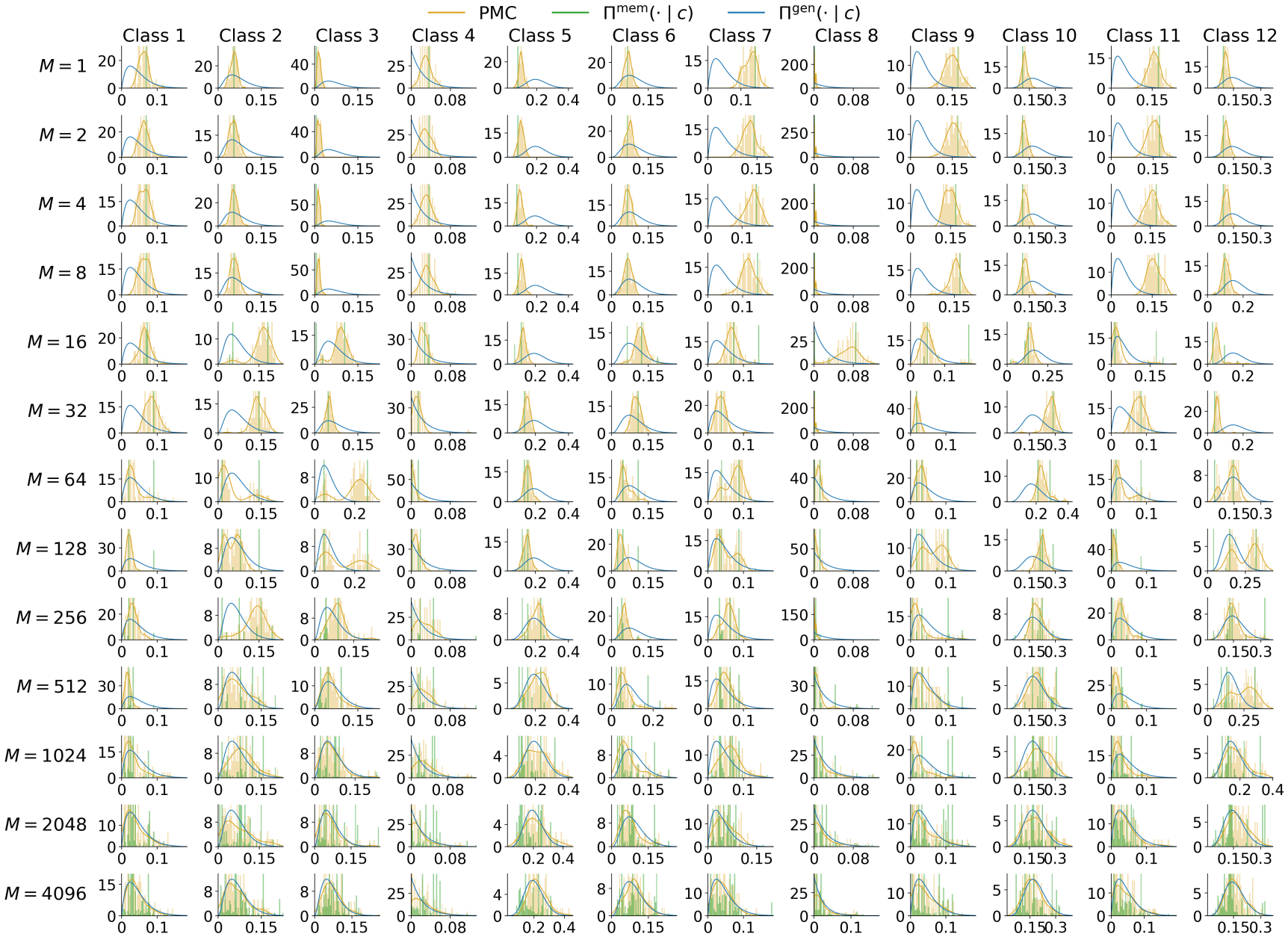

Figure 34: **Balls-and-urns: PMC posterior, marginal density.** Out-of-distribution prompt of length 16. Rows sweep task diversity M.

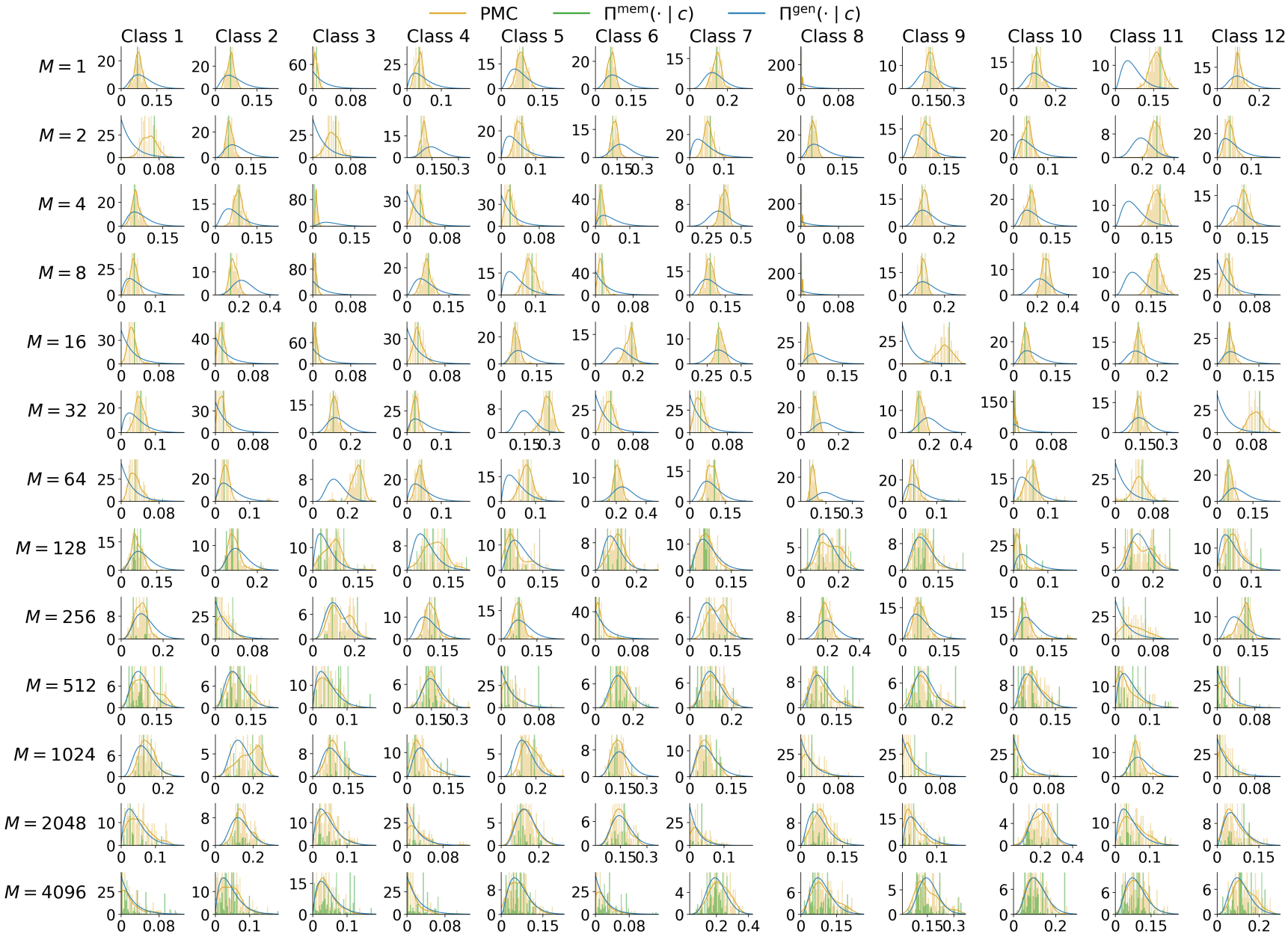

Figure 35: **Balls-and-urns: PMC posterior, marginal density.** In-distribution prompt of length 16. Rows sweep task diversity M.

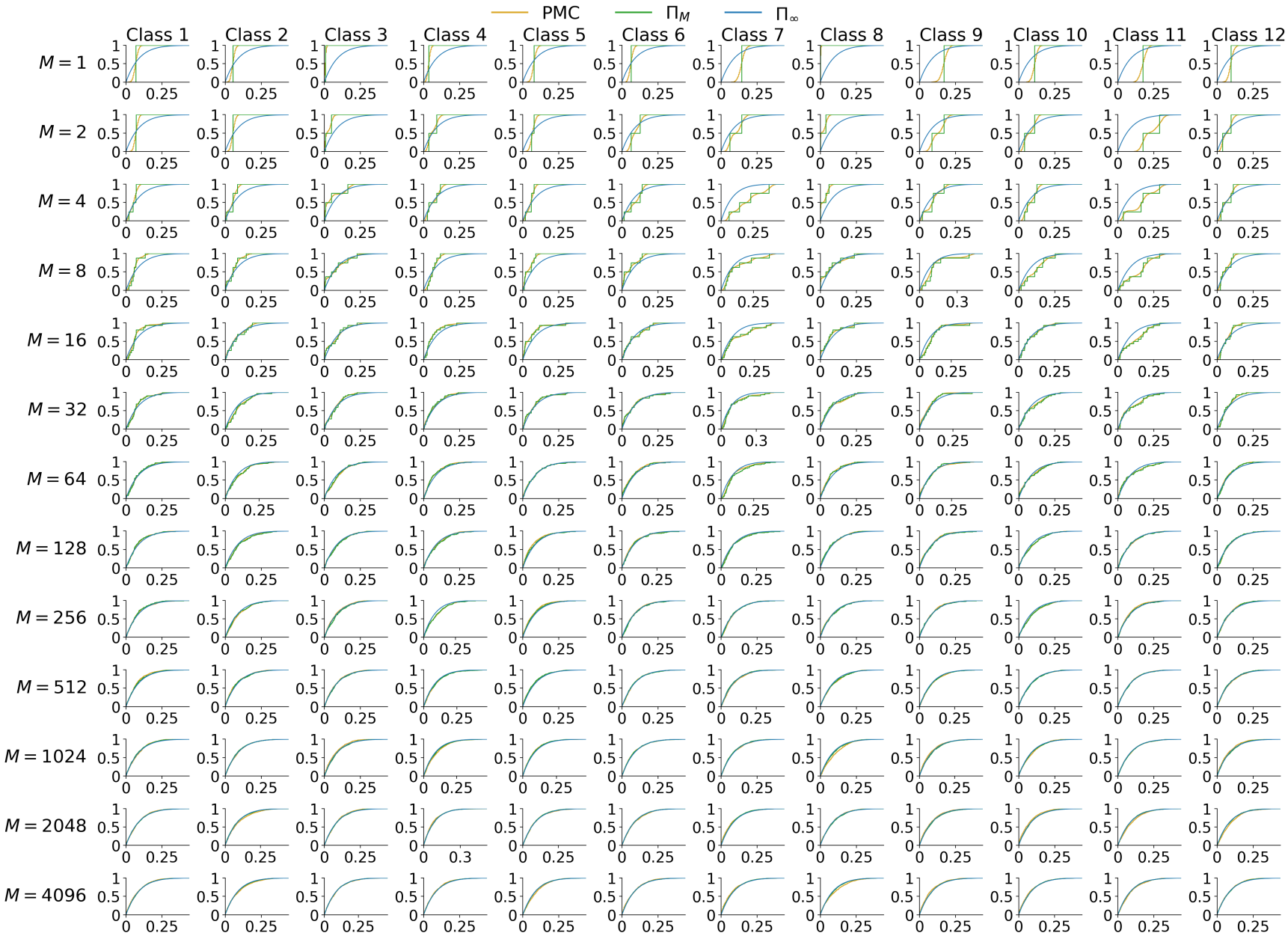

Figure 36: **Balls-and-urns: PMC prior, marginal CDF.** Rows sweep task diversity M.

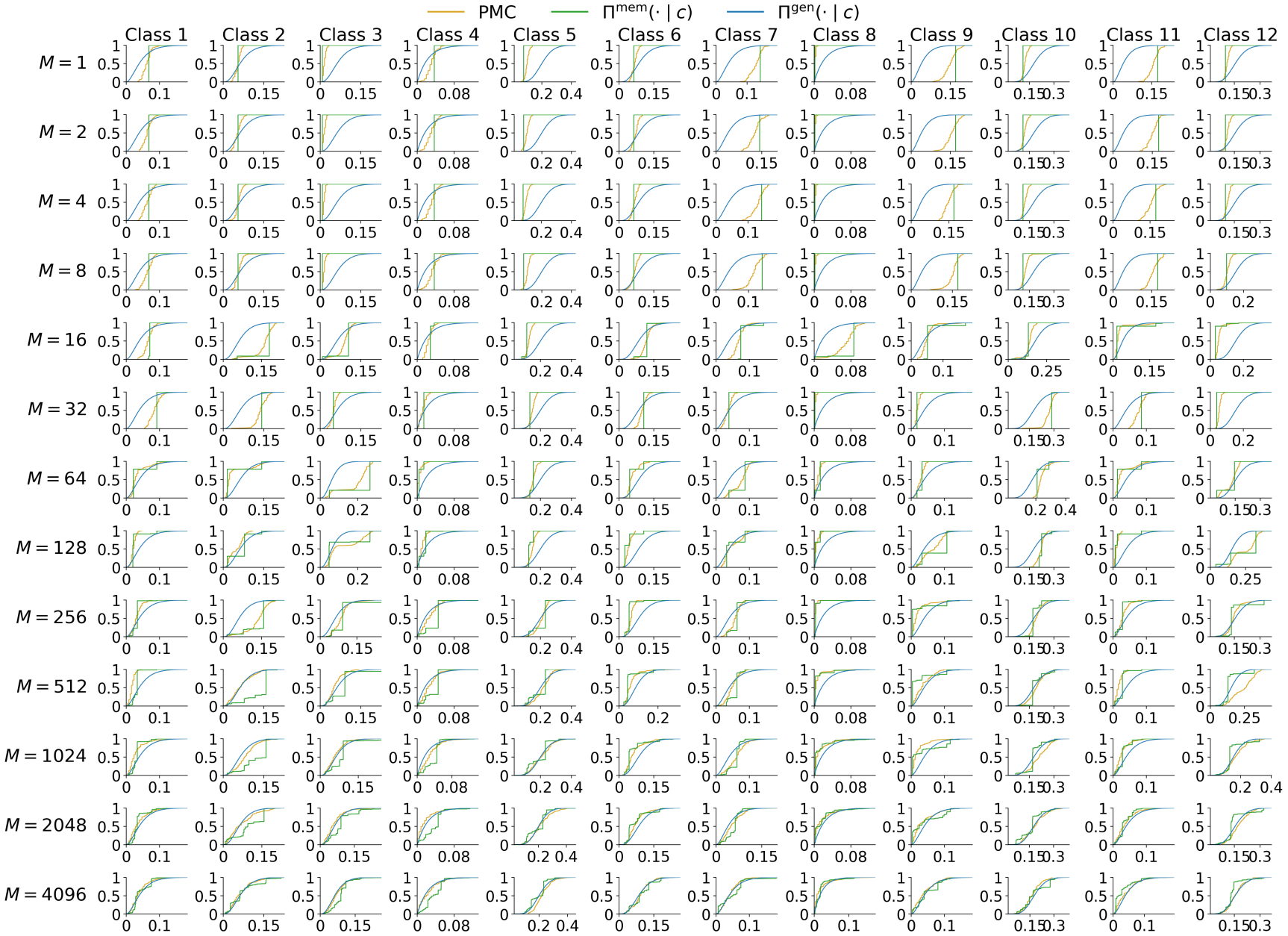

Figure 37: **Balls-and-urns: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 16. Rows sweep task diversity M.

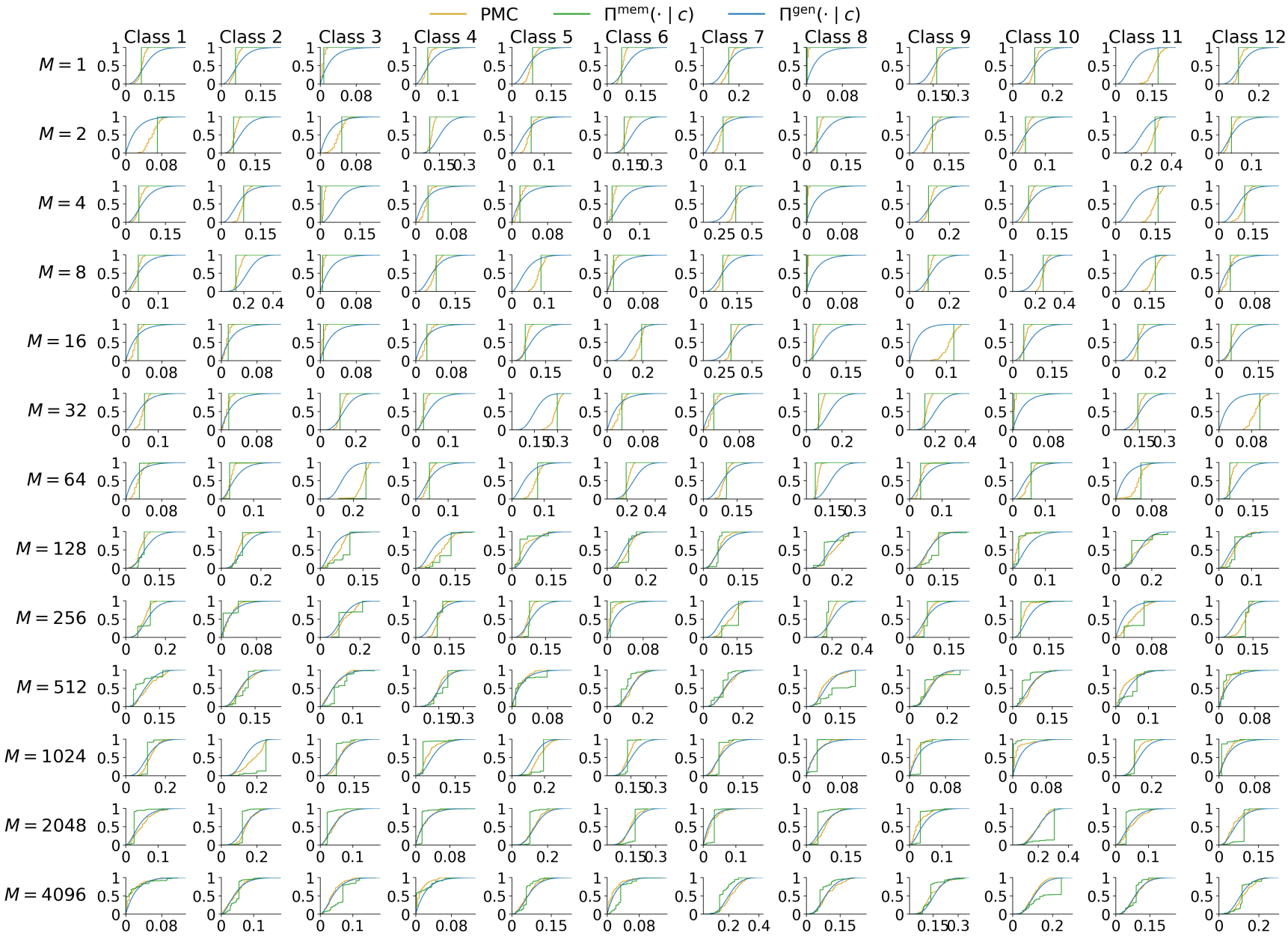

Figure 38: **Balls-and-urns: PMC posterior, marginal CDF.** In-distribution prompt of length 16. Rows sweep task diversity M.

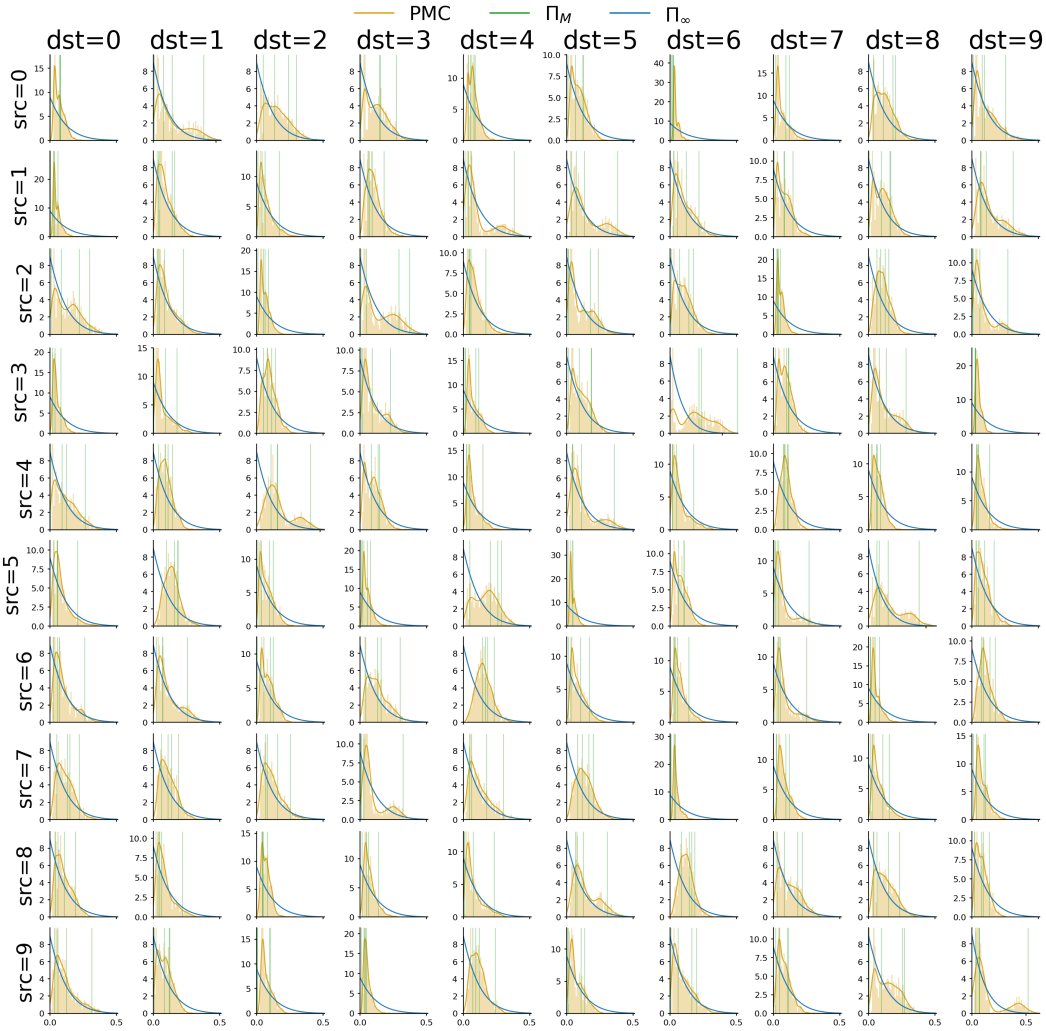

Figure 39: **Markov chains: PMC prior, marginal density at task diversity $M = 4$.**

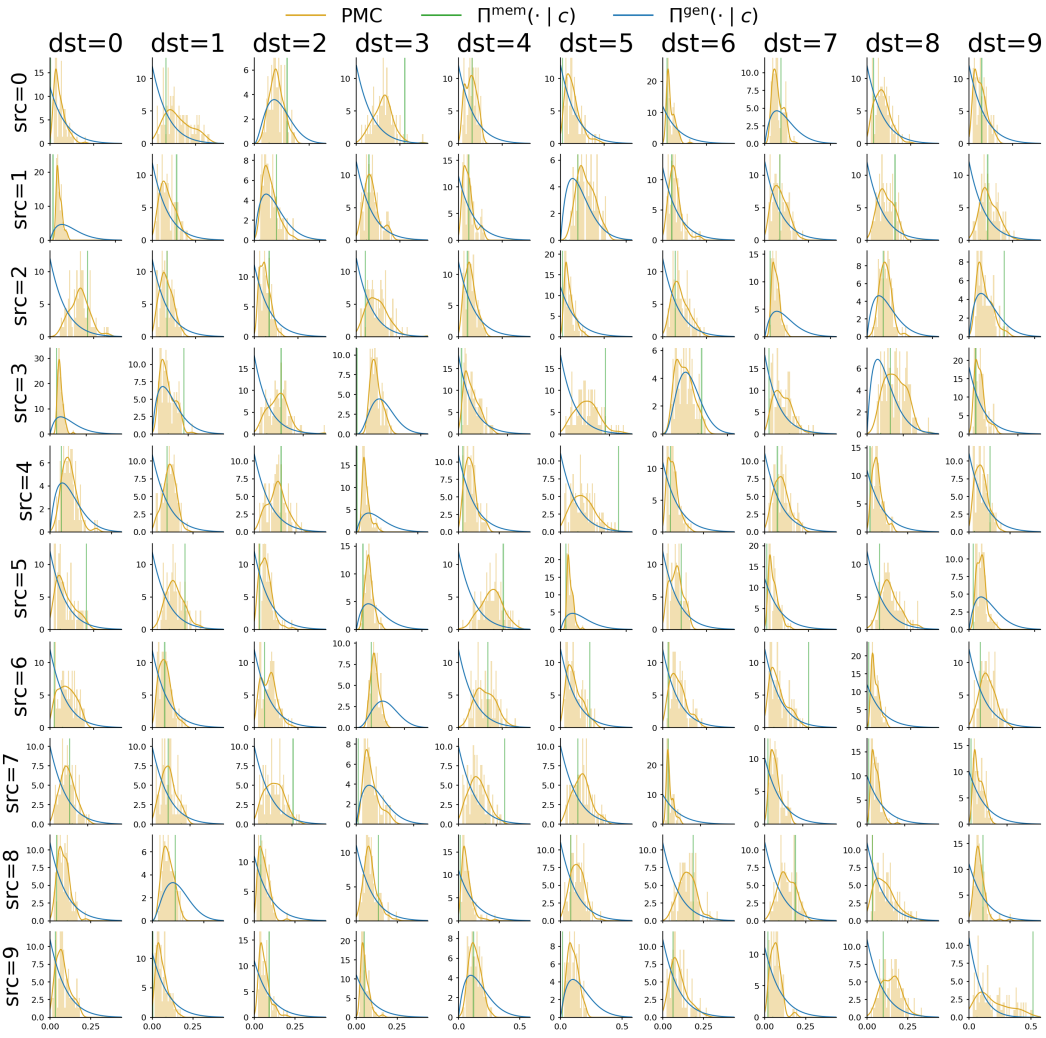

Figure 40: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 4$.

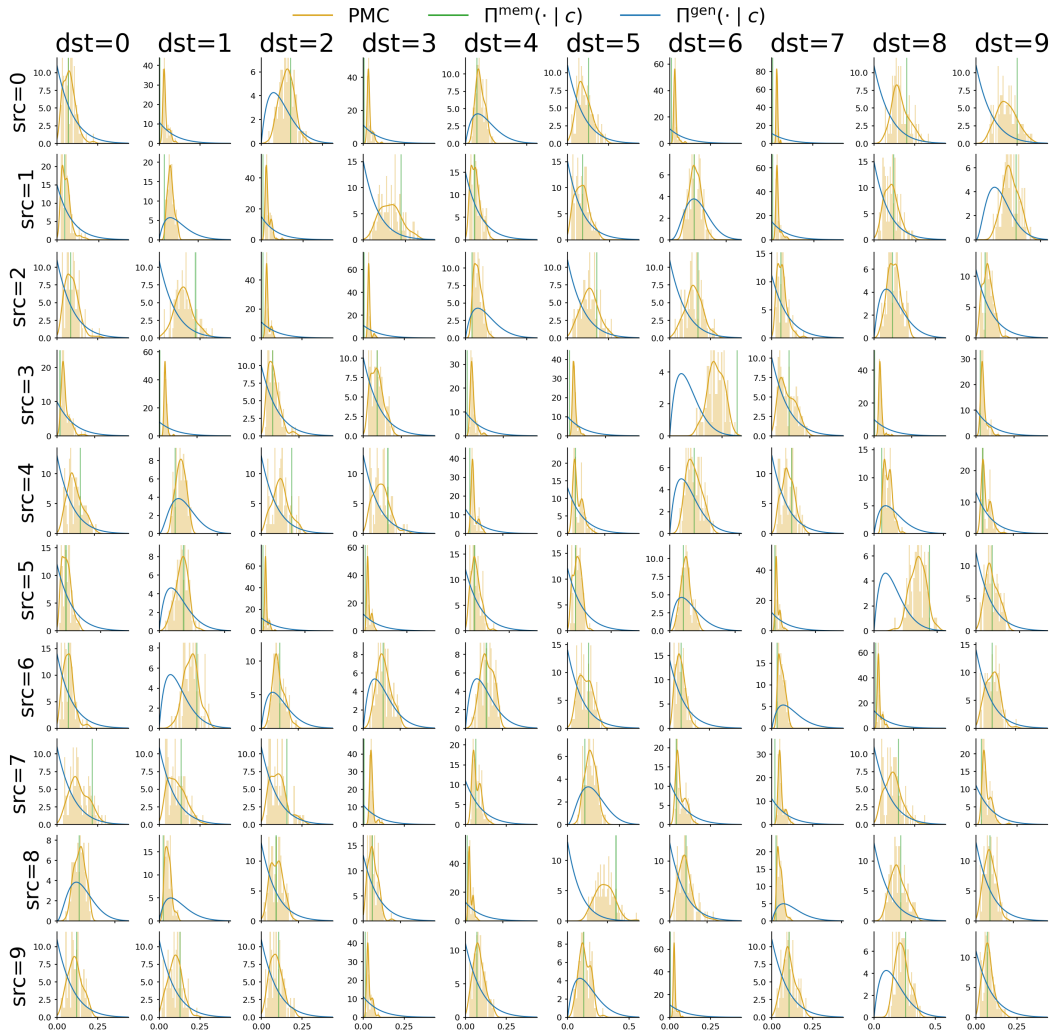

Figure 41: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 4$.

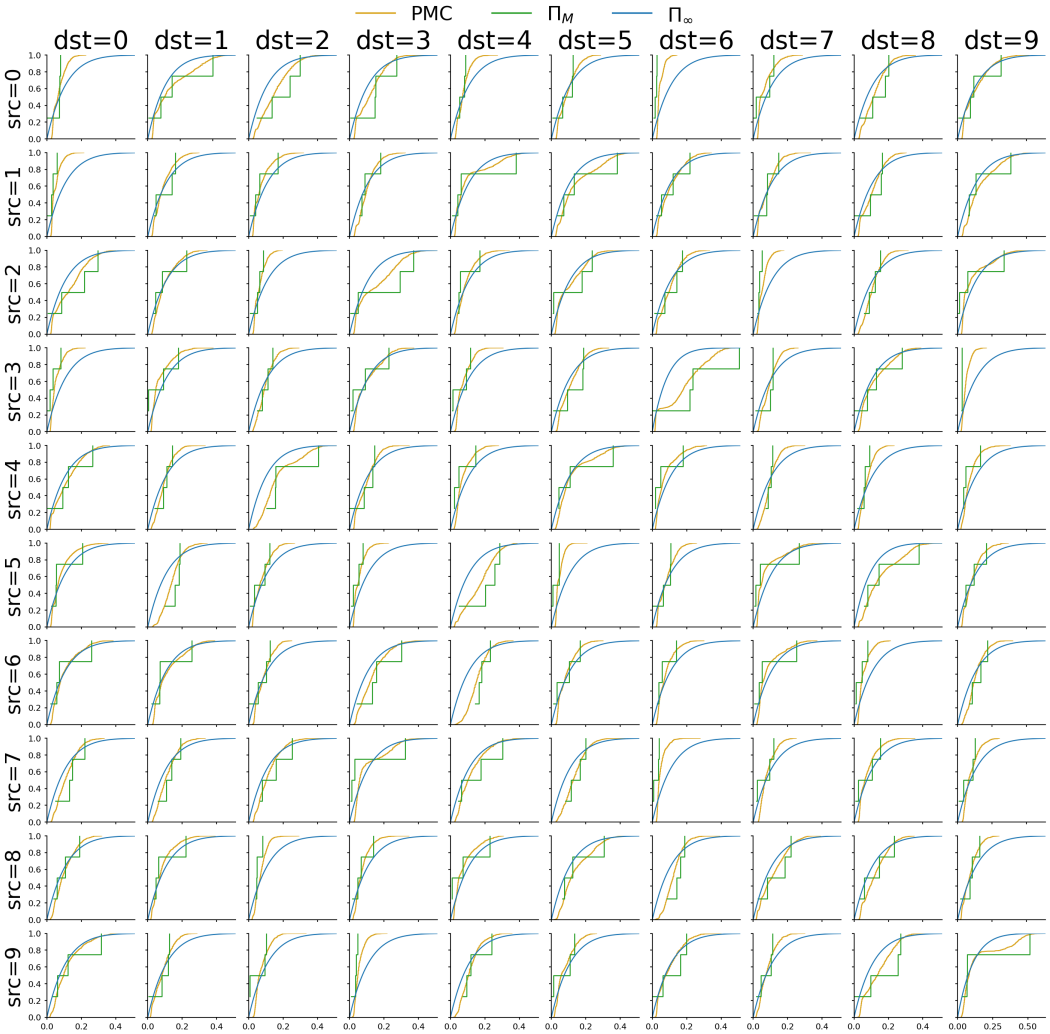

Figure 42: **Markov chains: PMC prior, marginal CDF at task diversity $M = 4$.**

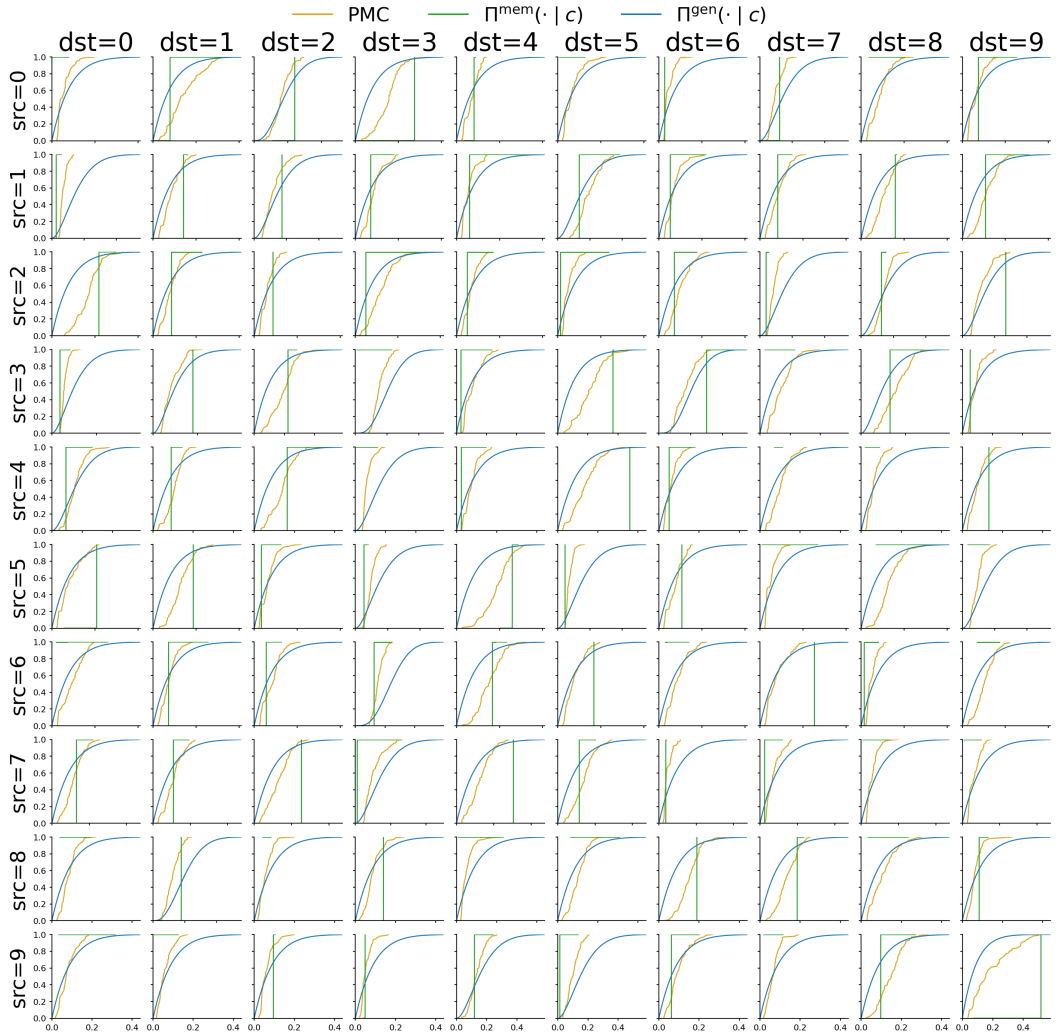

Figure 43: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 4$.

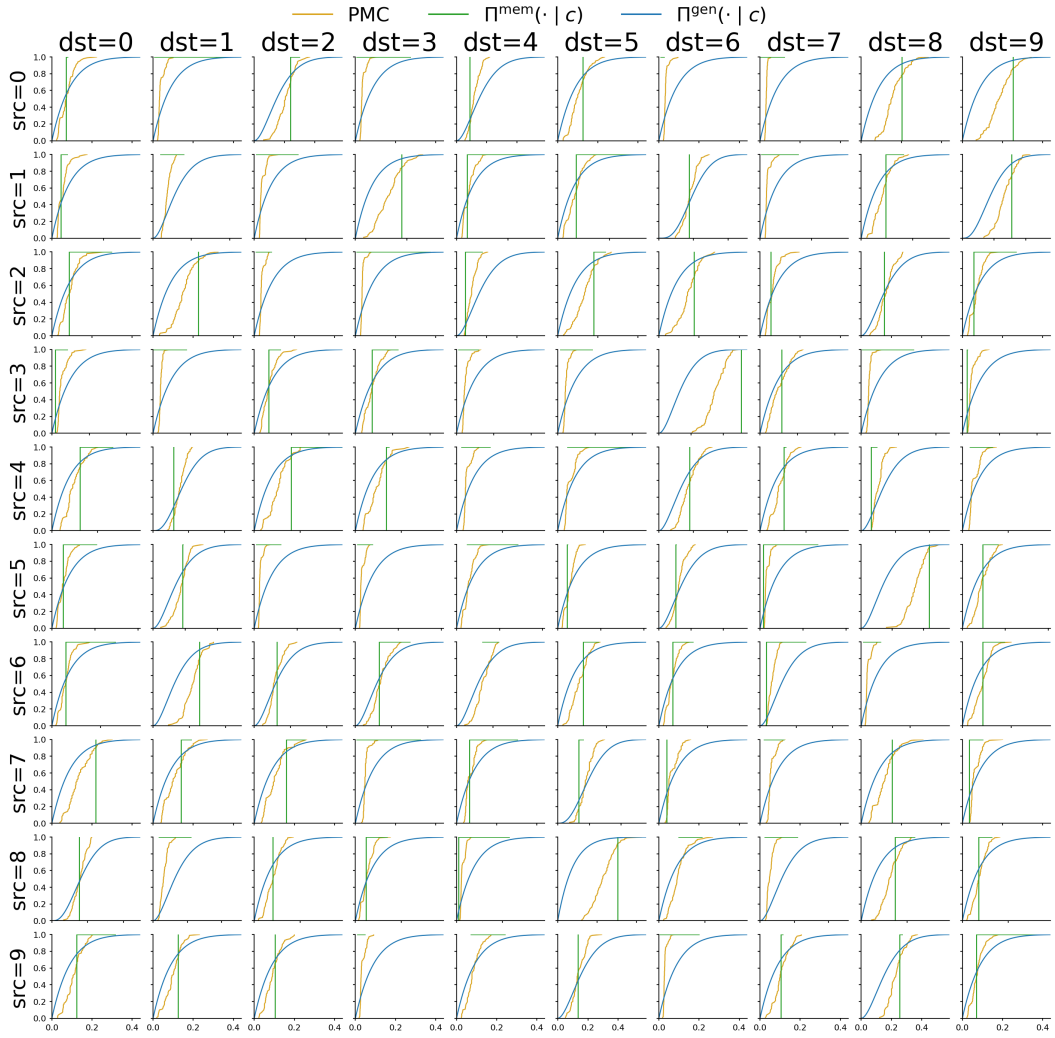

Figure 44: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 4$.

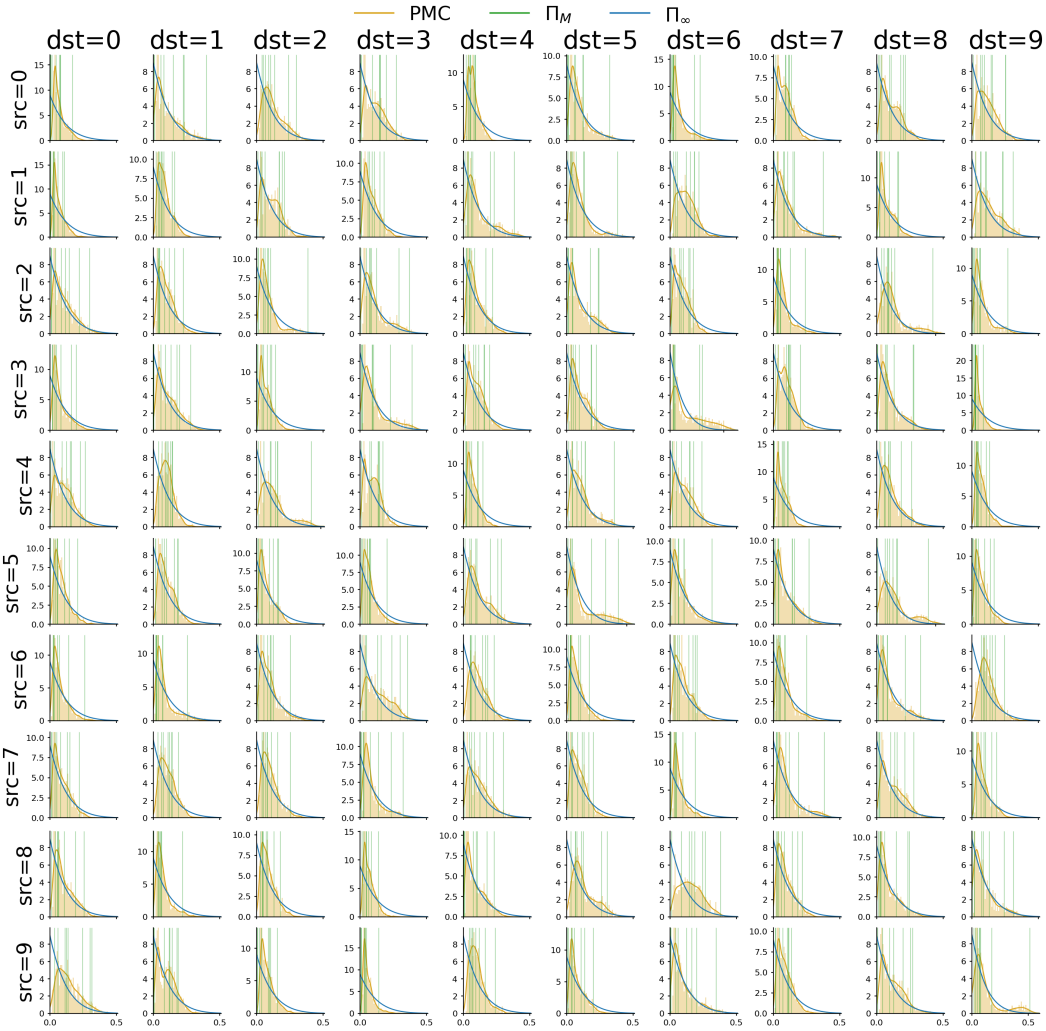

Figure 45: **Markov chains: PMC prior, marginal density at task diversity $M = 8$.**

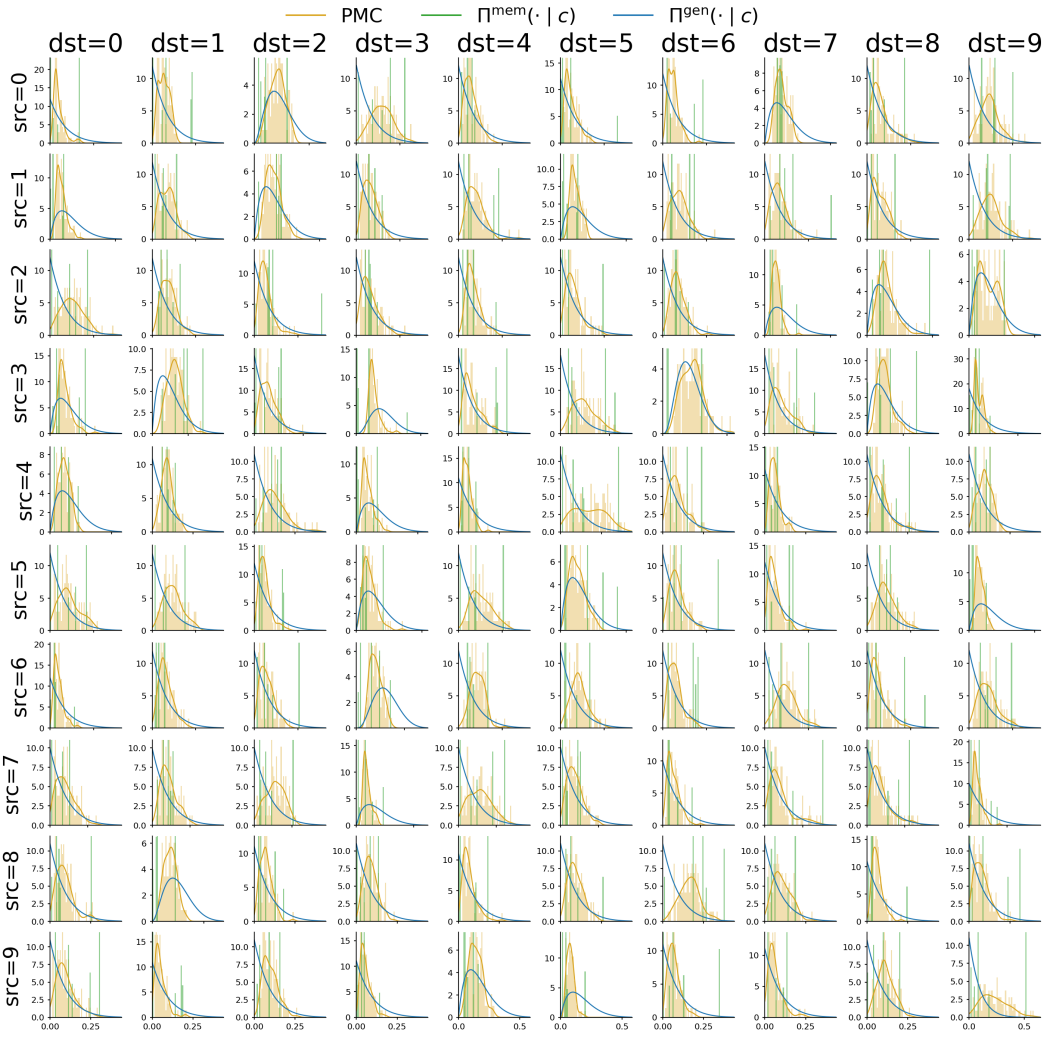

Figure 46: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 8$.

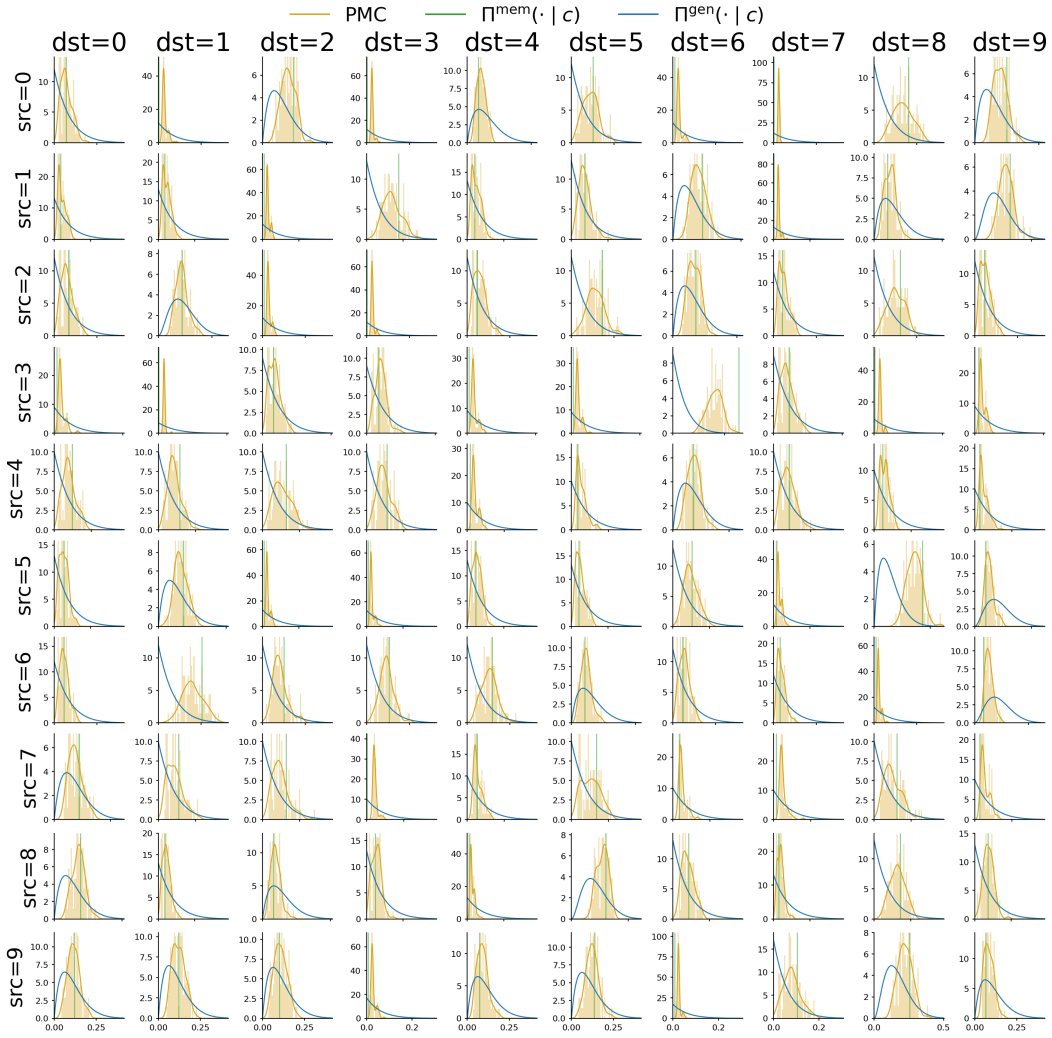

Figure 47: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 8$.

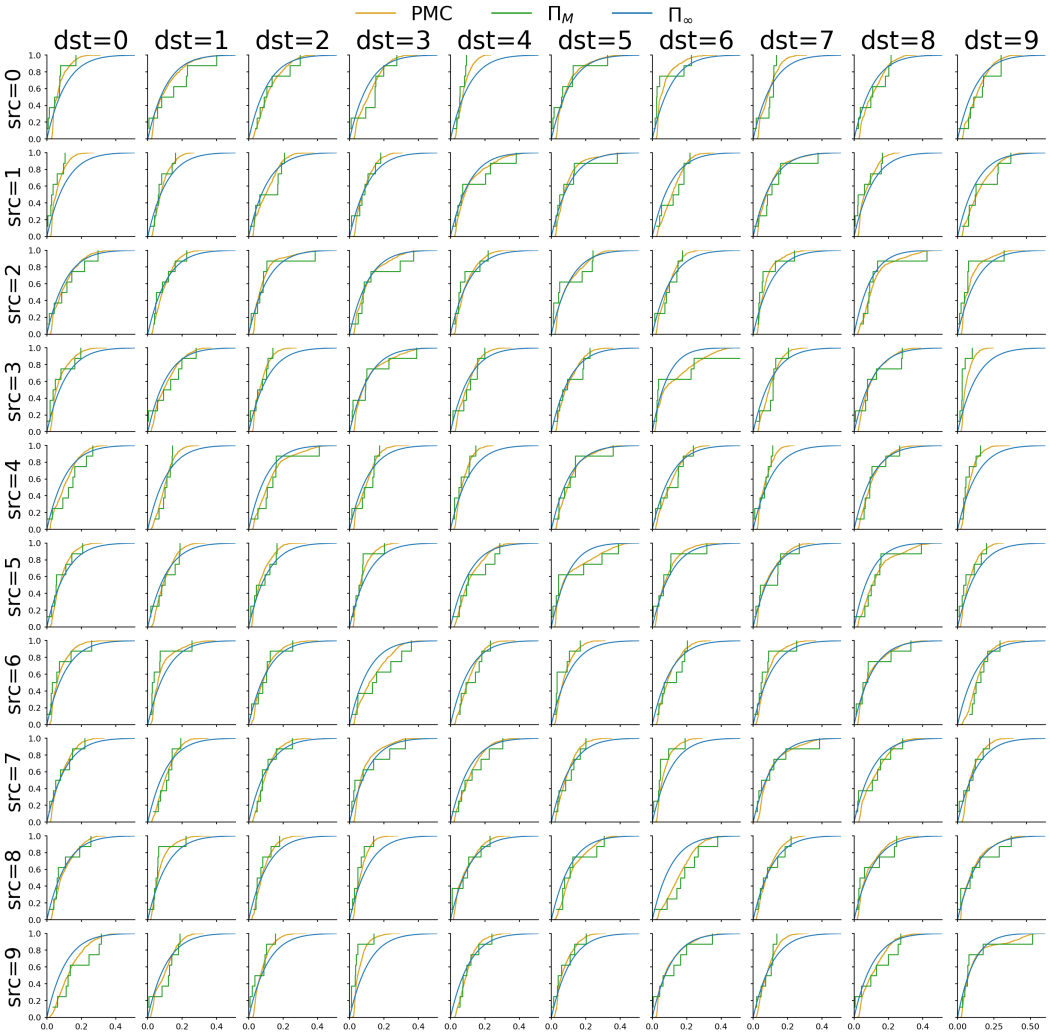

Figure 48: **Markov chains: PMC prior, marginal CDF at task diversity $M = 8$.**

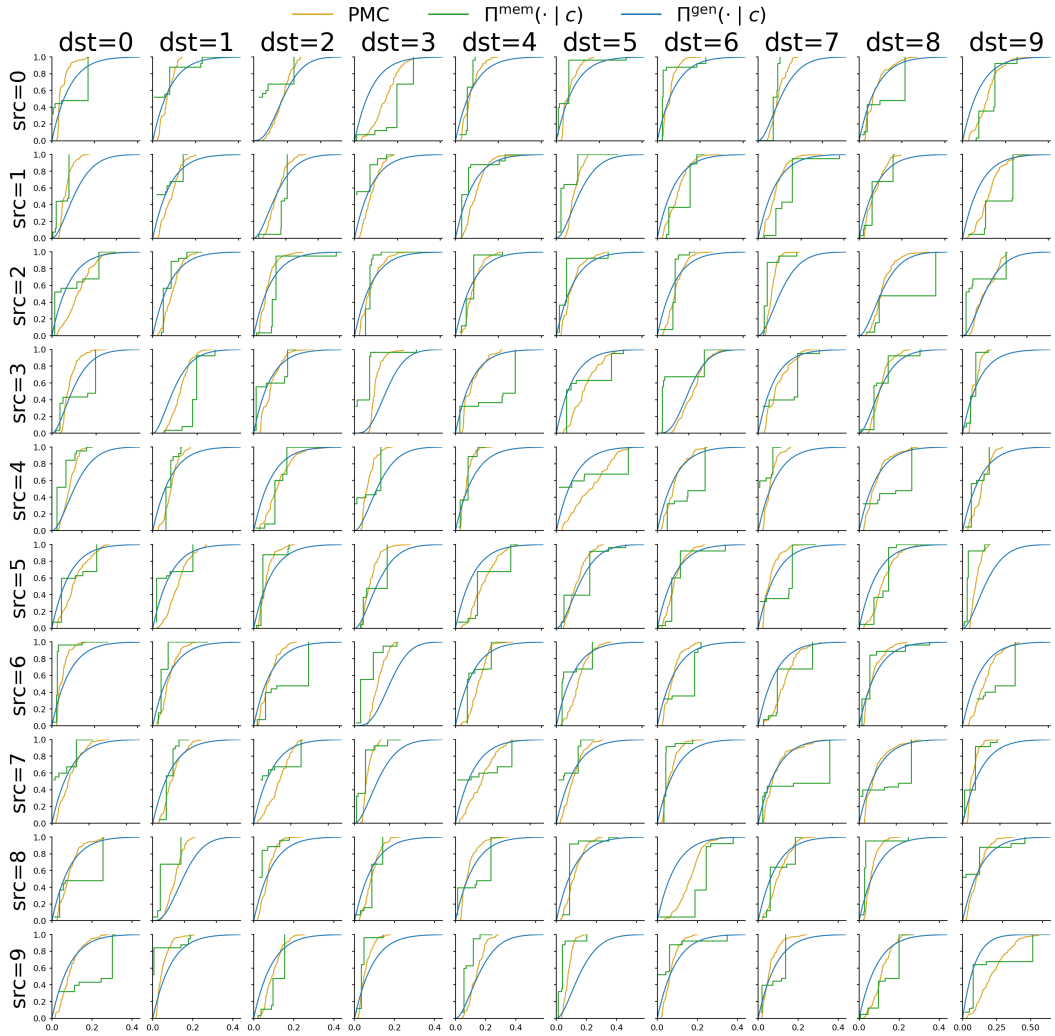

Figure 49: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 8$.

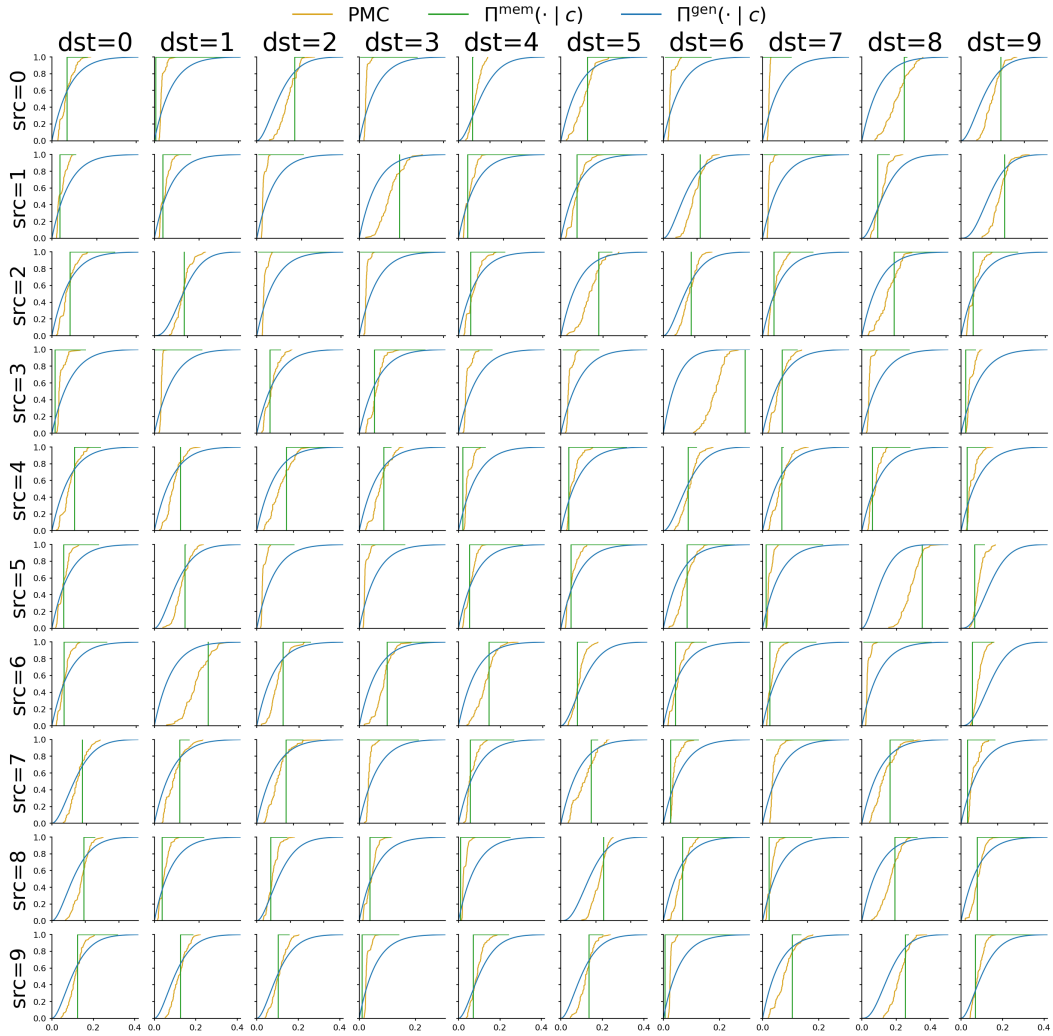

Figure 50: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 8$.

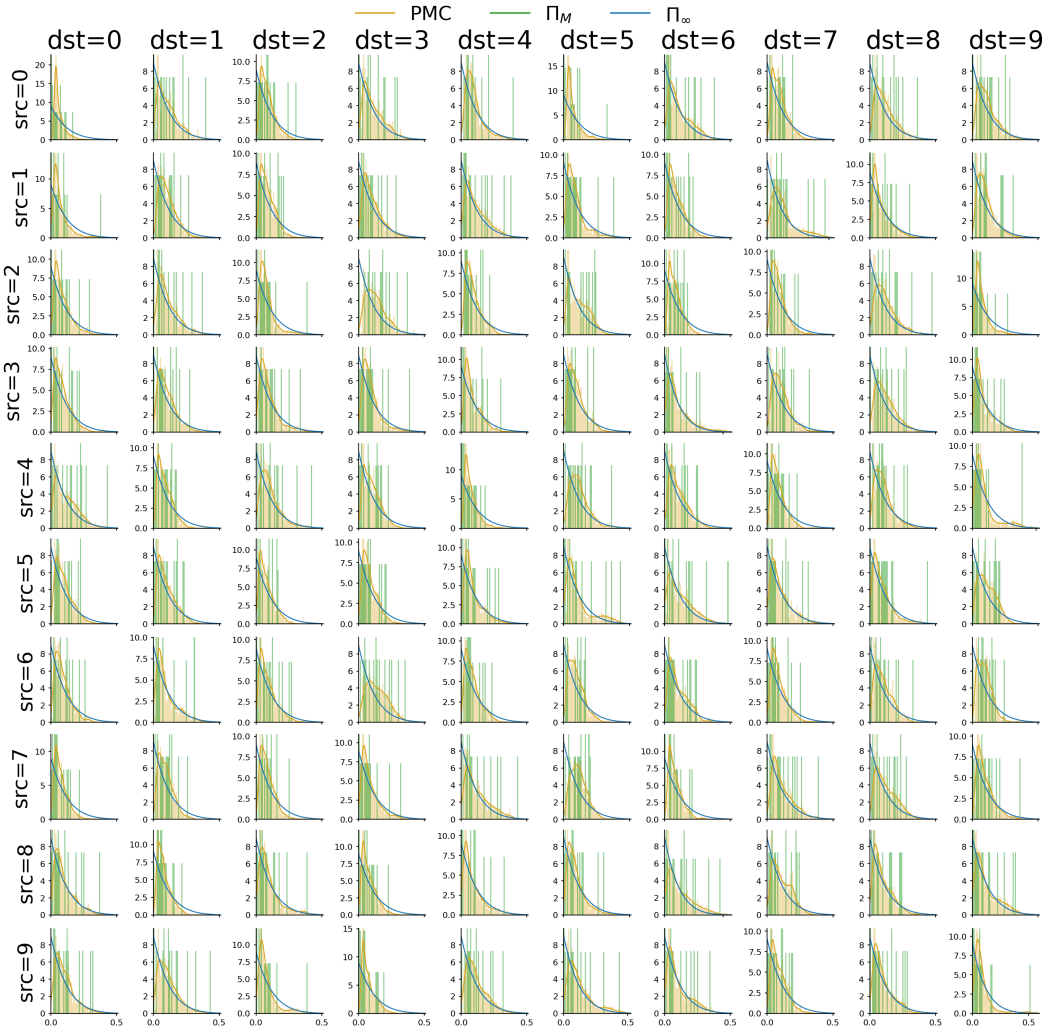

Figure 51: **Markov chains: PMC prior, marginal density at task diversity $M = 16$.**

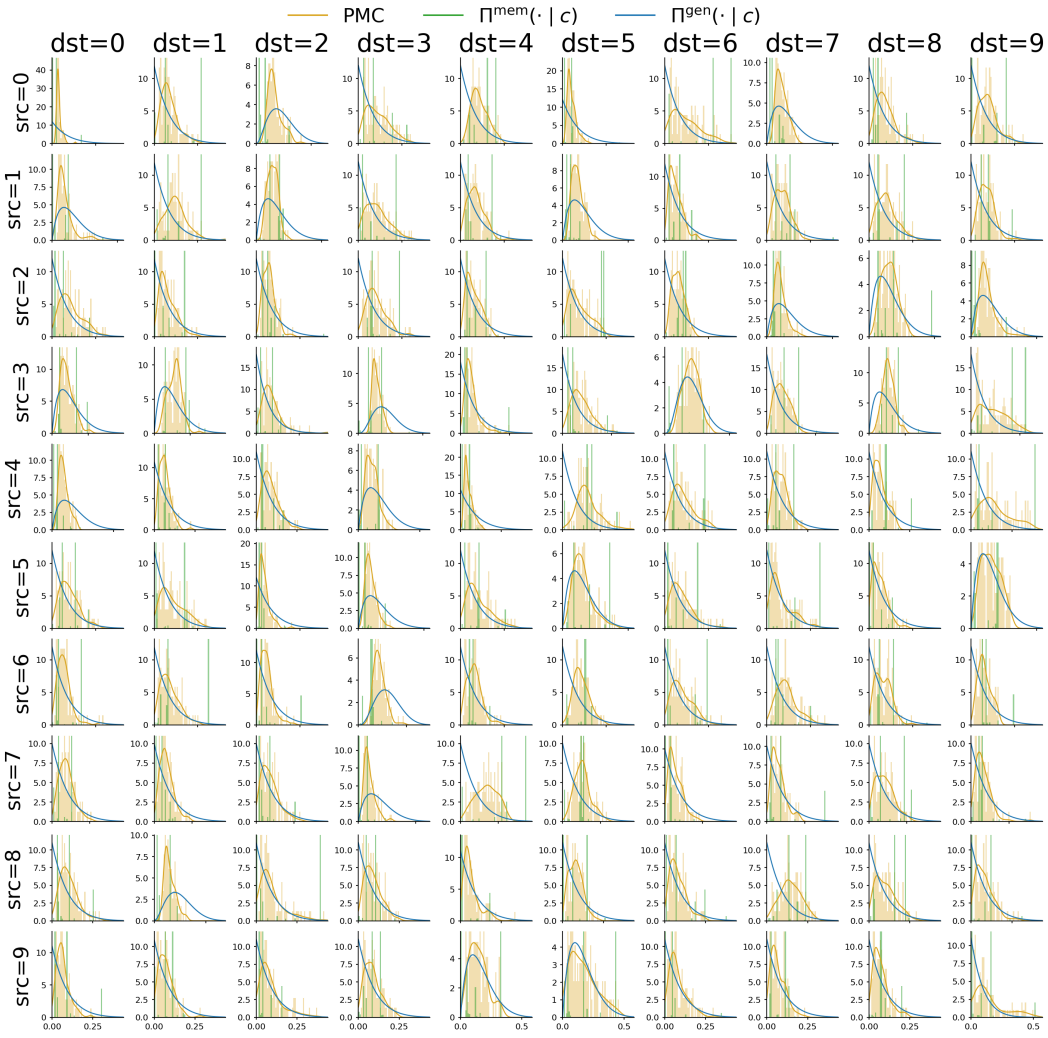

Figure 52: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 16$.

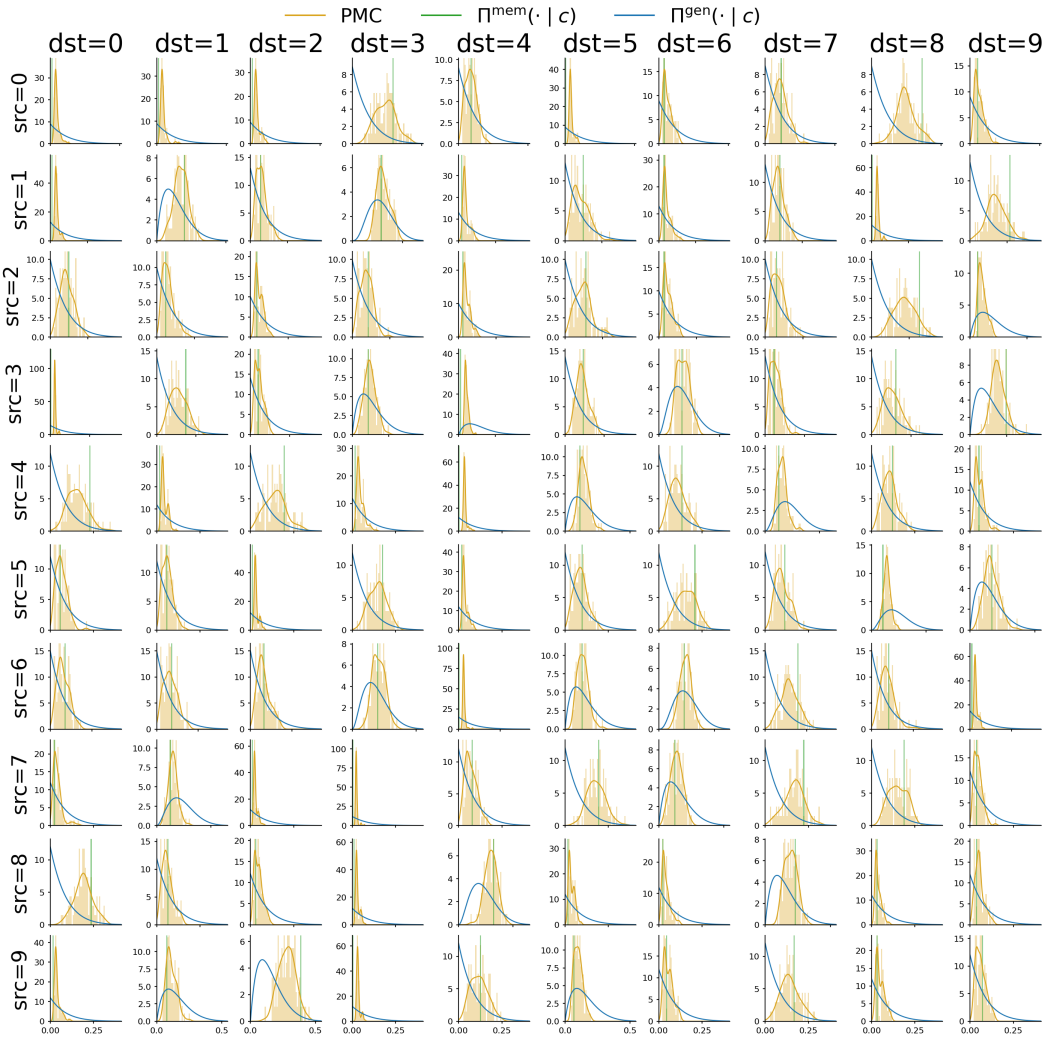

Figure 53: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 16$.

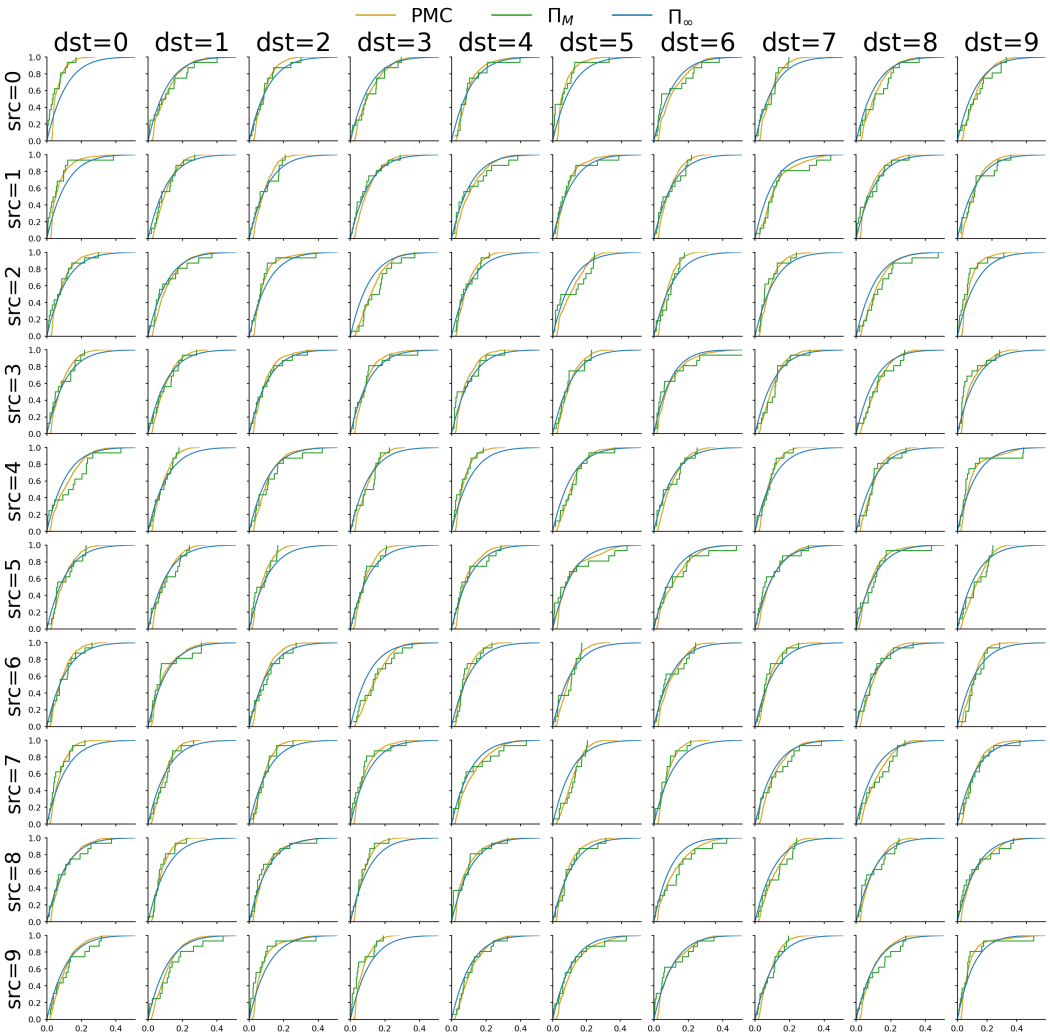

Figure 54: **Markov chains: PMC prior, marginal CDF at task diversity $M = 16$.**

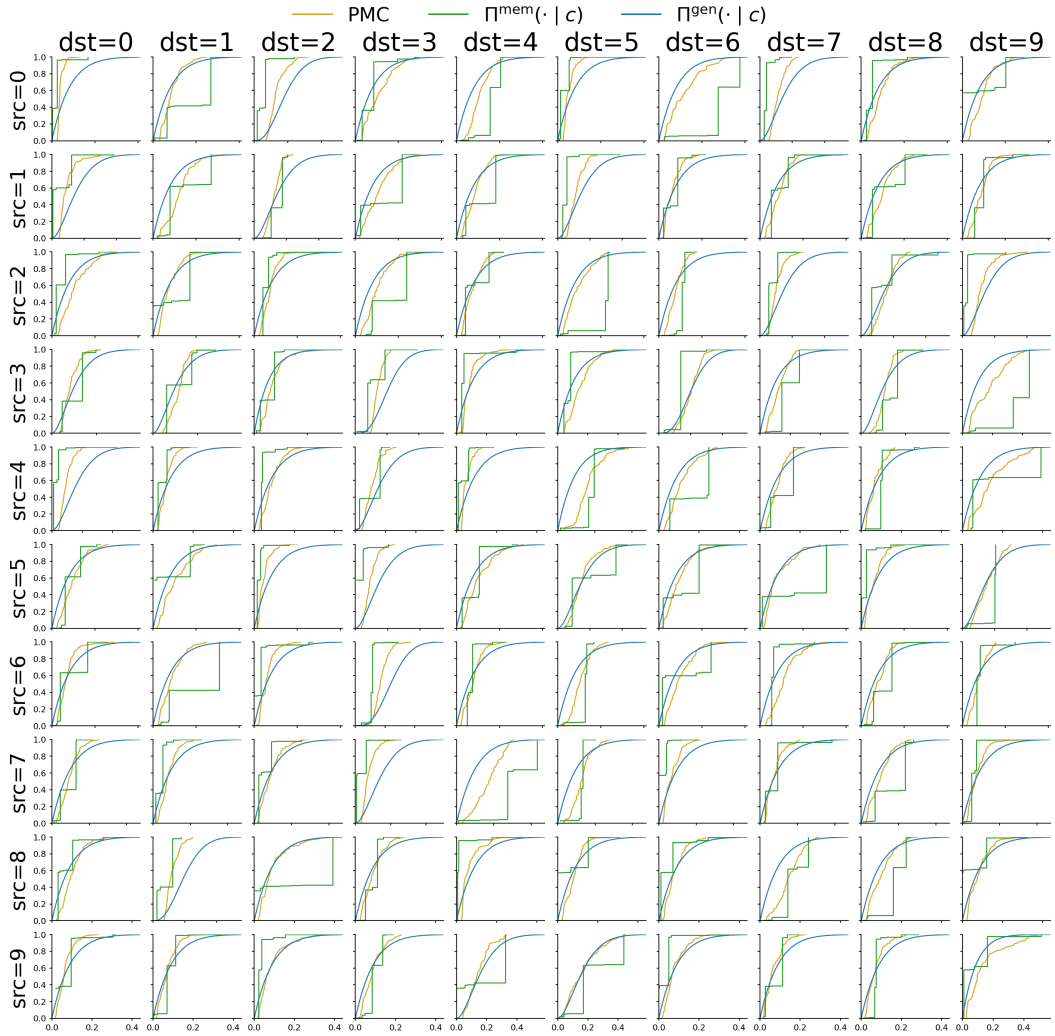

Figure 55: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 16$.

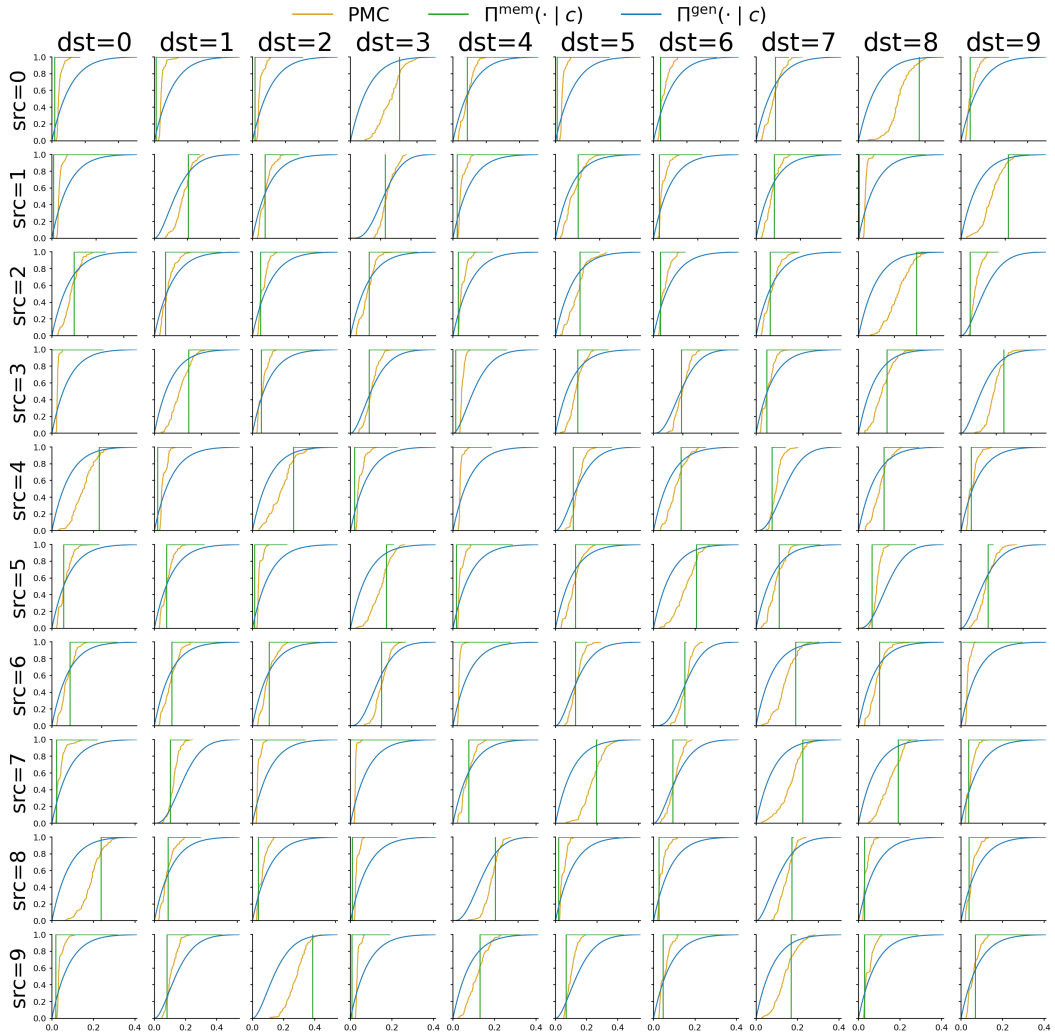

Figure 56: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 16$.

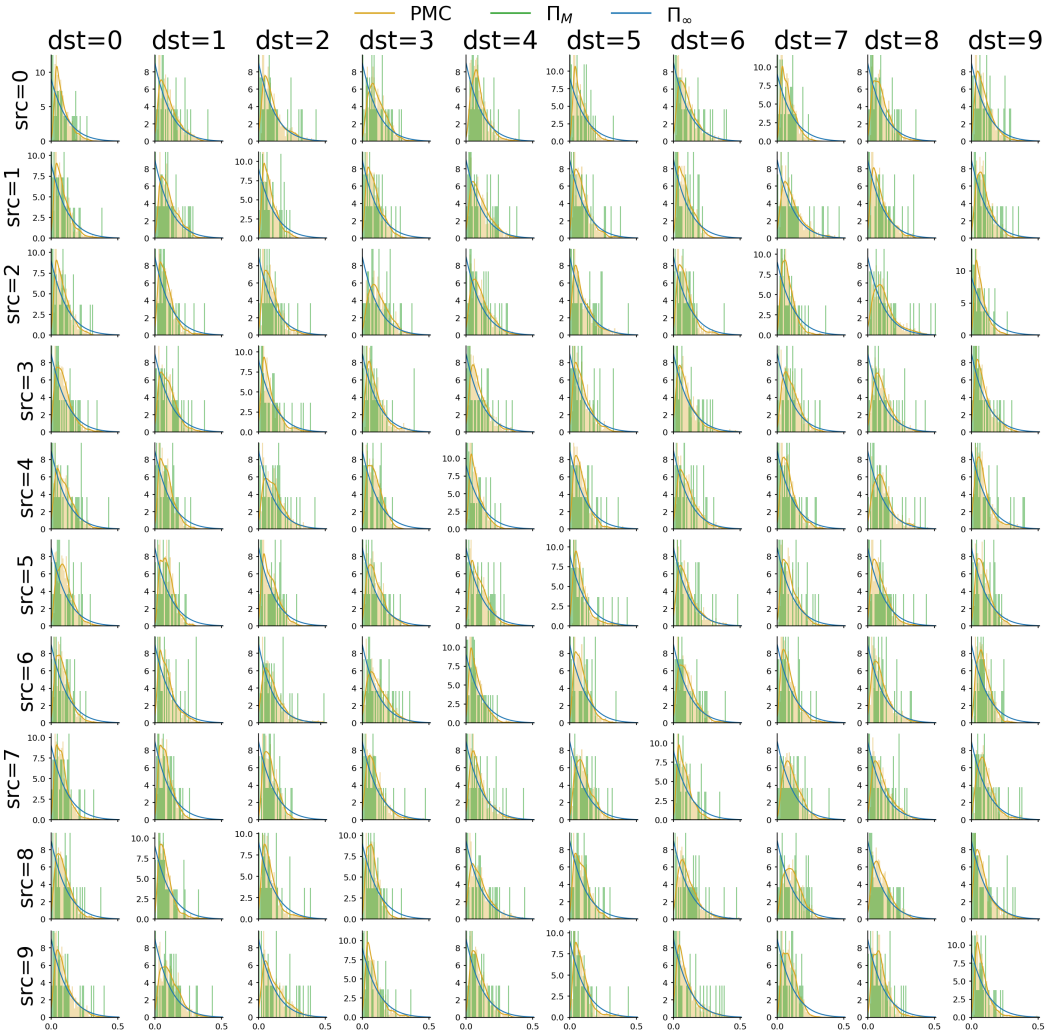

Figure 57: **Markov chains: PMC prior, marginal density at task diversity $M = 32$.**

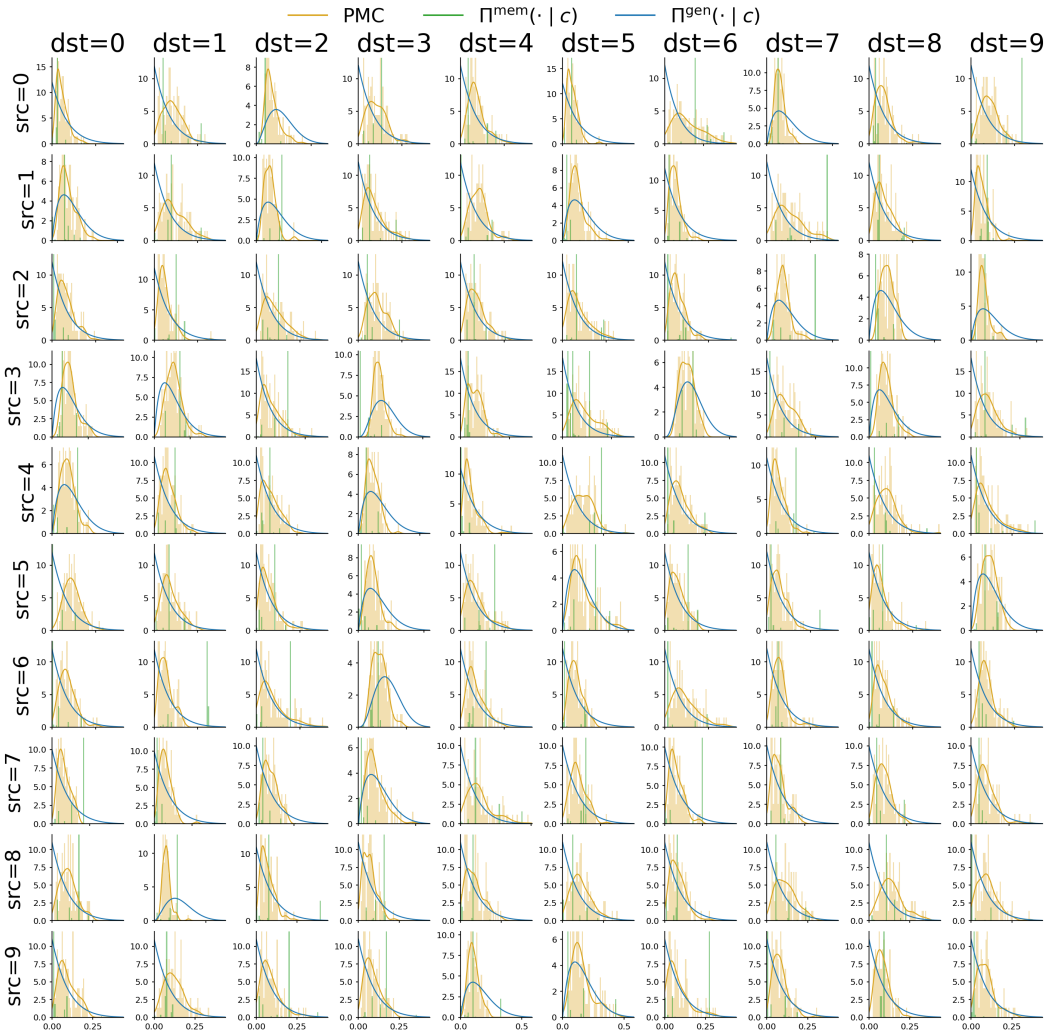

Figure 58: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 32$.

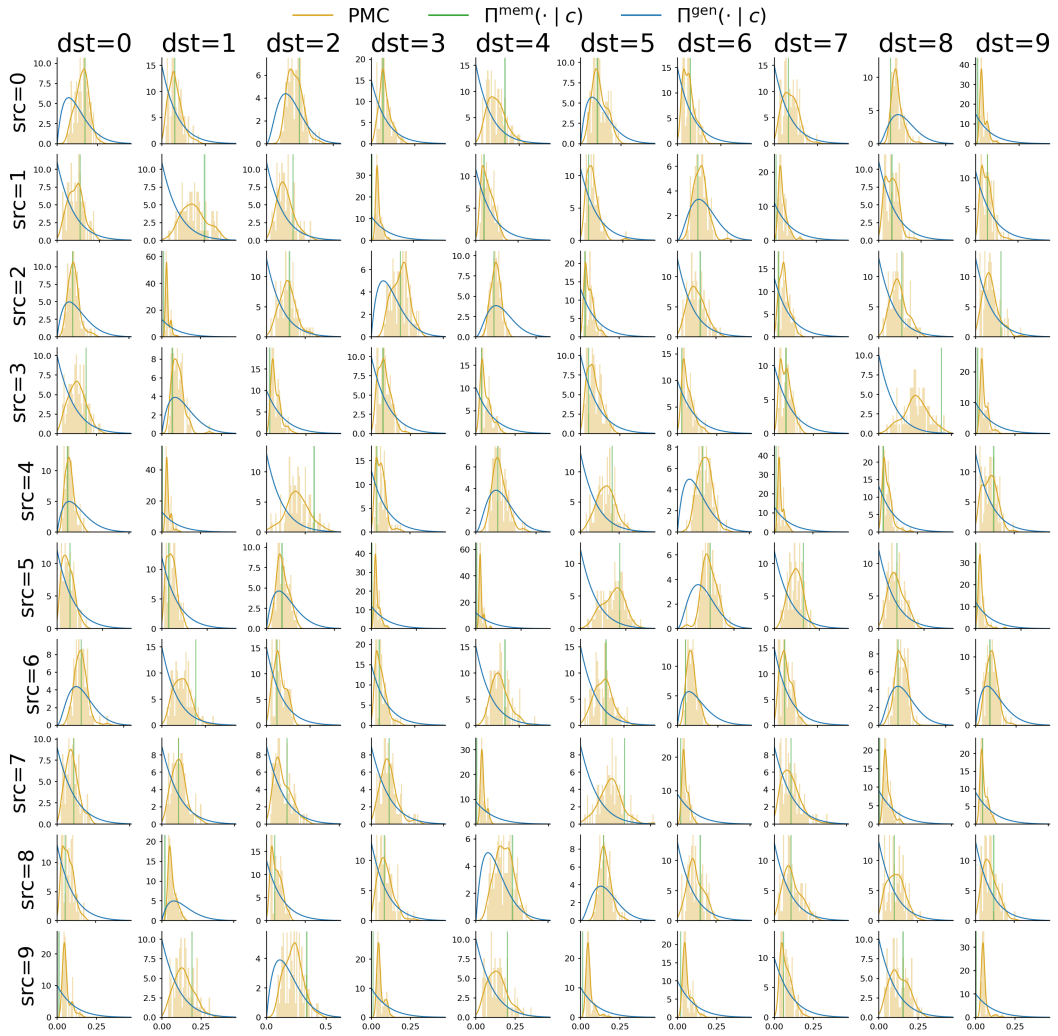

Figure 59: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 32$.

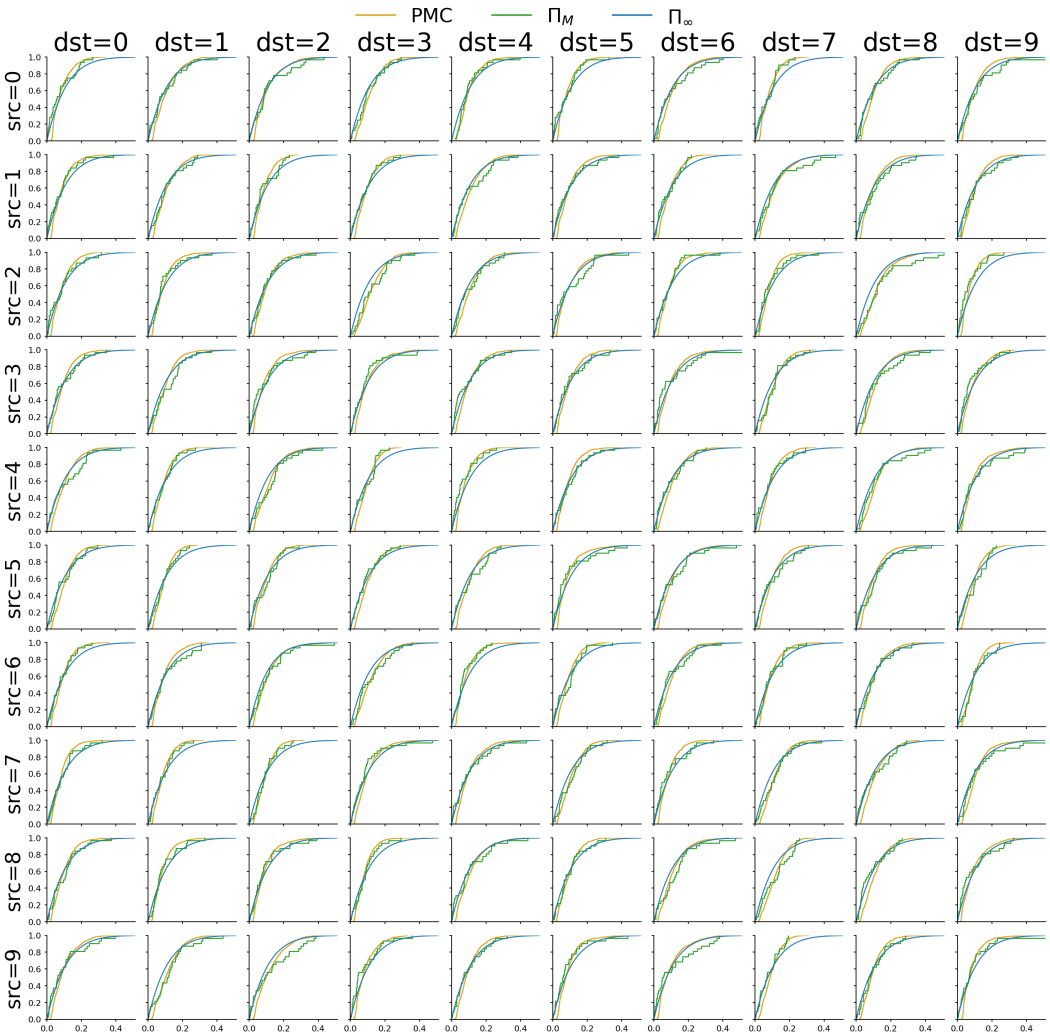

Figure 60: **Markov chains: PMC prior, marginal CDF at task diversity $M = 32$.**

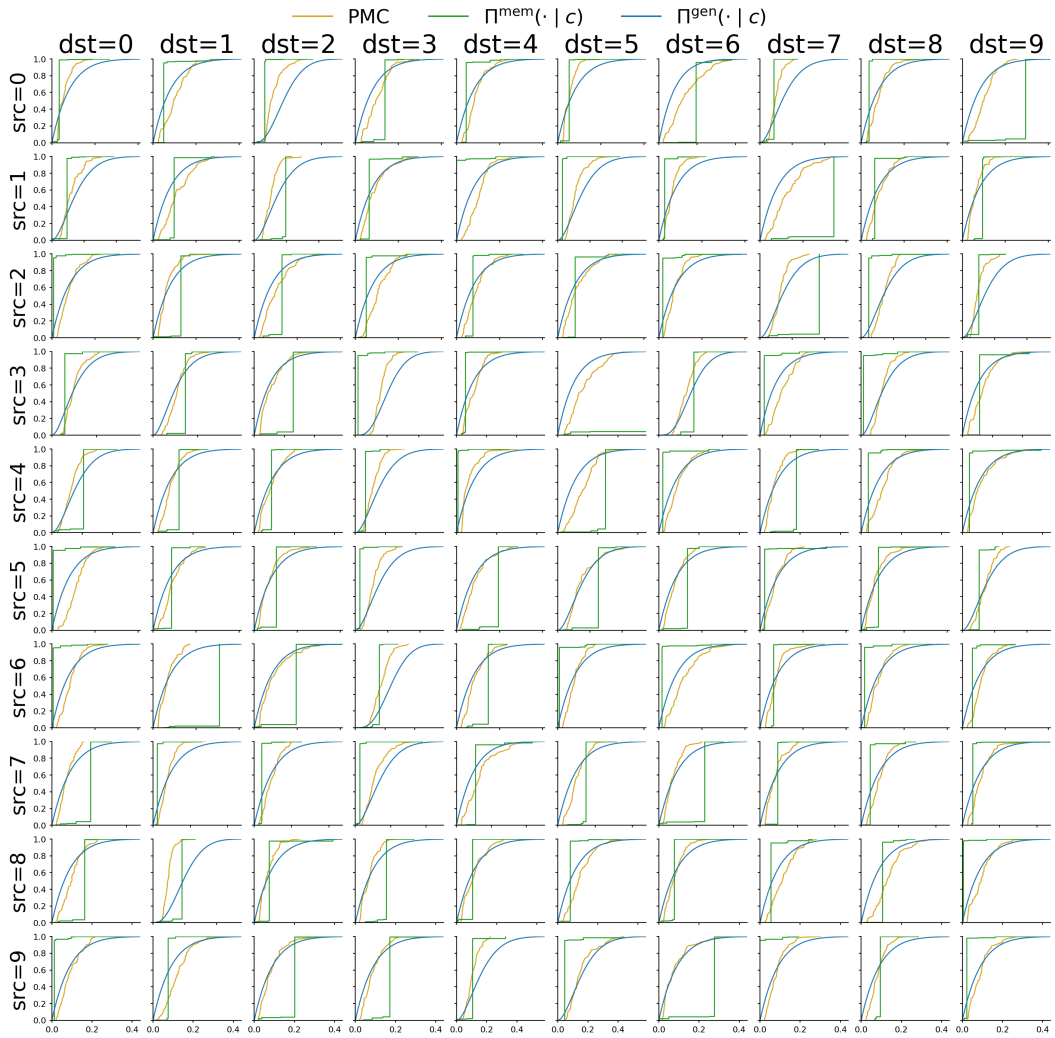

Figure 61: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 32$.

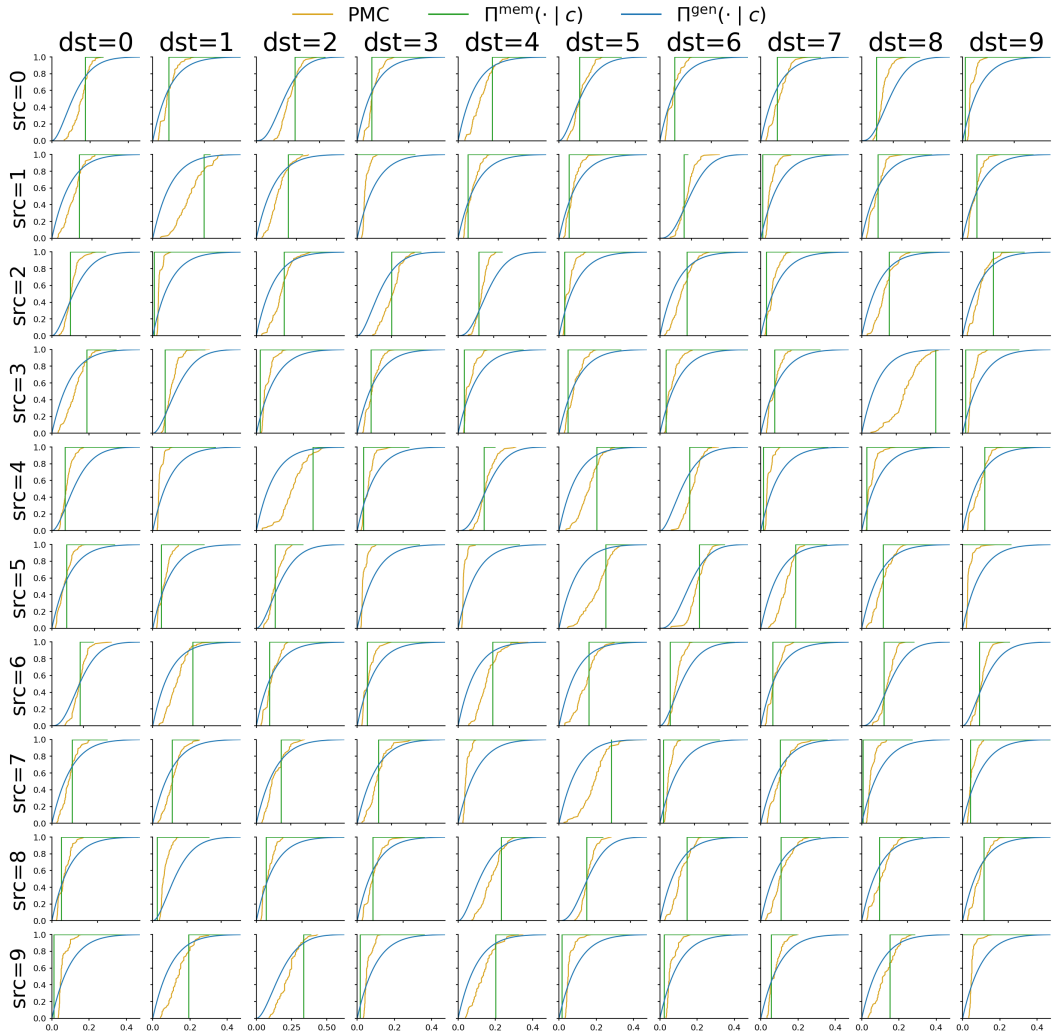

Figure 62: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 32$.

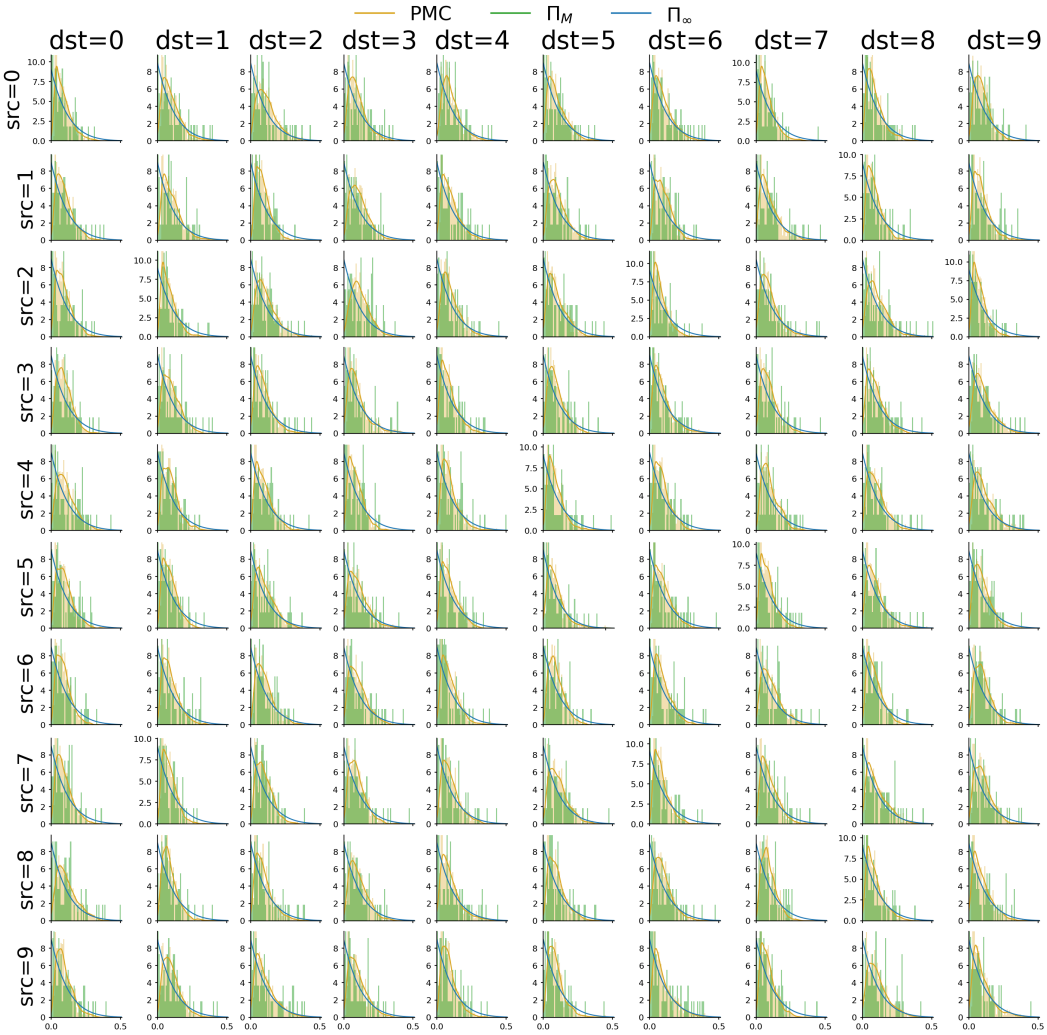

Figure 63: **Markov chains: PMC prior, marginal density at task diversity $M = 64$.**

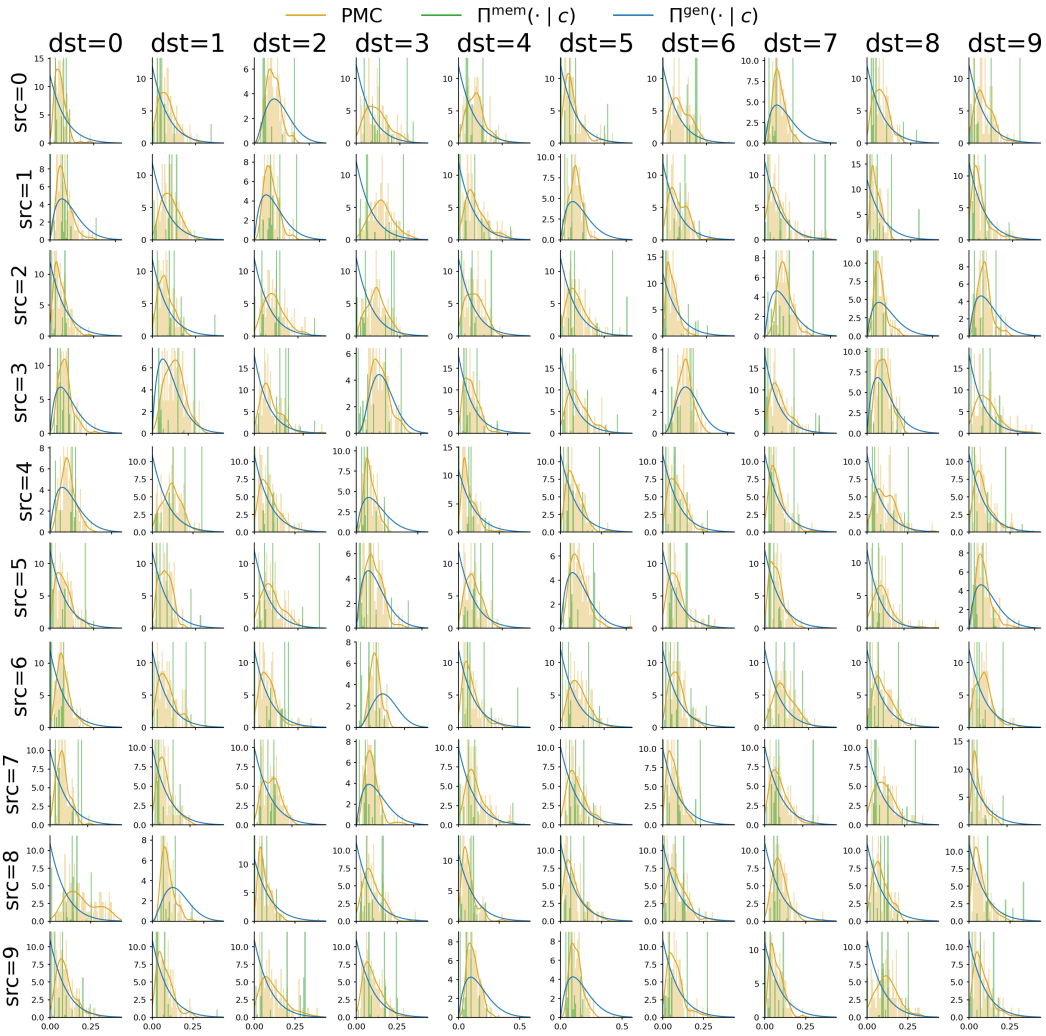

Figure 64: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 64$.

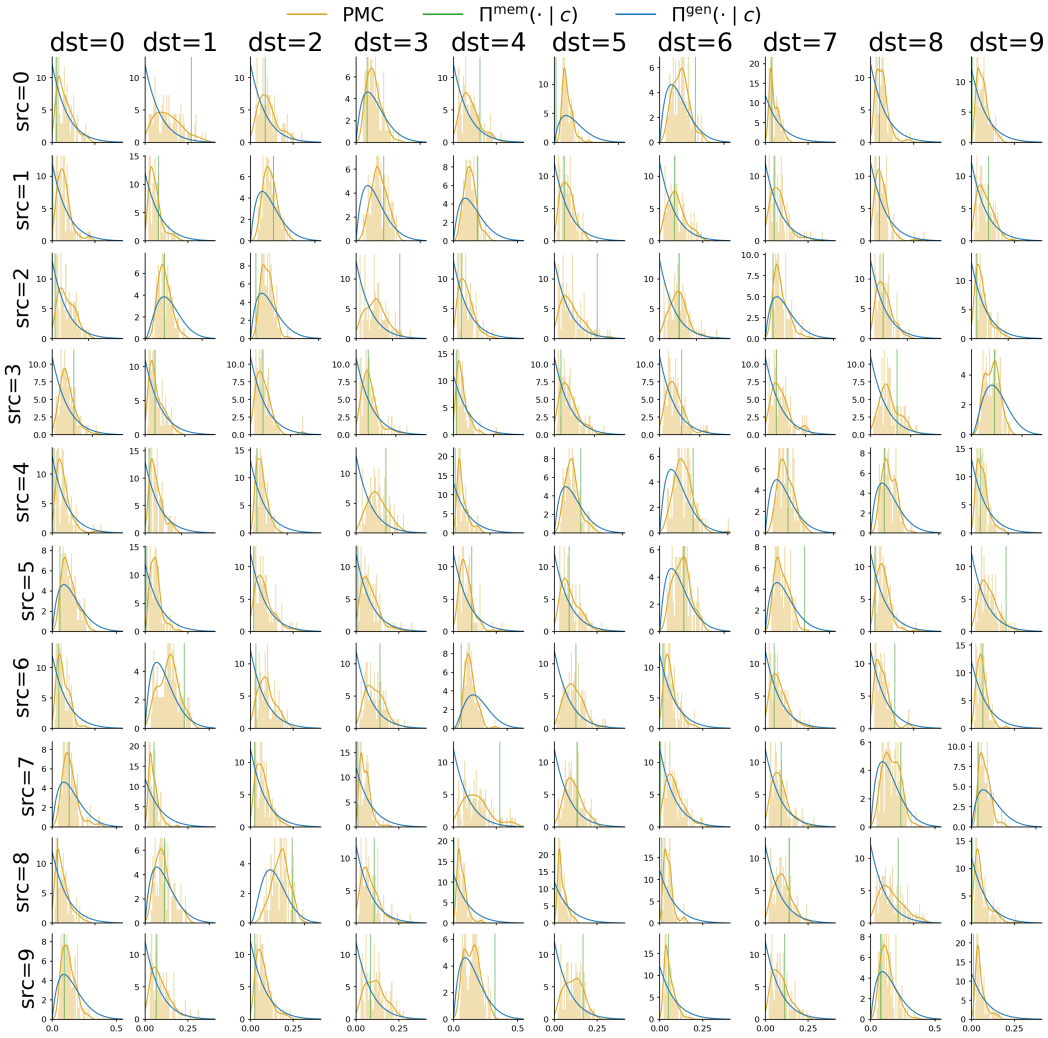

Figure 65: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 64$.

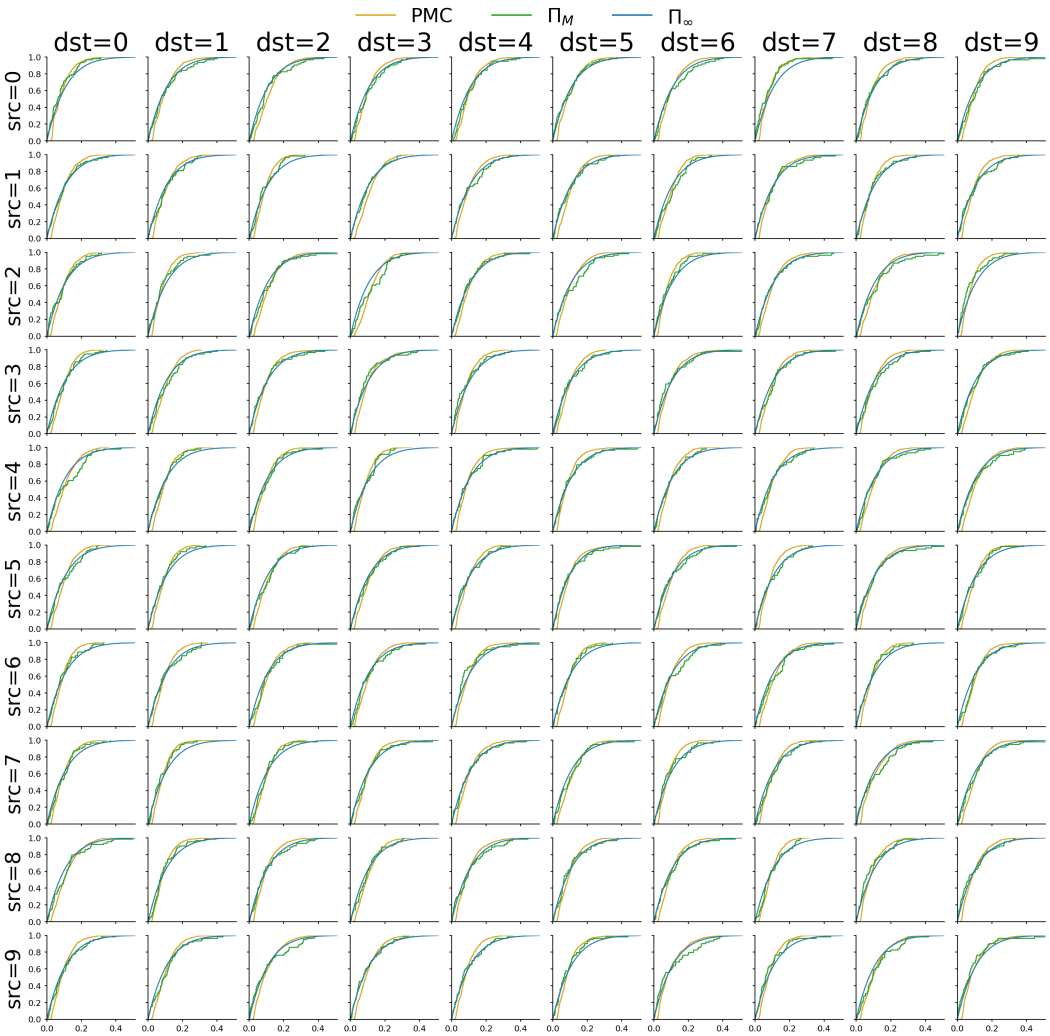

Figure 66: **Markov chains: PMC prior, marginal CDF at task diversity $M = 64$.**

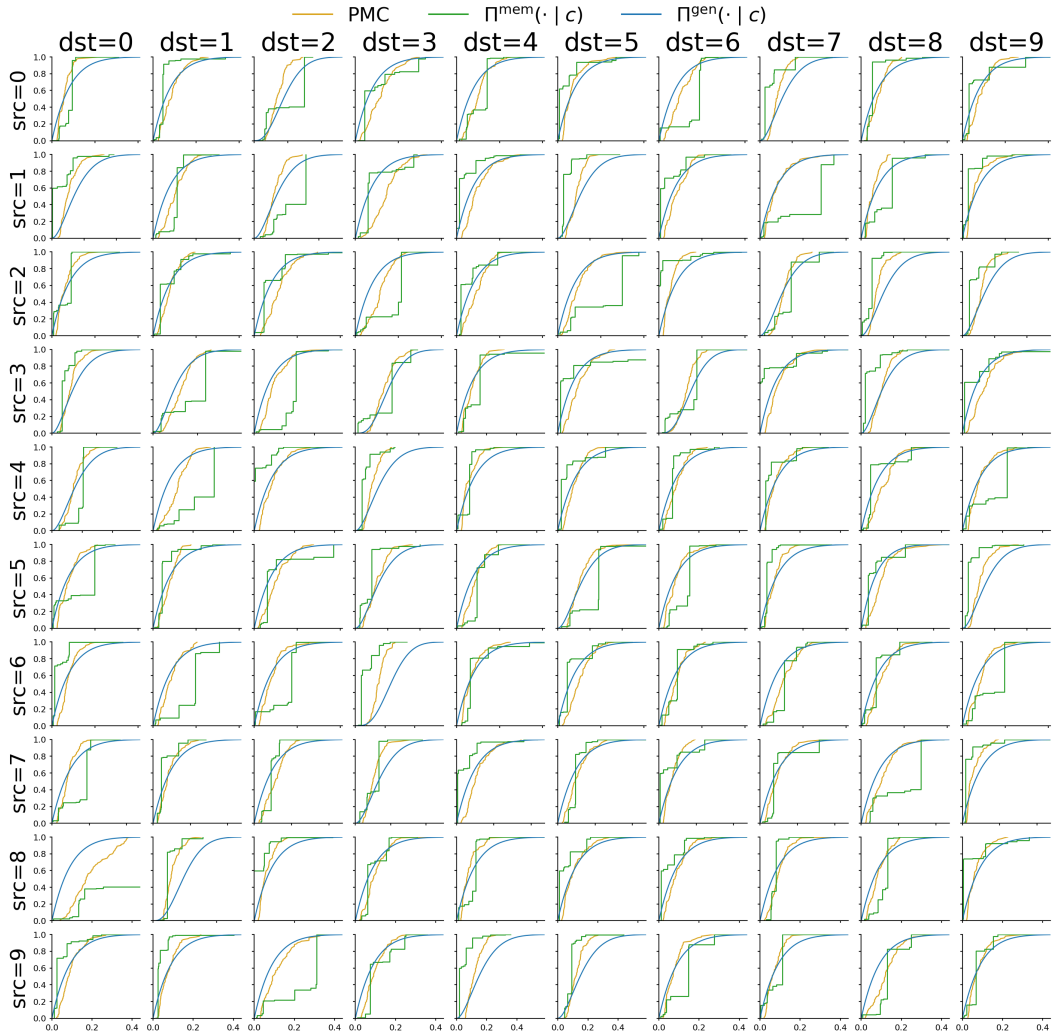

Figure 67: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 64$.

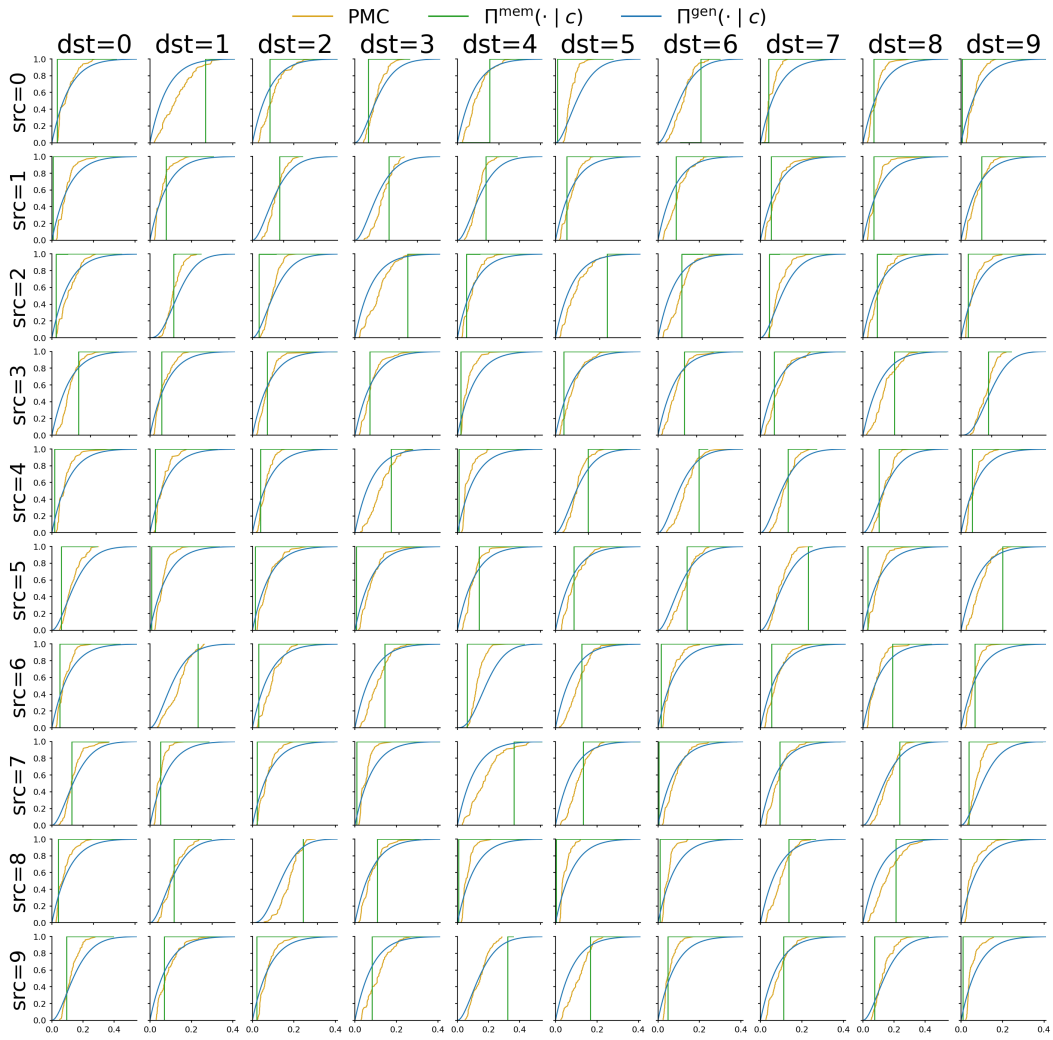

Figure 68: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 64$.

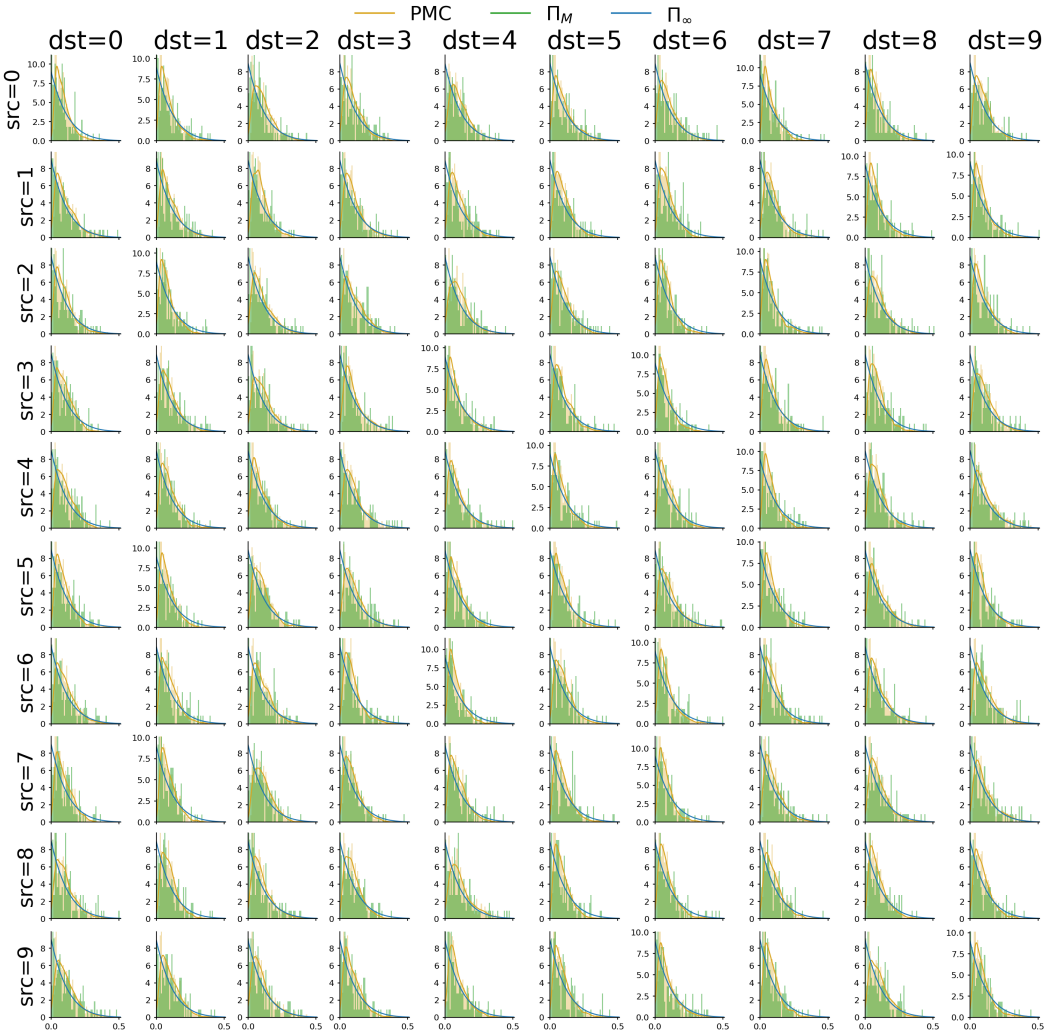

Figure 69: **Markov chains: PMC prior, marginal density at task diversity $M = 128$.**

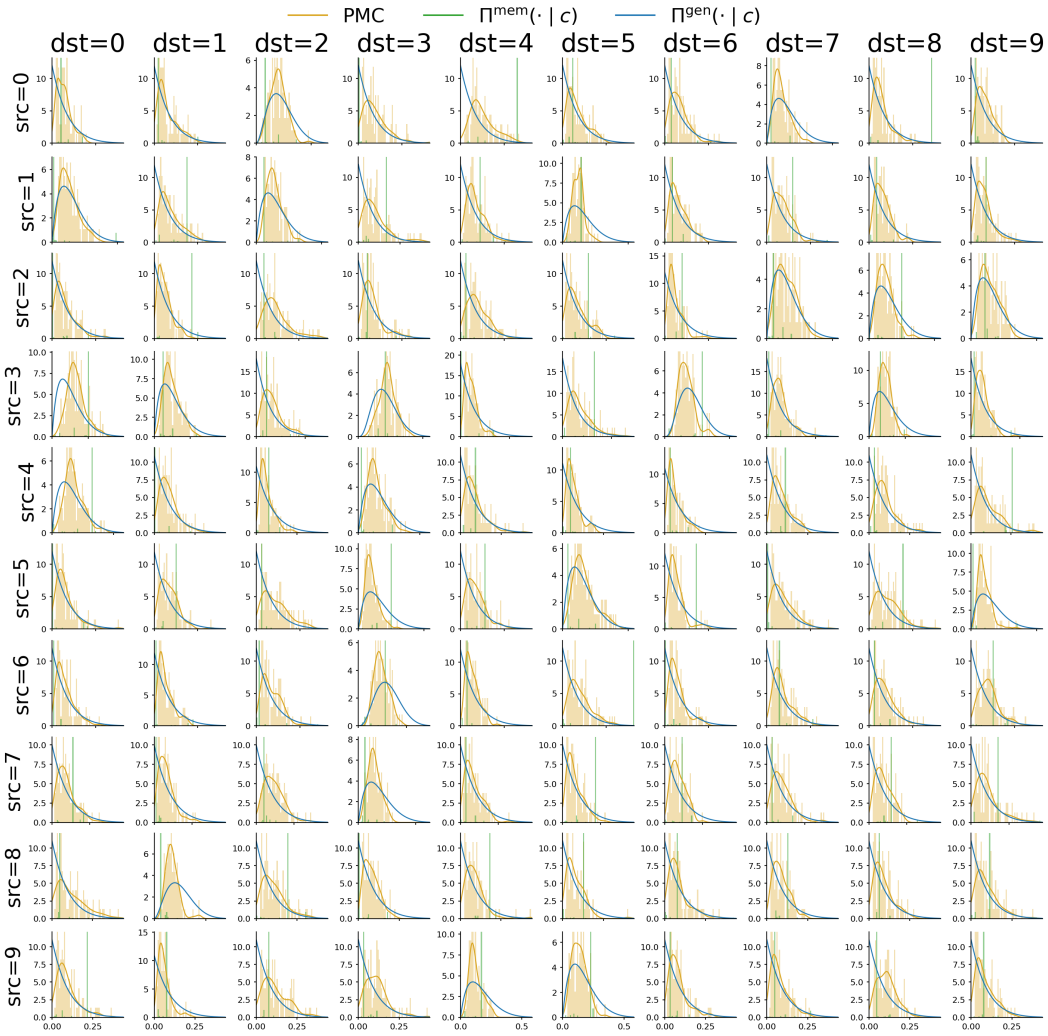

Figure 70: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 128$.

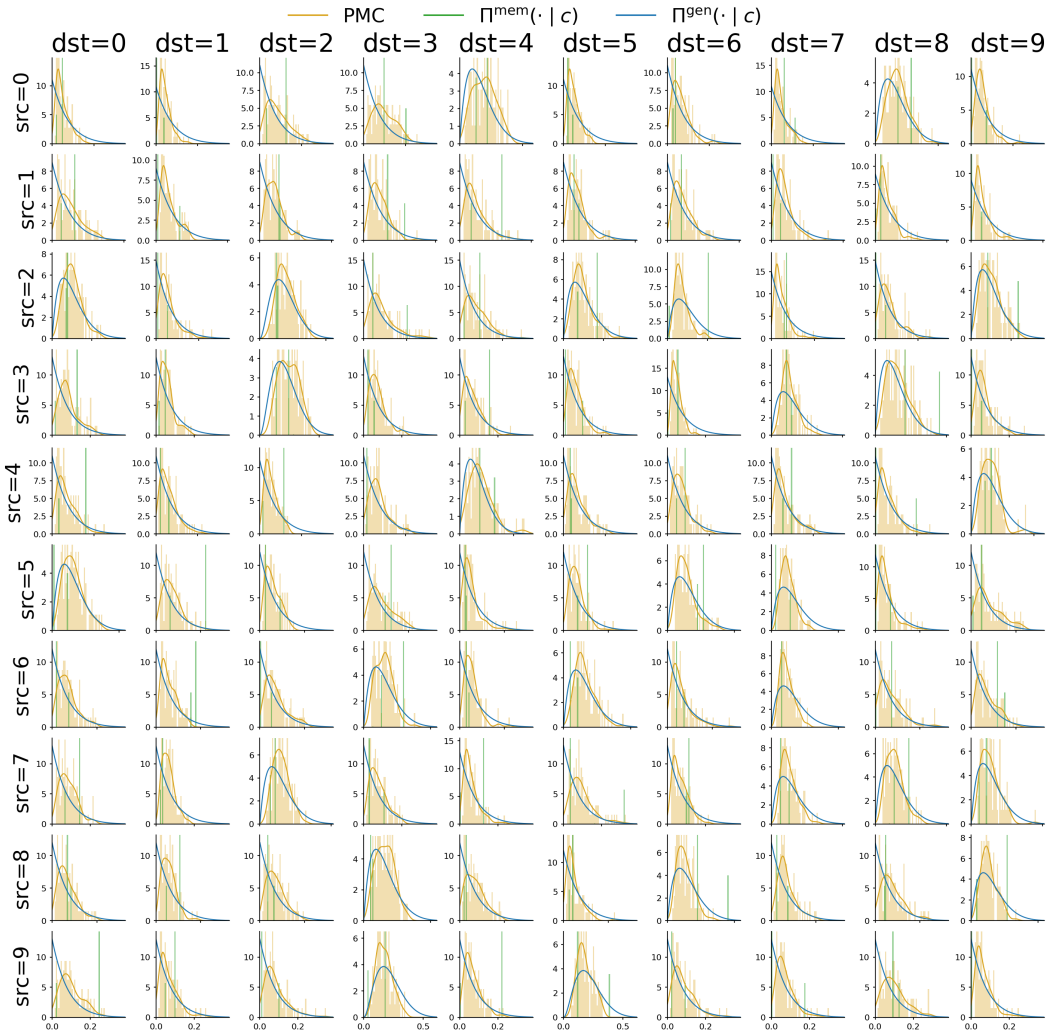

Figure 71: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 128$.

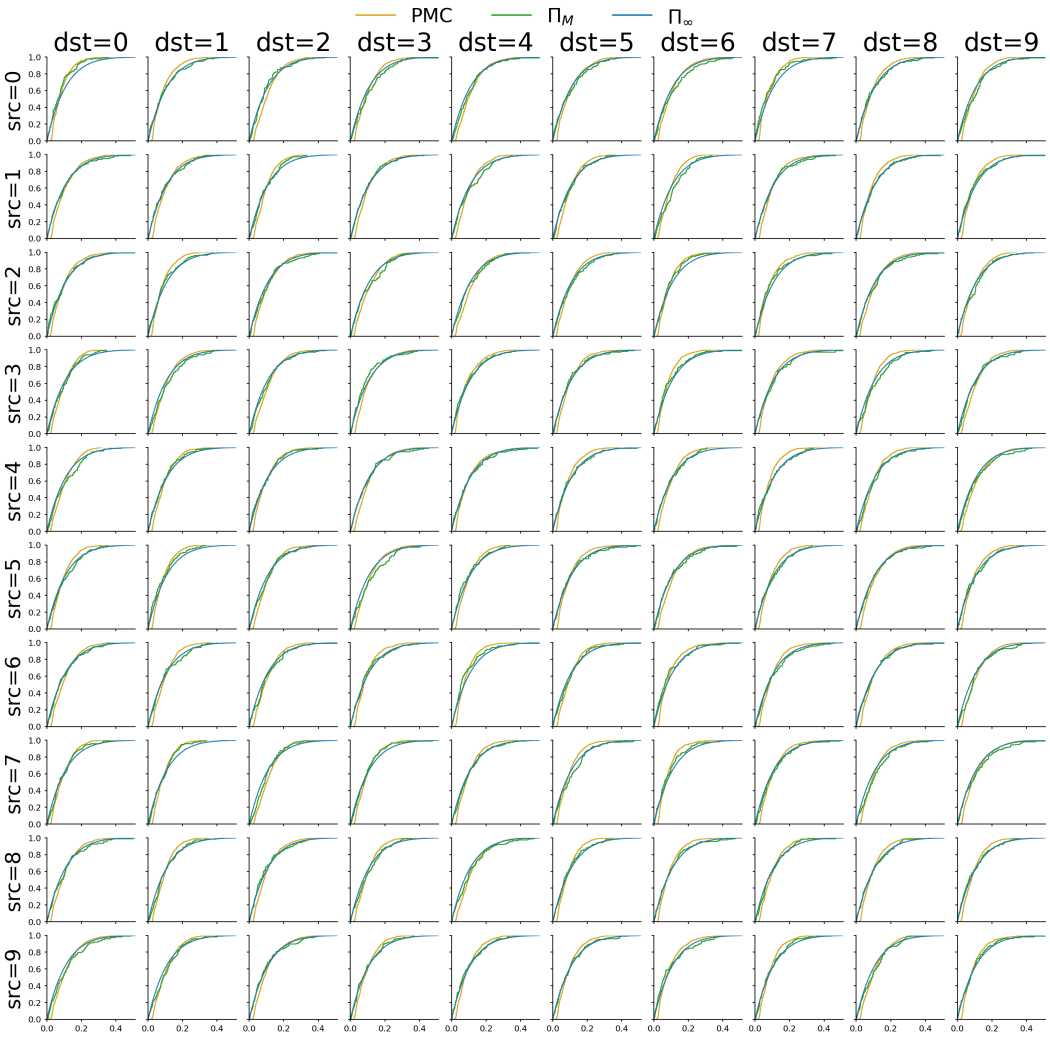

Figure 72: **Markov chains: PMC prior, marginal CDF at task diversity $M = 128$.**

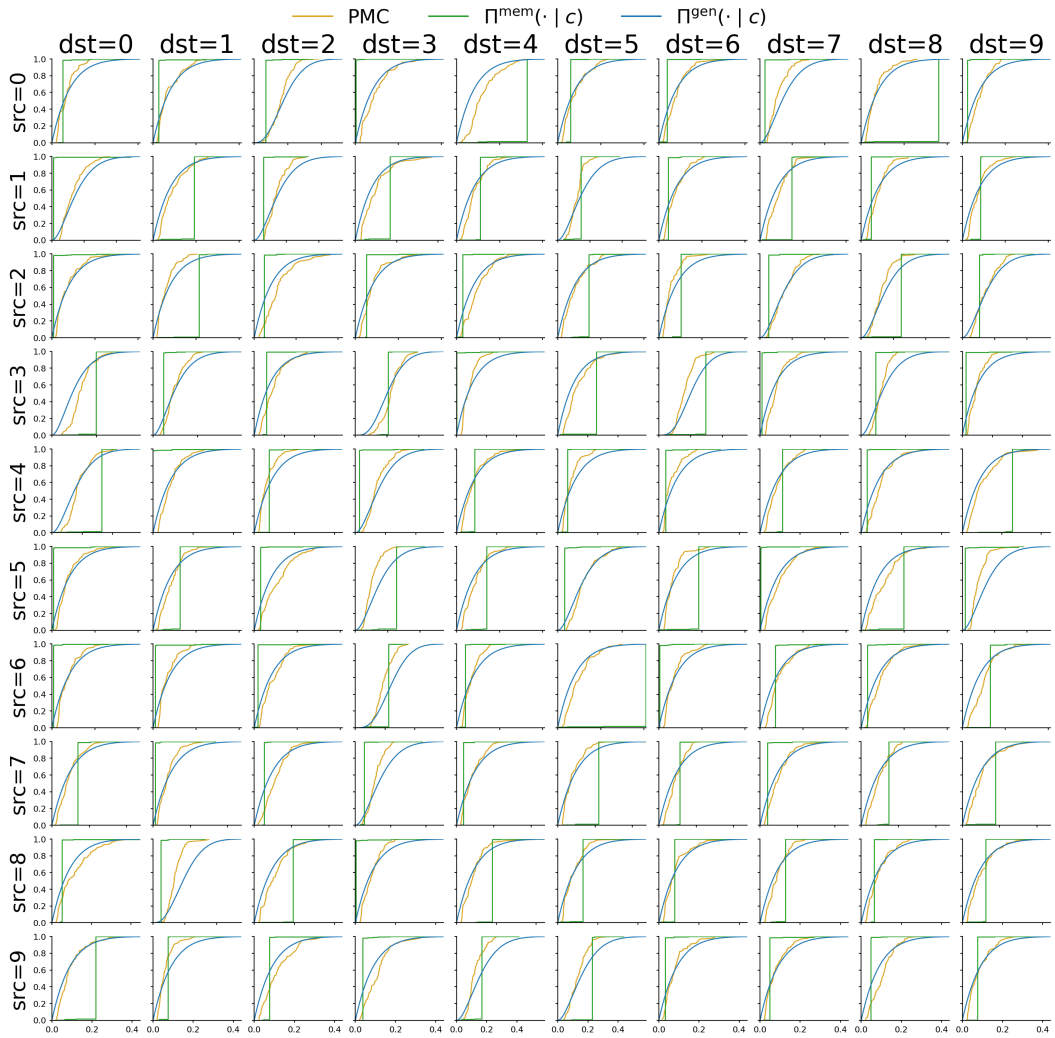

Figure 73: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 128$.

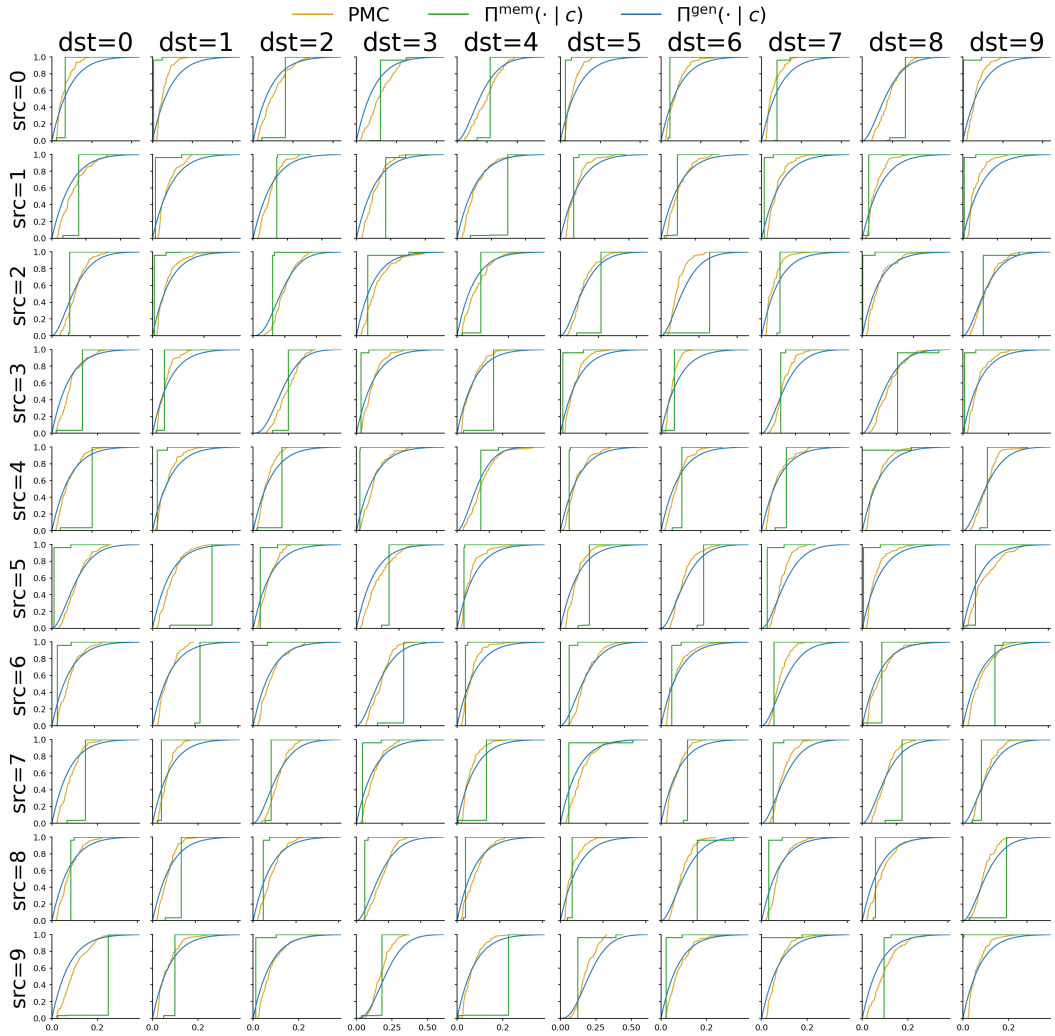

Figure 74: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 128$.

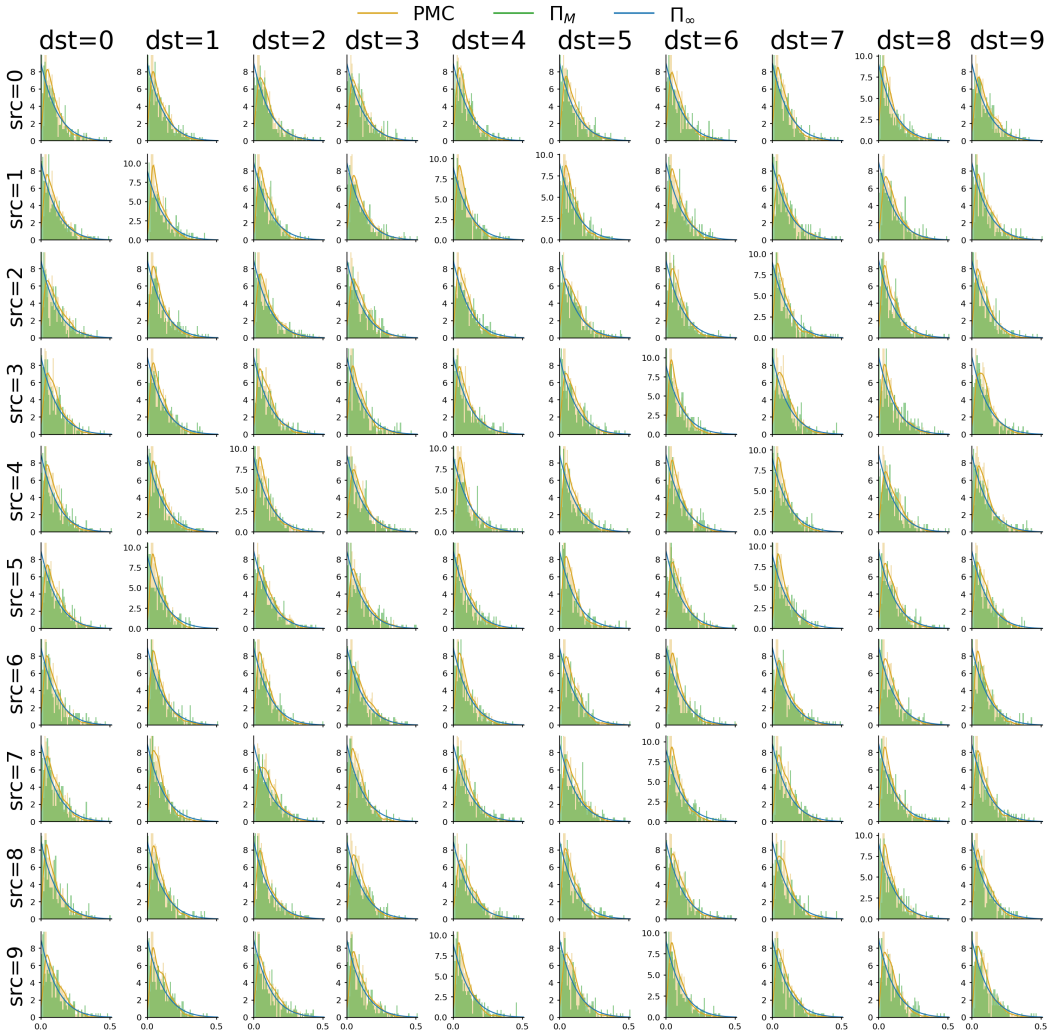

Figure 75: **Markov chains: PMC prior, marginal density at task diversity $M = 256$.**

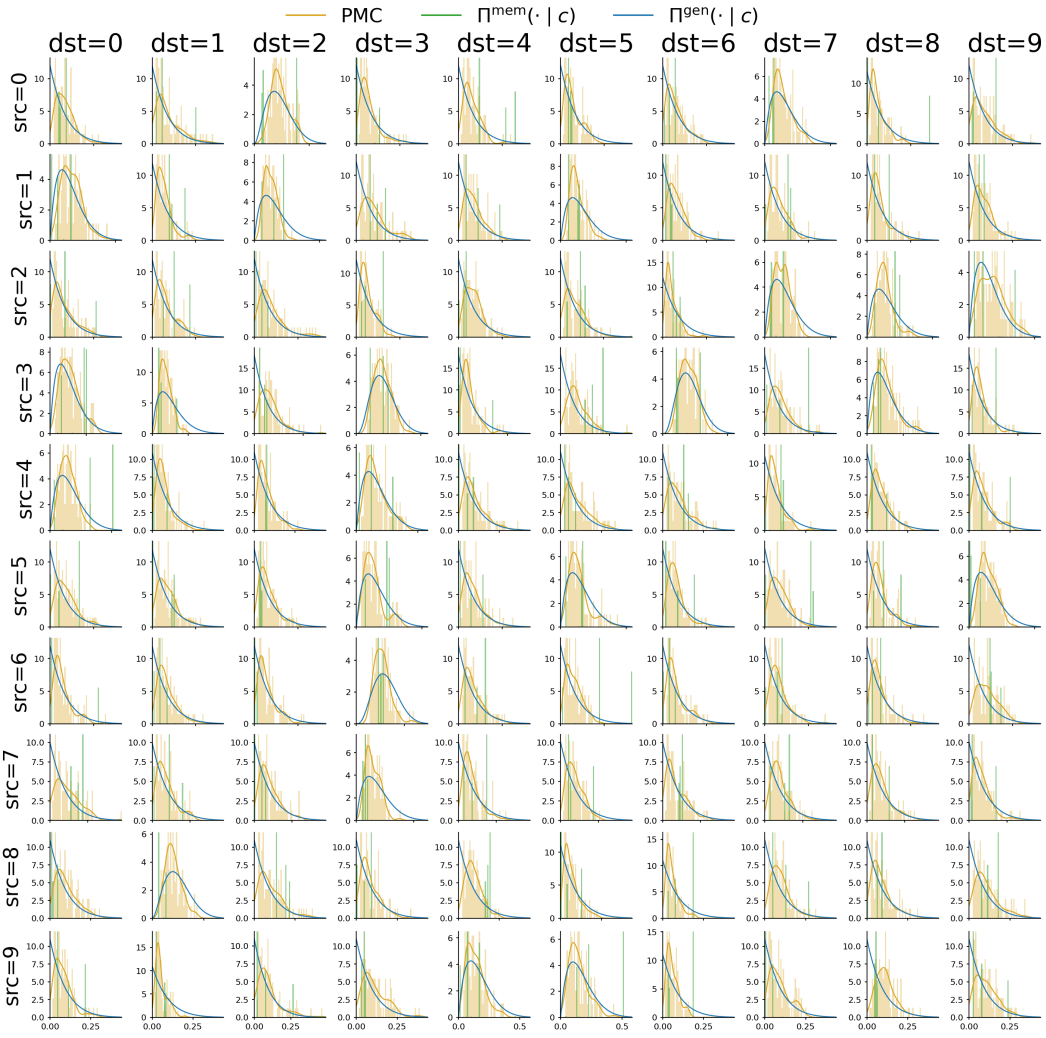

Figure 76: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 256$.

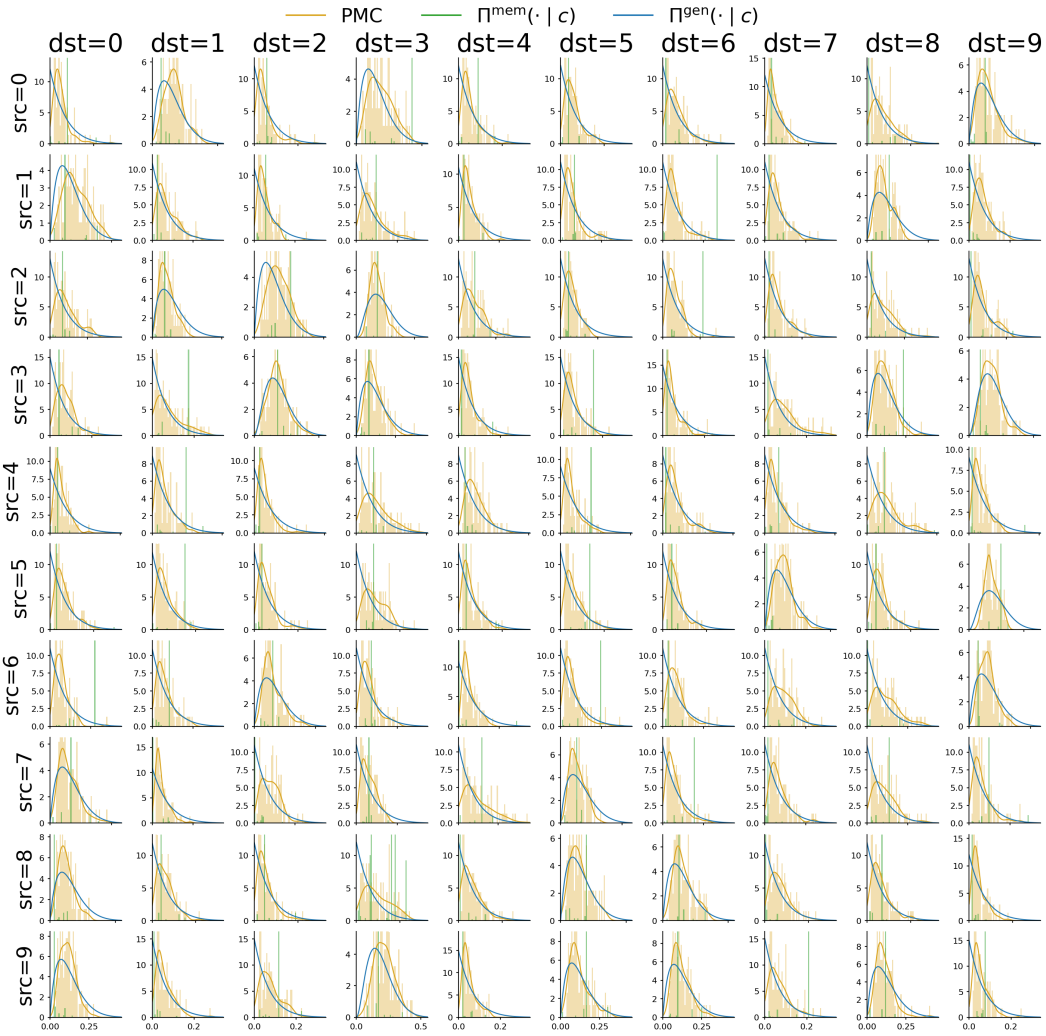

Figure 77: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 256$.

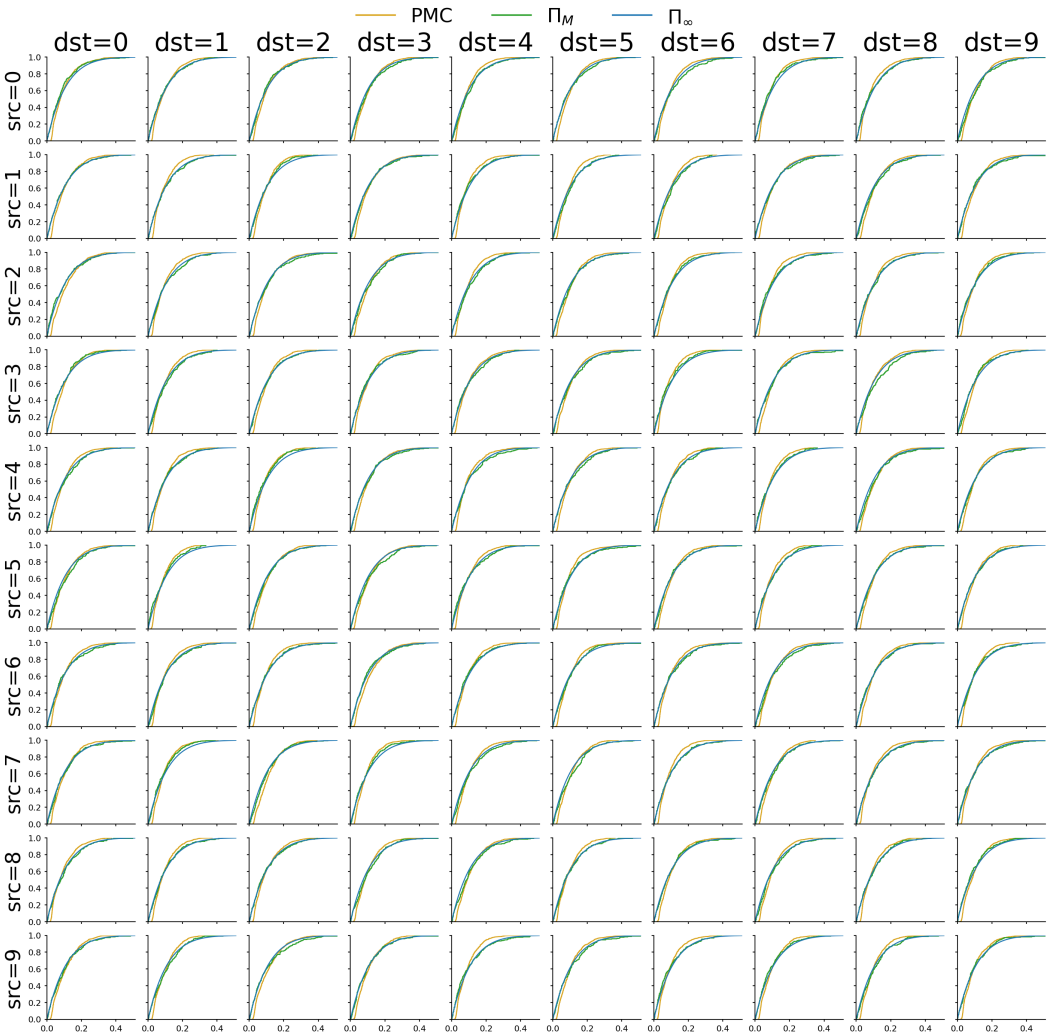

Figure 78: **Markov chains: PMC prior, marginal CDF at task diversity $M = 256$.**

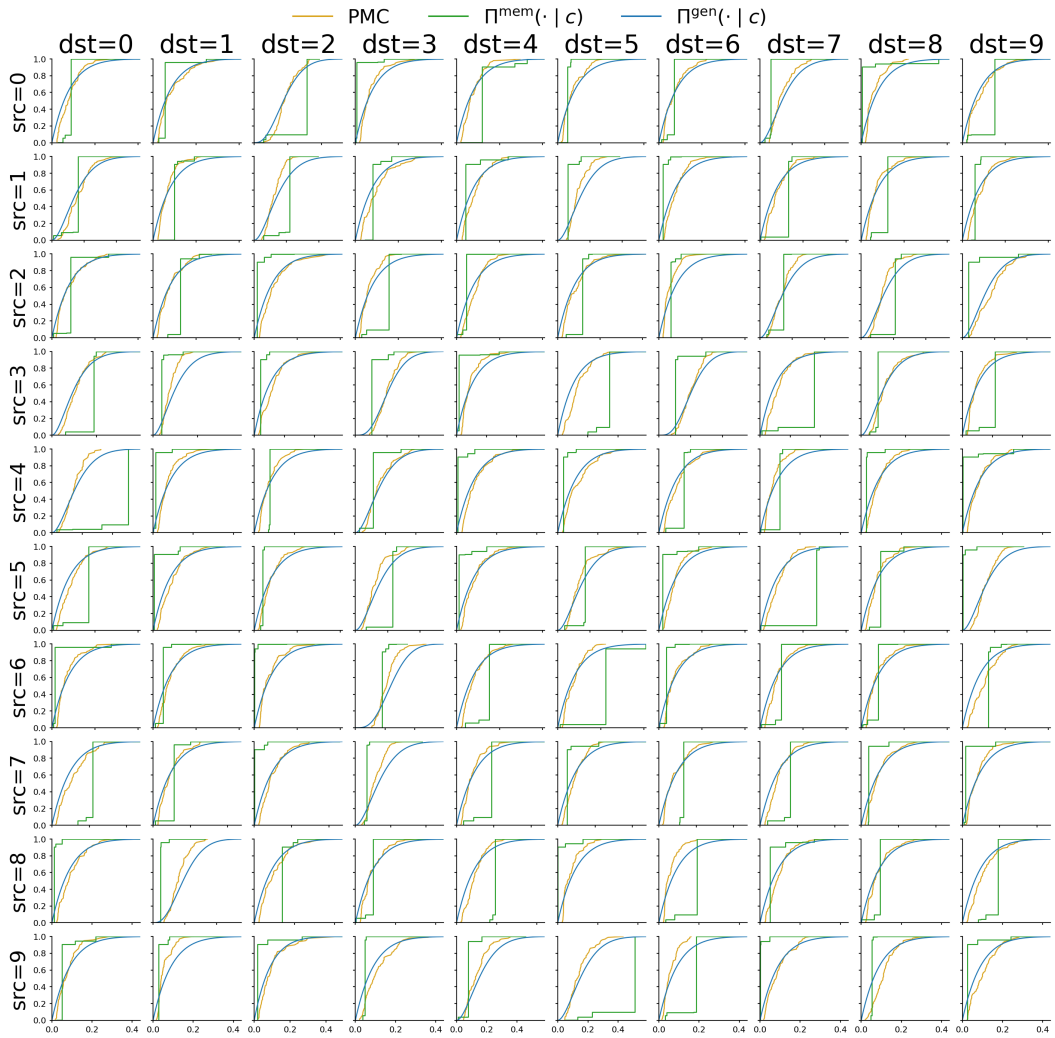

Figure 79: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 256$.

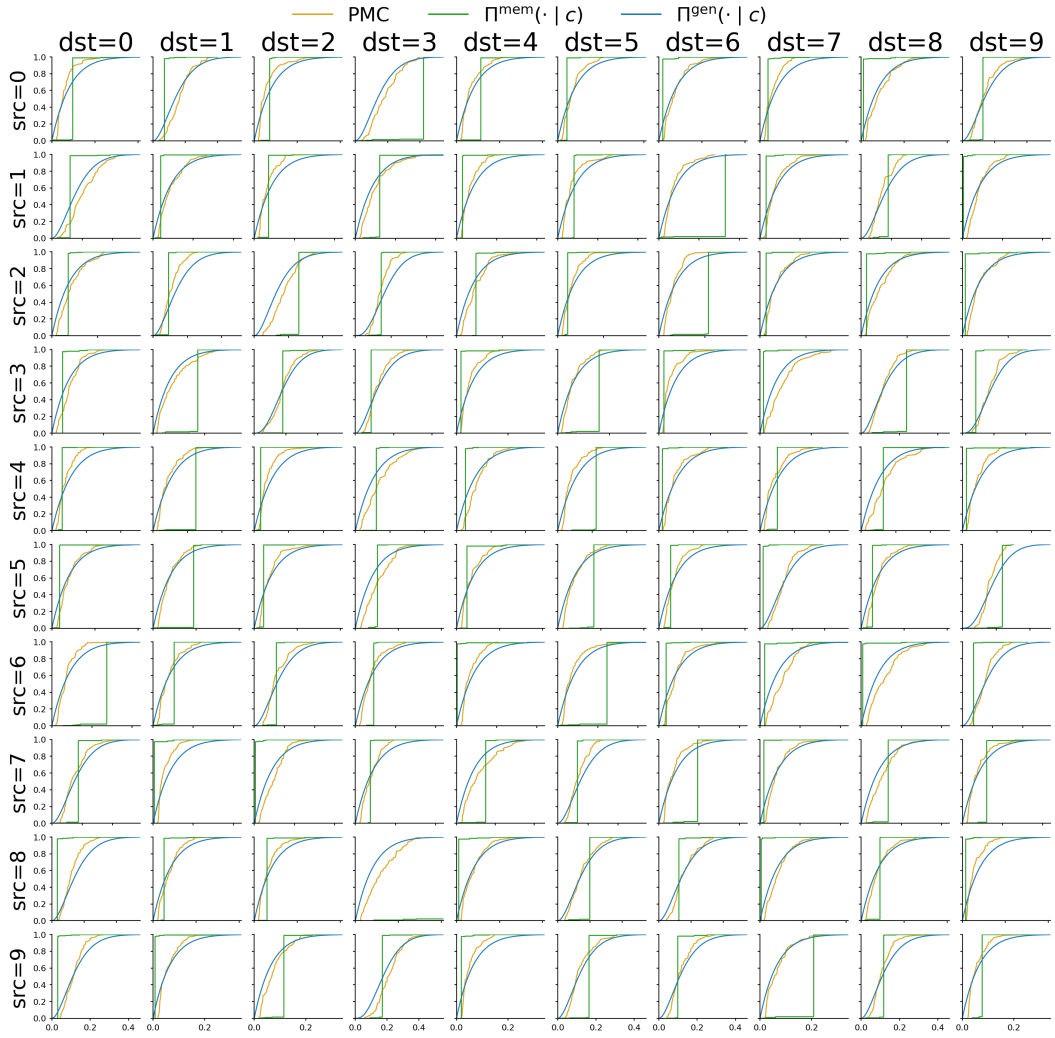

Figure 80: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 256$.

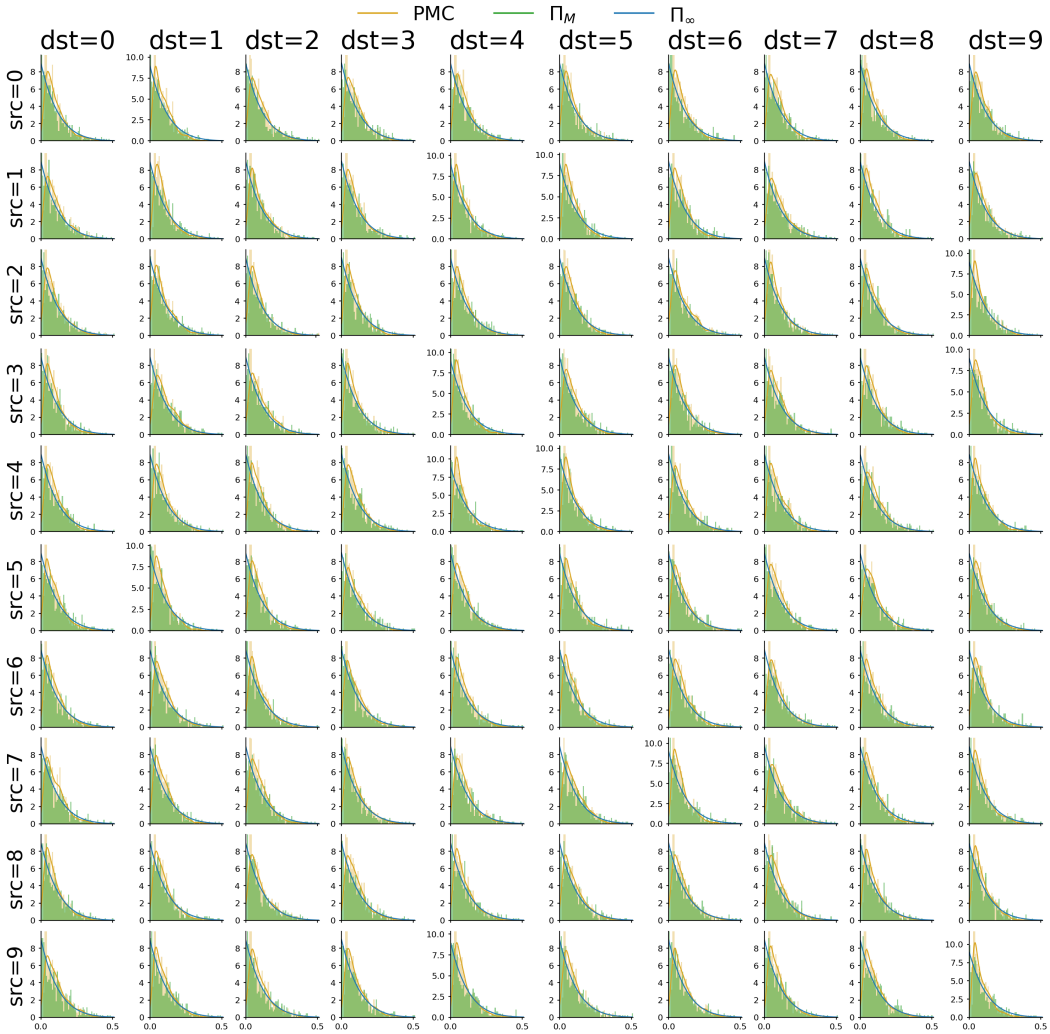

Figure 81: **Markov chains: PMC prior, marginal density at task diversity $M = 512$.**

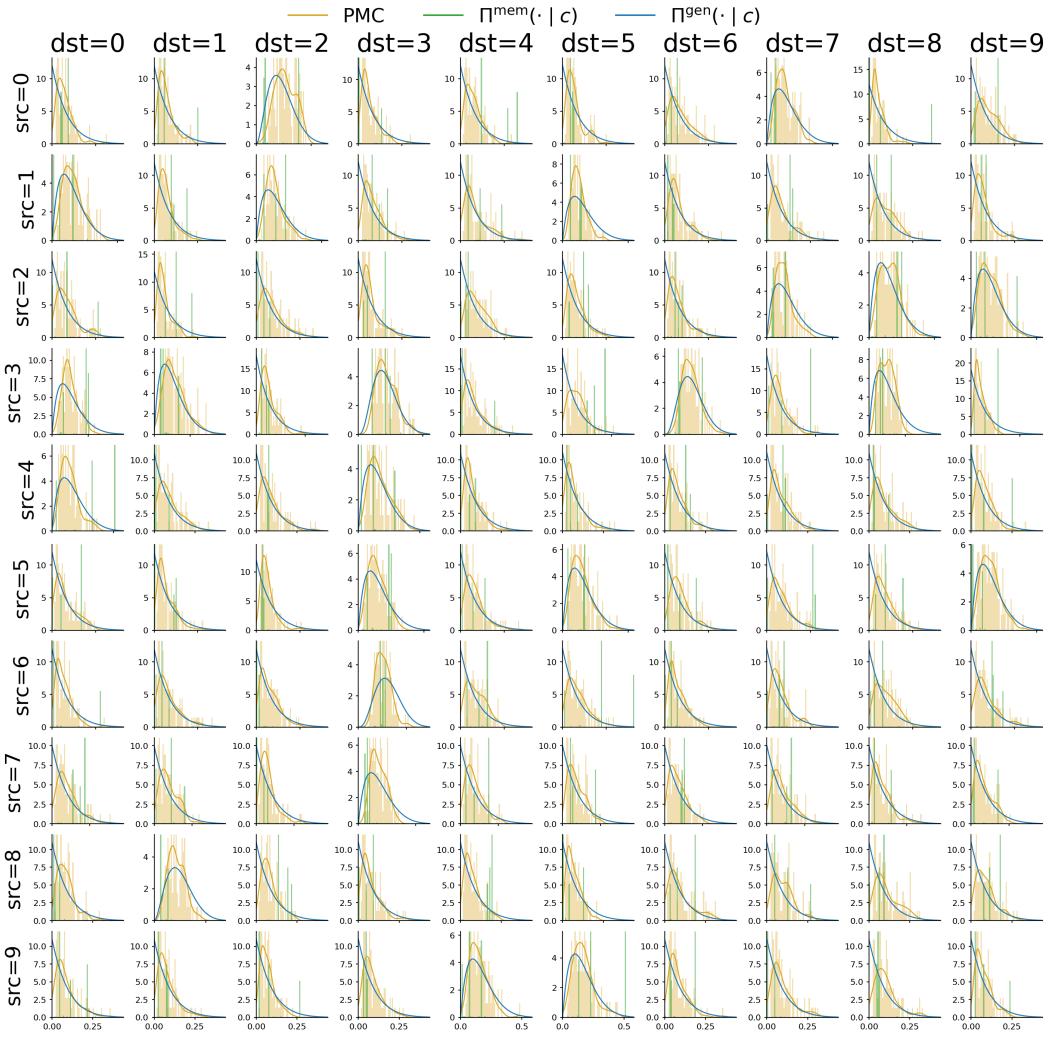

Figure 82: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 512$.

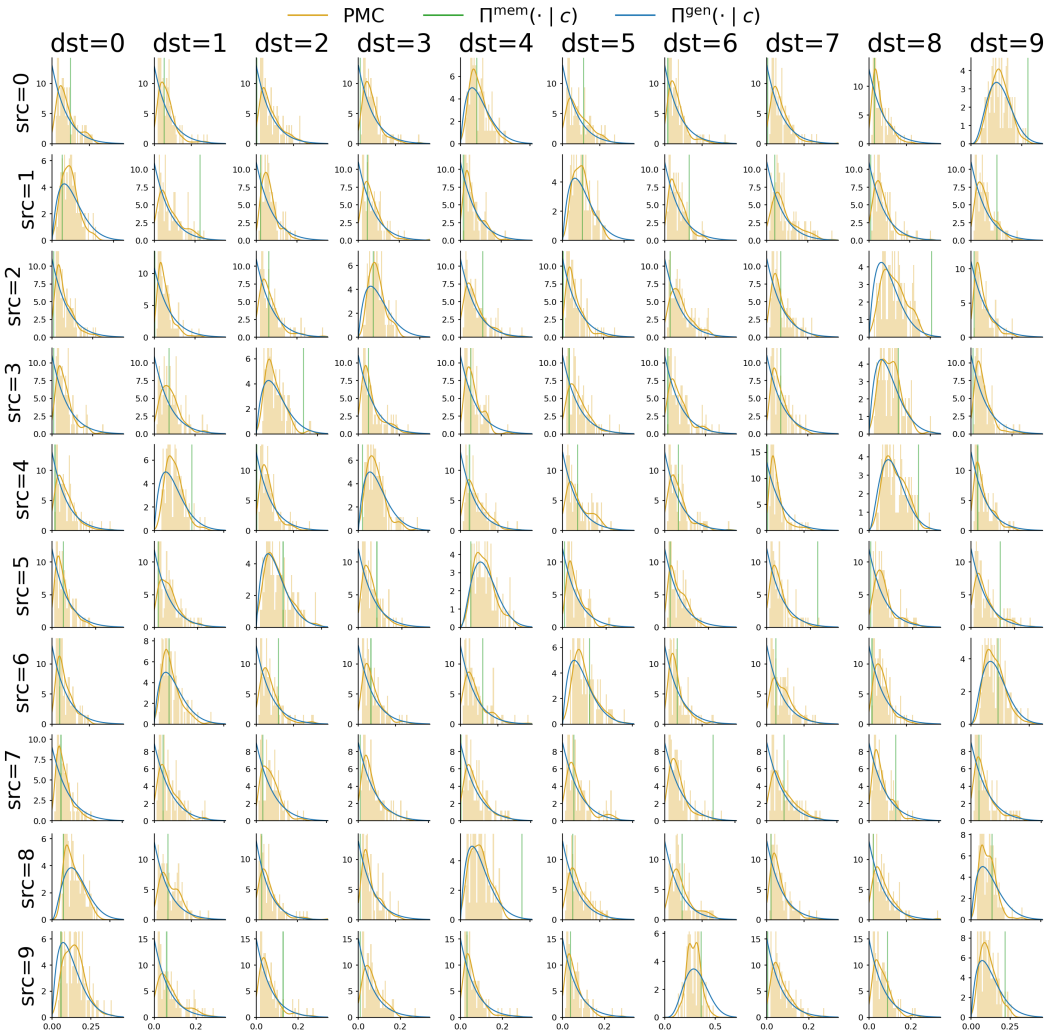

Figure 83: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 512$.

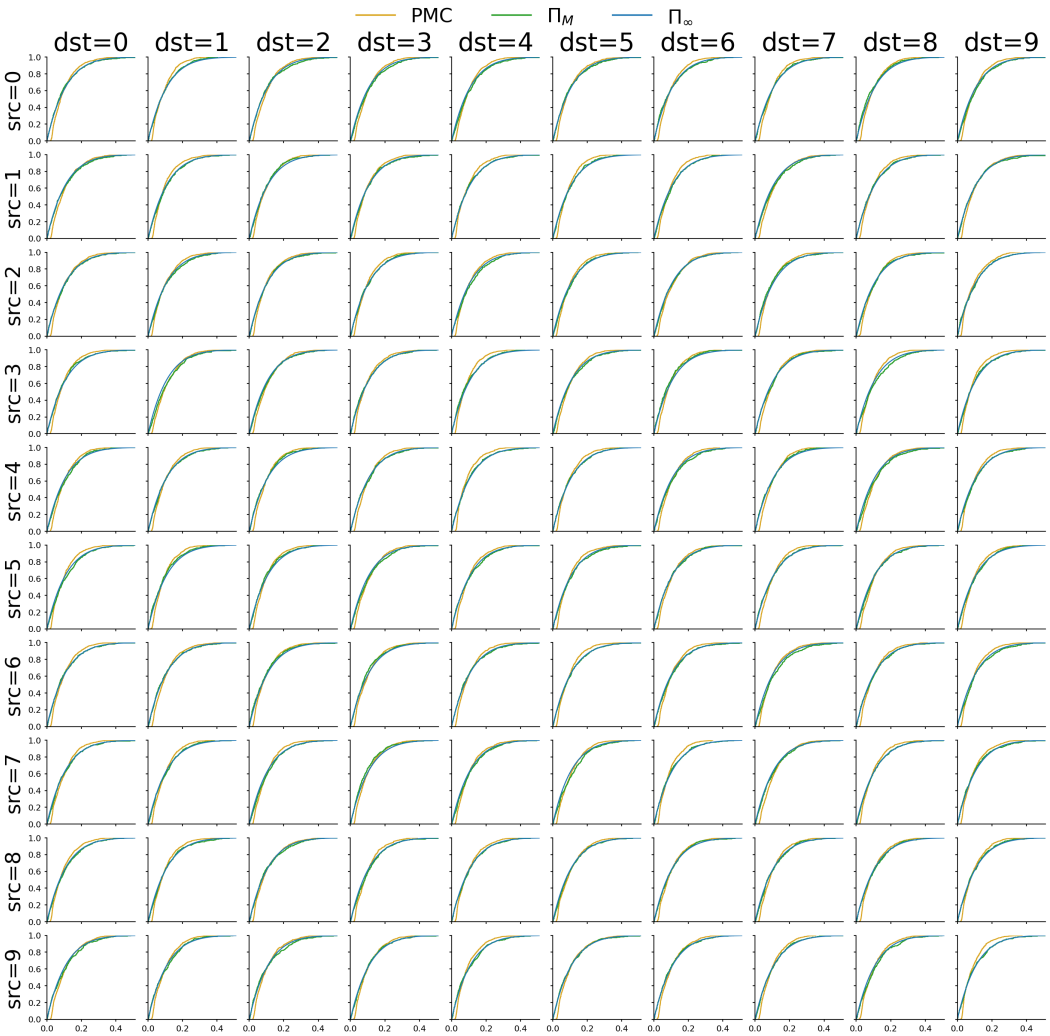

Figure 84: **Markov chains: PMC prior, marginal CDF at task diversity $M = 512$.**

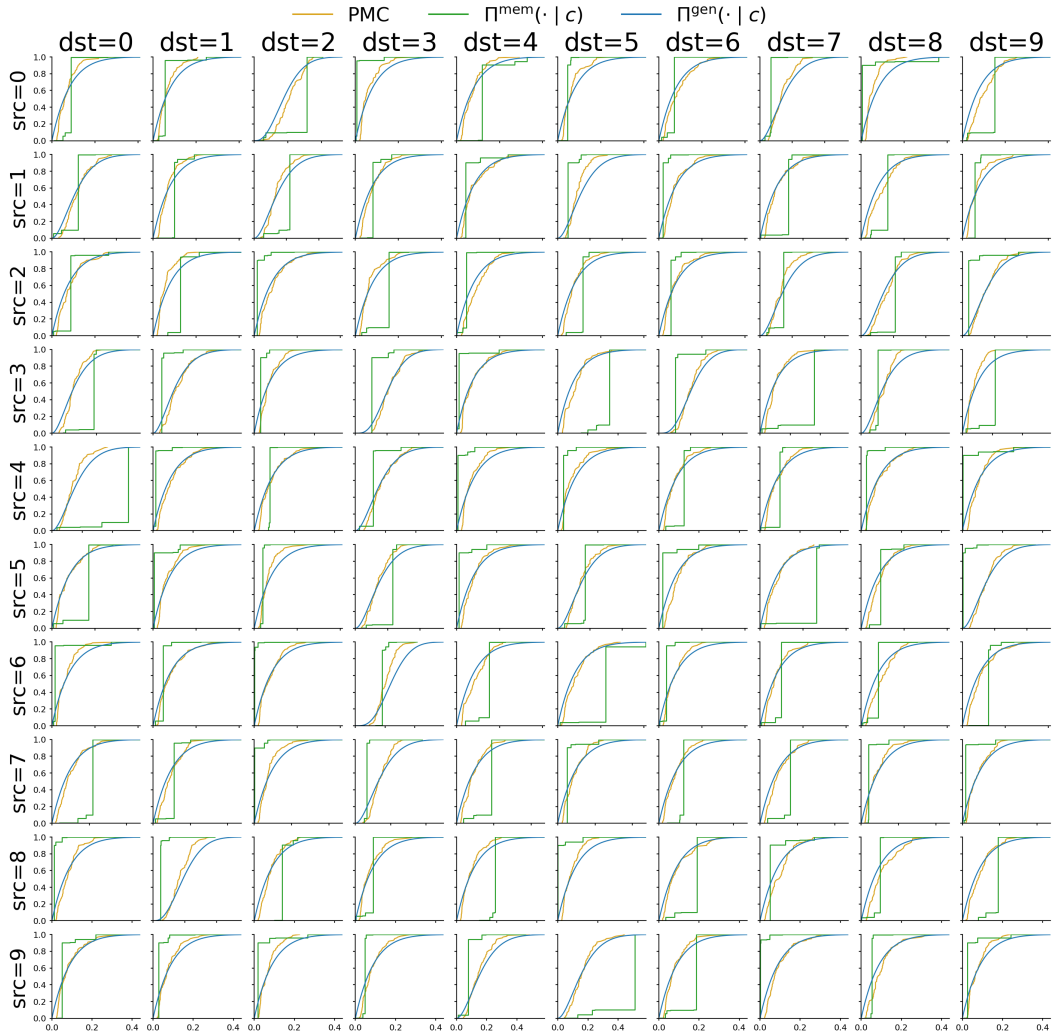

Figure 85: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 512$.

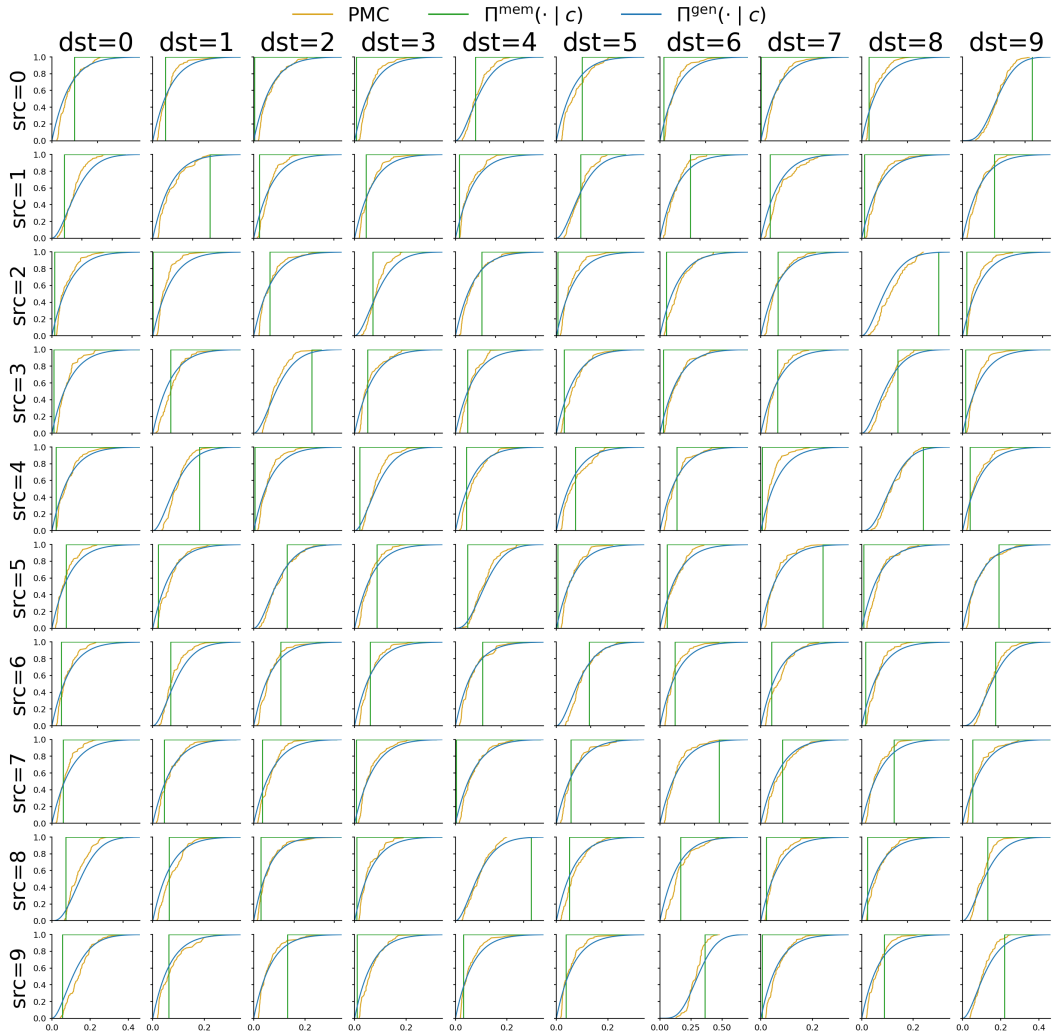

Figure 86: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 512$.

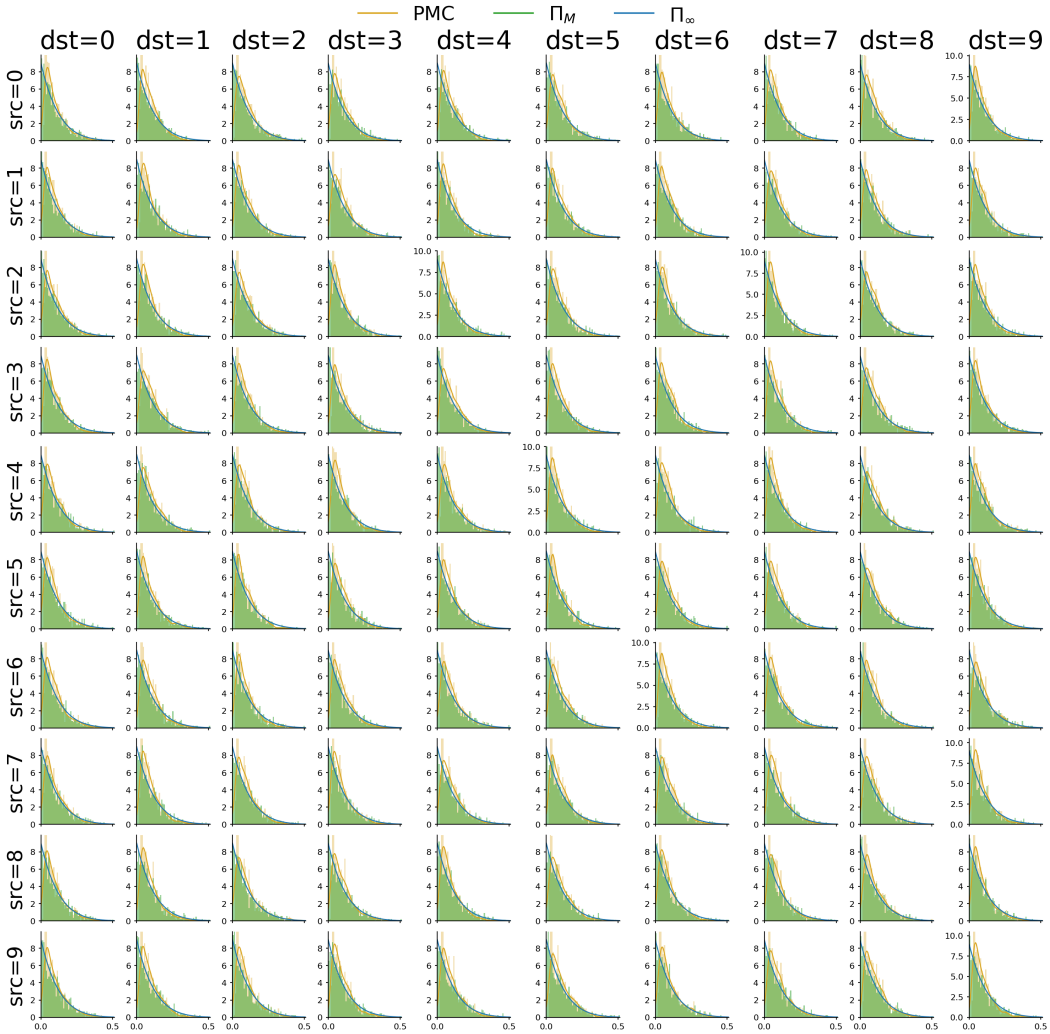

Figure 87: **Markov chains: PMC prior, marginal density at task diversity $M = 1024$.**

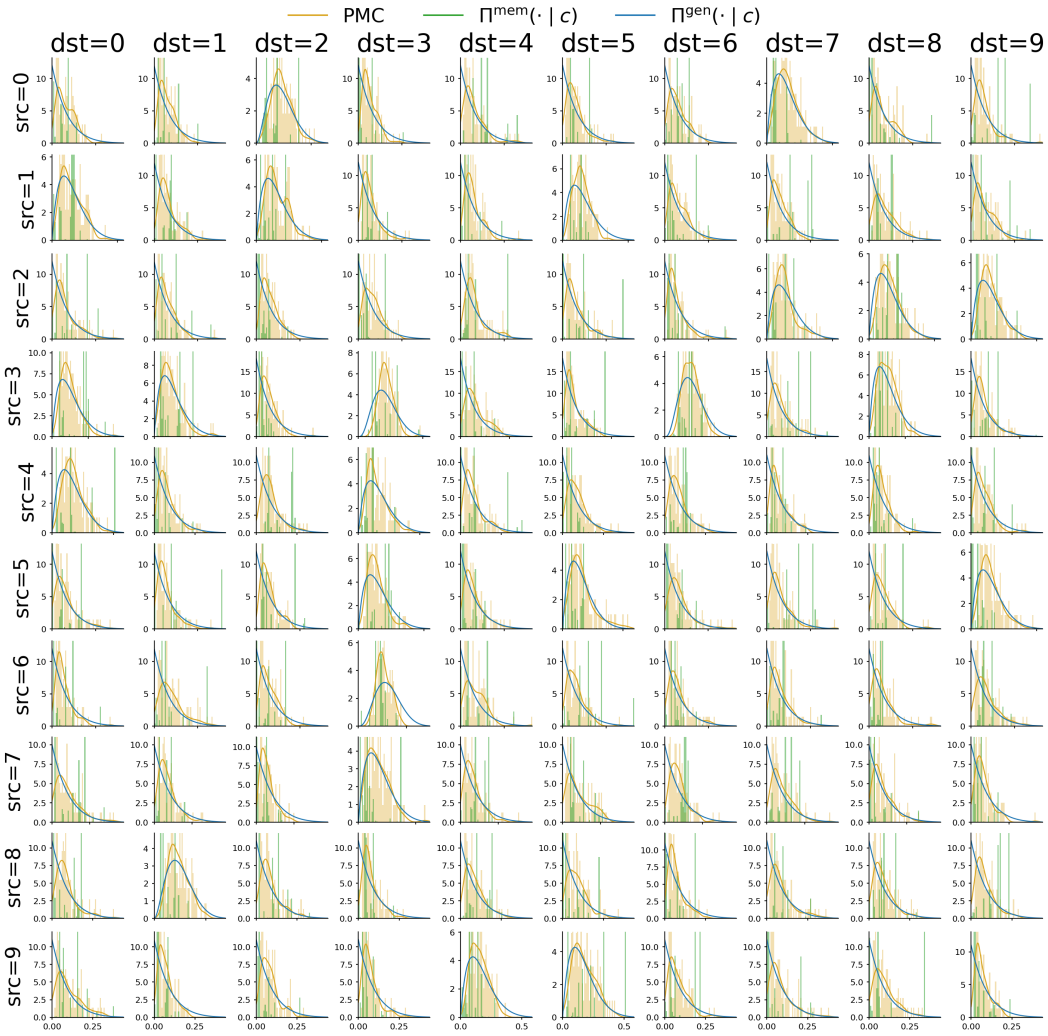

Figure 88: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 1024$.

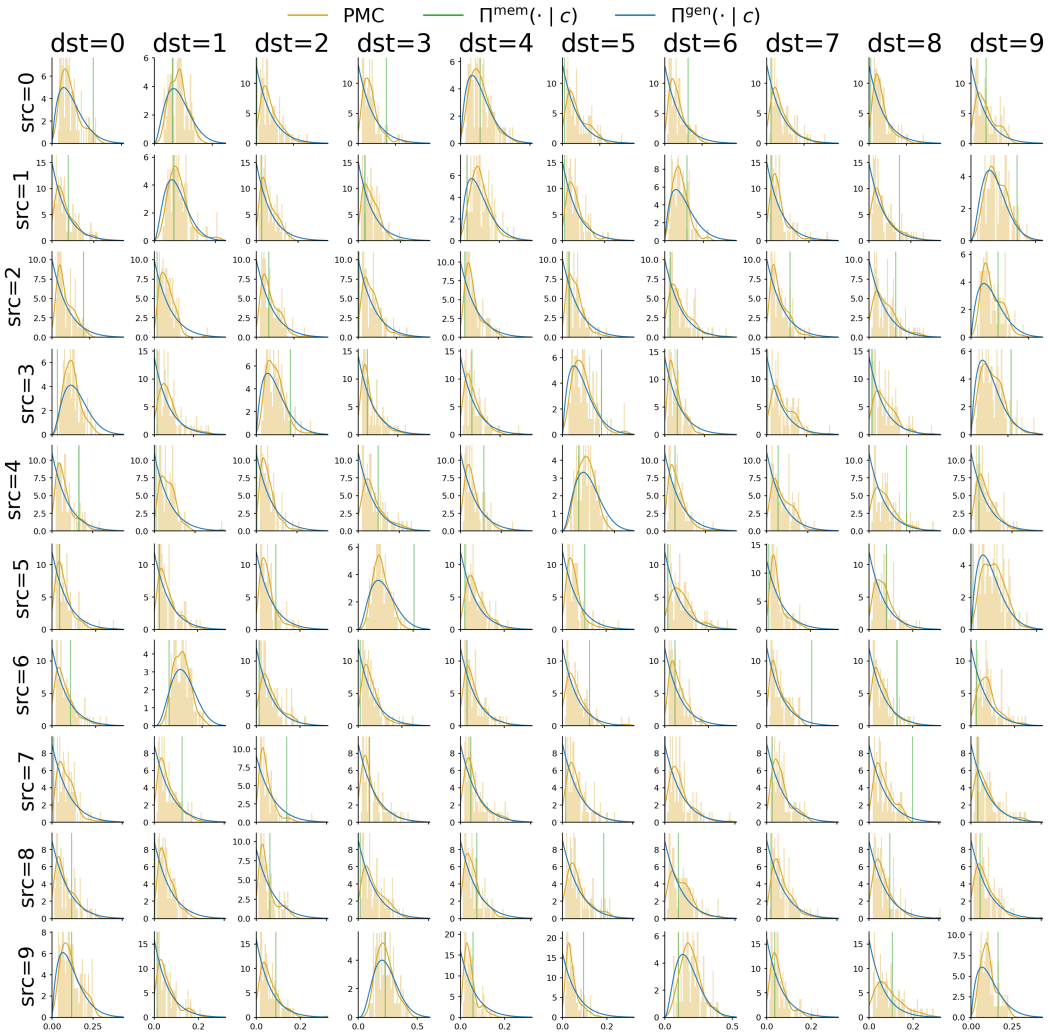

Figure 89: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 1024$.

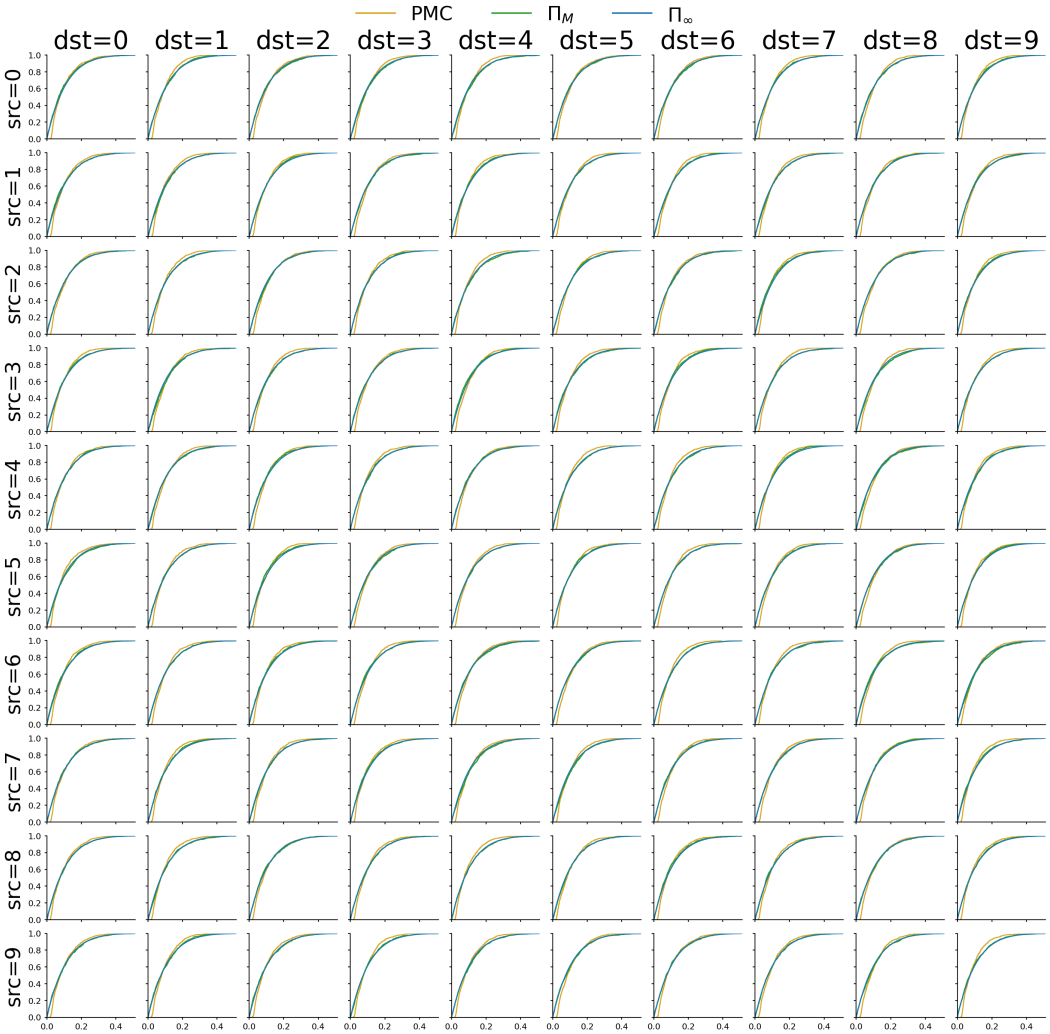

Figure 90: **Markov chains: PMC prior, marginal CDF at task diversity $M = 1024$.**

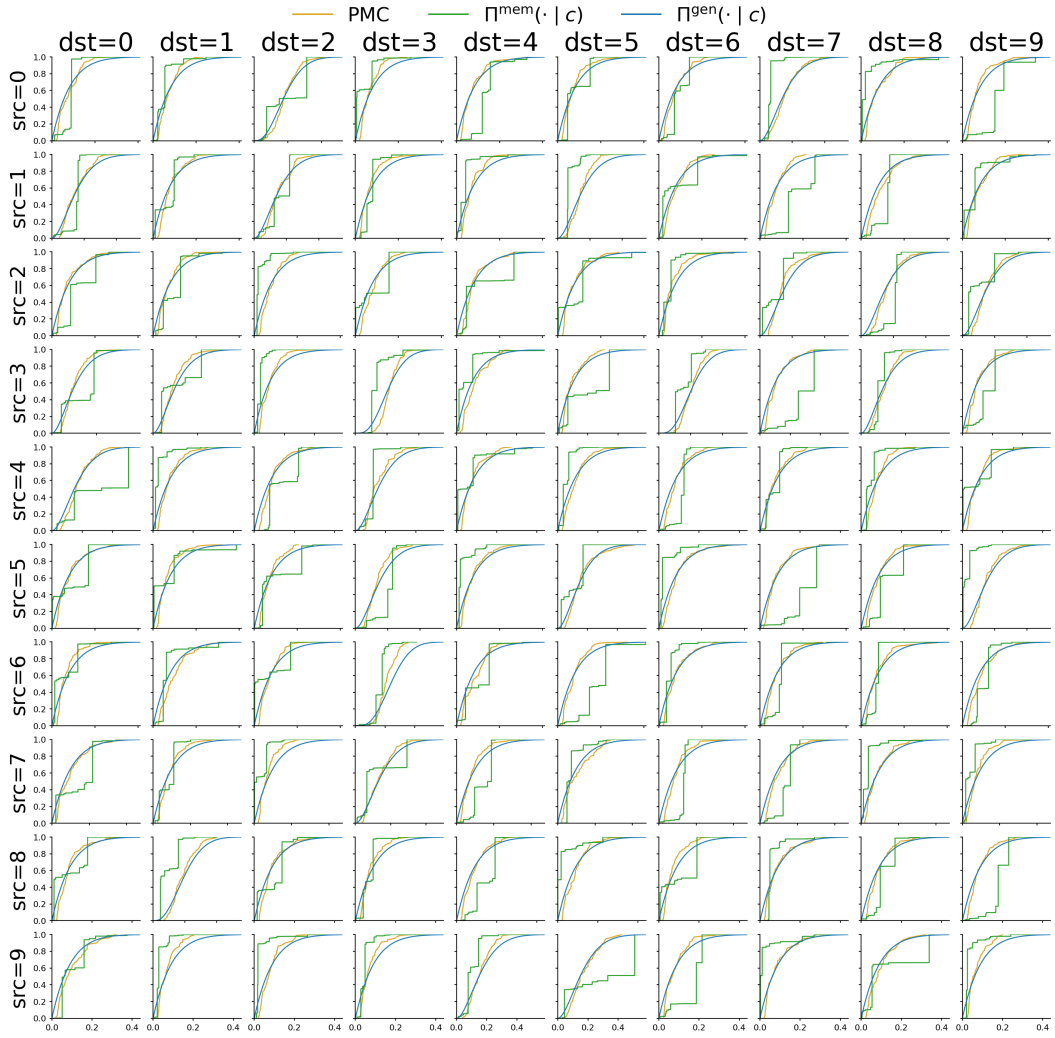

Figure 91: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 1024$.

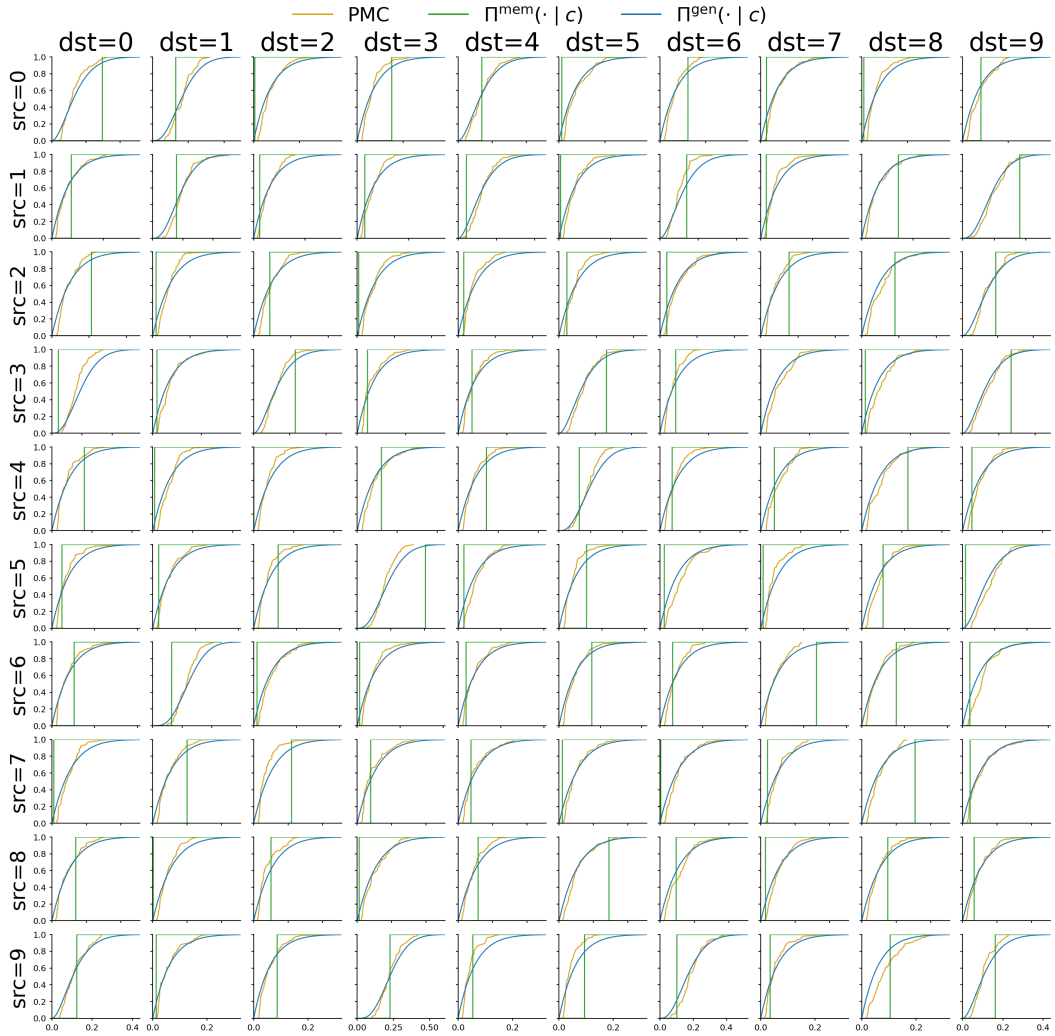

Figure 92: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 1024$.

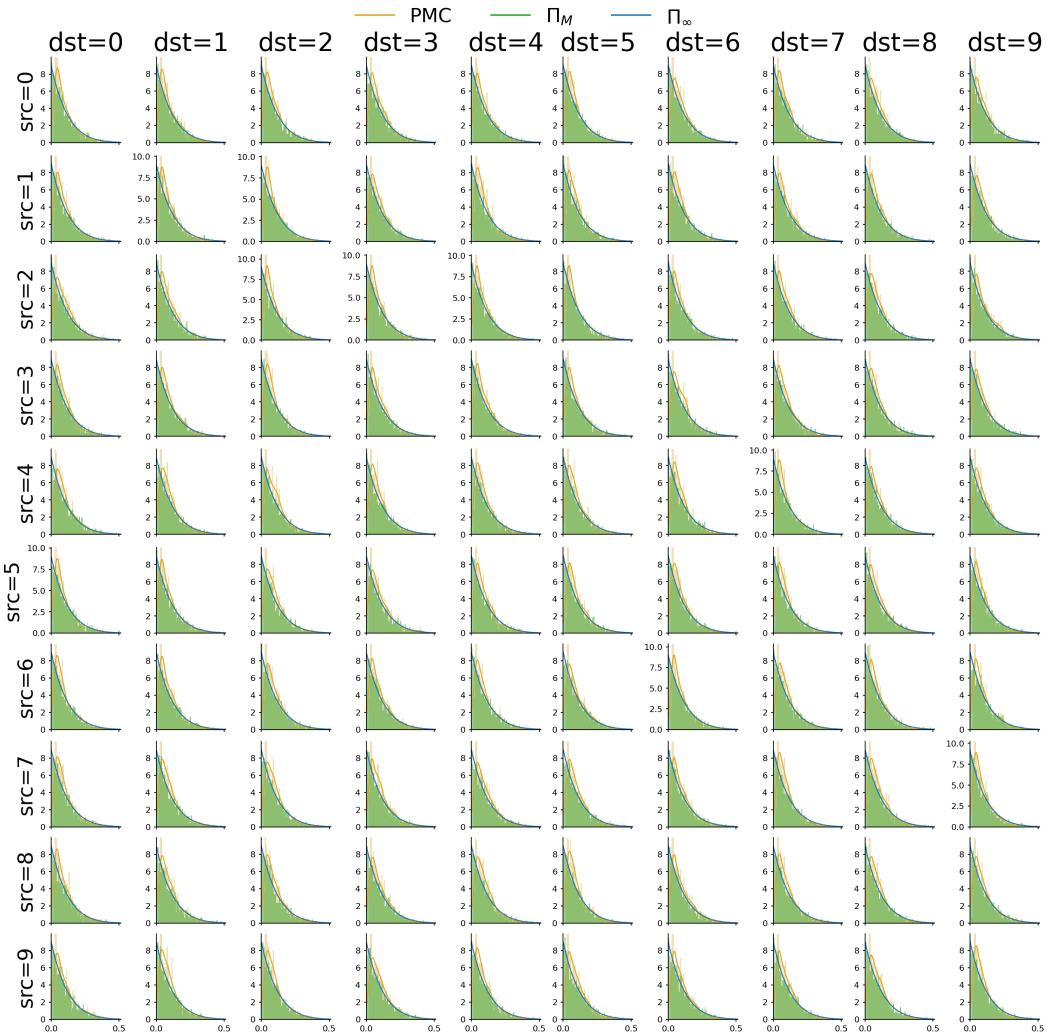

Figure 93: **Markov chains: PMC prior, marginal density at task diversity $M = 2048$.**

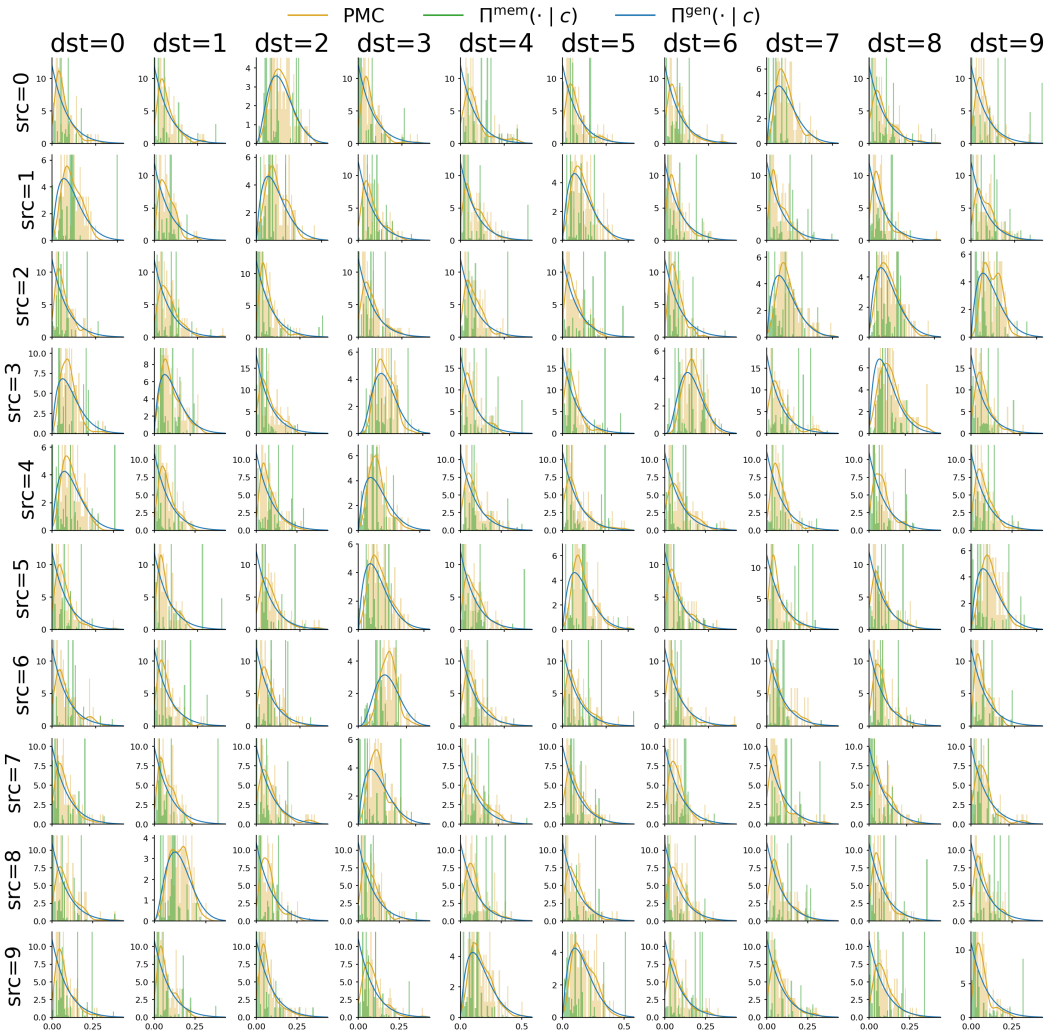

Figure 94: **Markov chains: PMC posterior, marginal density.** Out-of-distribution prompt of length 8 at task diversity $M = 2048$.

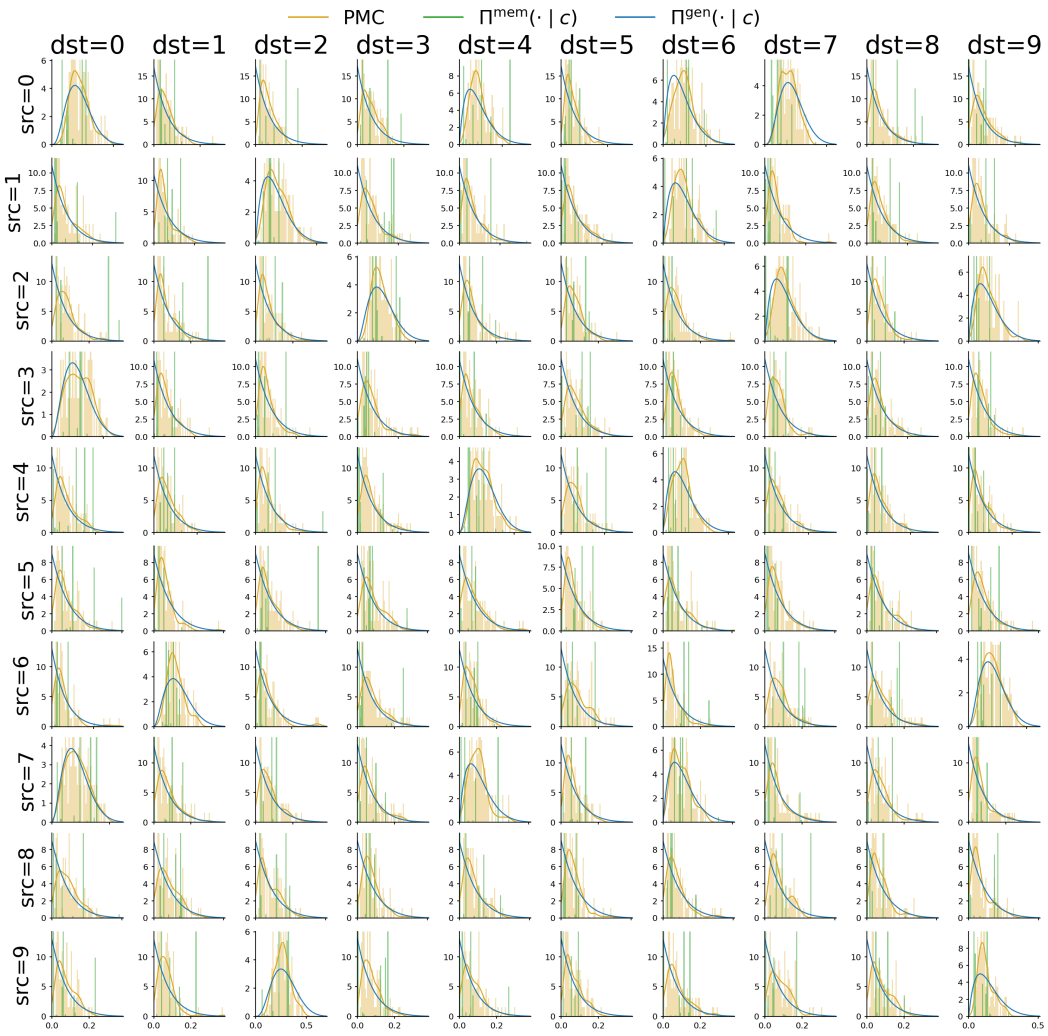

Figure 95: **Markov chains: PMC posterior, marginal density.** In-distribution prompt of length 8 at task diversity $M = 2048$.

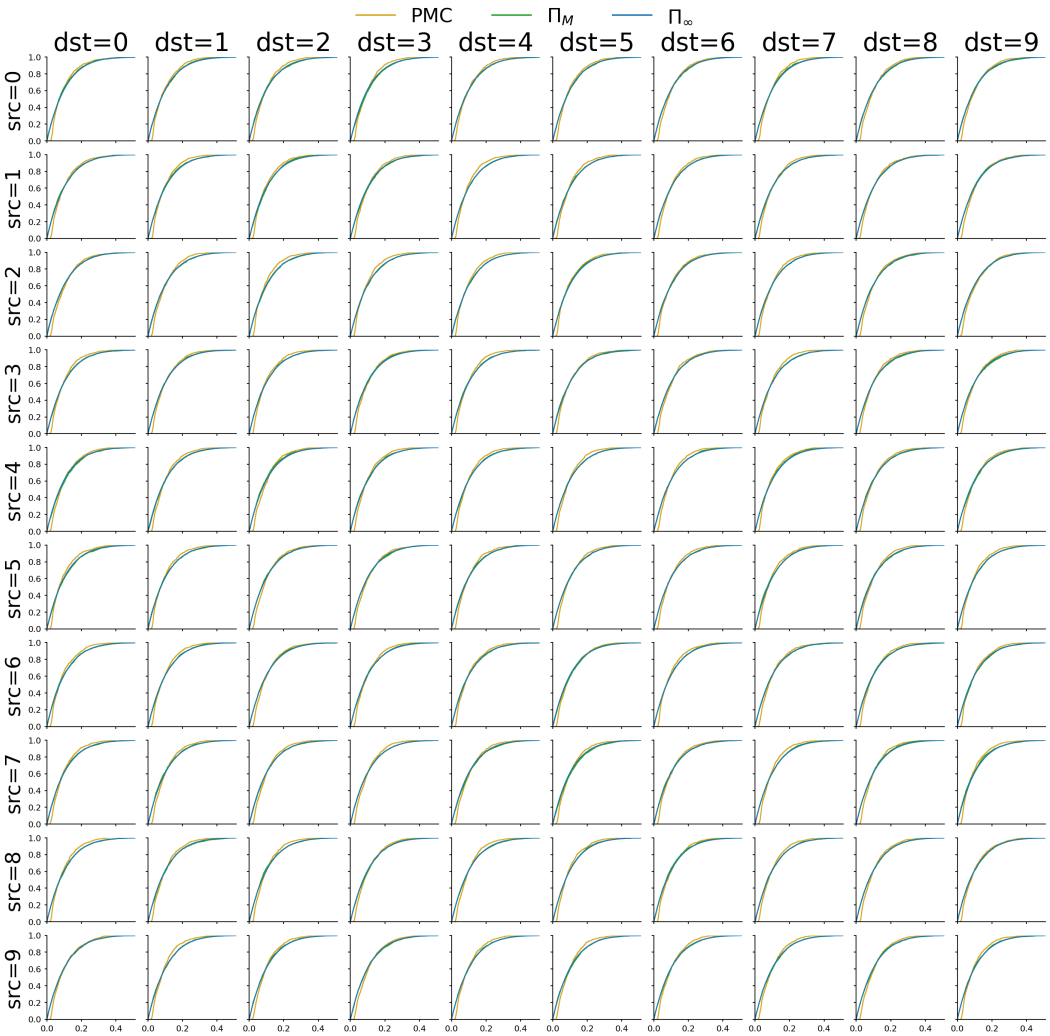

Figure 96: **Markov chains: PMC prior, marginal CDF at task diversity $M = 2048$.**

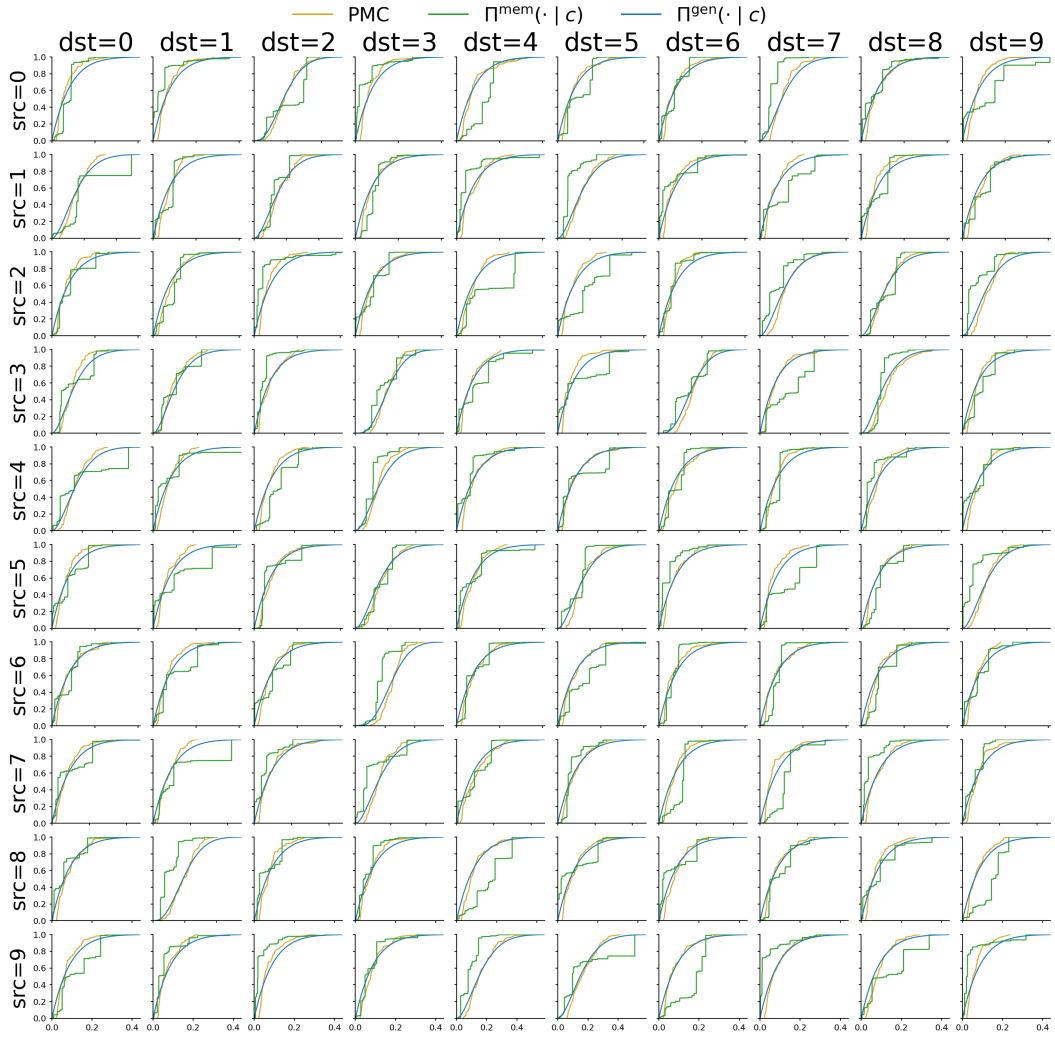

Figure 97: **Markov chains: PMC posterior, marginal CDF.** Out-of-distribution prompt of length 8 at task diversity $M = 2048$.

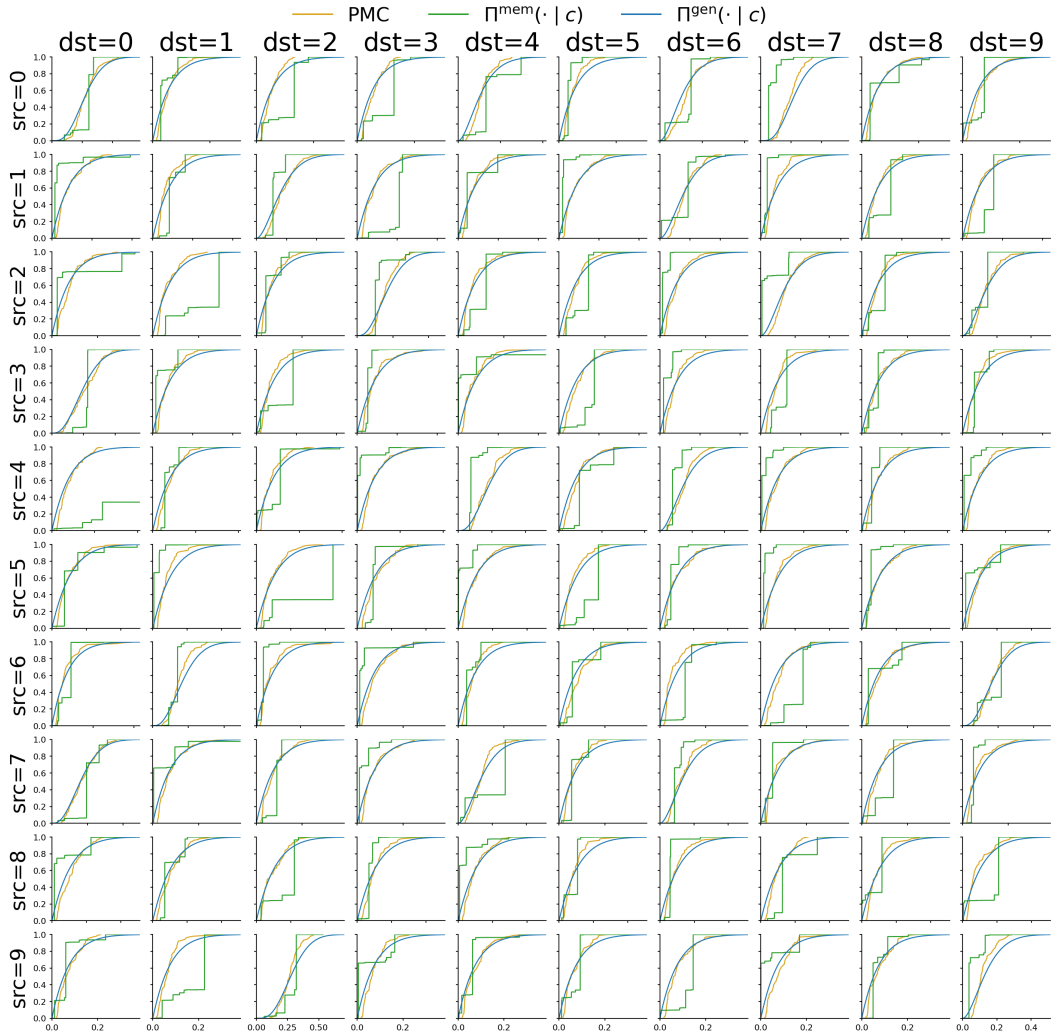

Figure 98: **Markov chains: PMC posterior, marginal CDF.** In-distribution prompt of length 8 at task diversity $M = 2048$.